Supply Chain Management for Smart Home Security System in Honeywell

Group 6

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Executive Summary

Proposal - Honeywell Pre-Processing

The Project Proposal created the foundation for designing the supply chain of the Honeywell Emergency Ready Lock, a safety smart home security device. The team defined core objectives including refining the product concept from TIM 172P, identifying major subsystems with a FAST diagram and Morphological Matrix, and constructing a preliminary supply chain infrastructure through stage and network diagrams. The product's differentiation strategy, centered on reliability (MEMS sensors), redundancy (hardwired and battery power), and high retail availability, was used to guide decisions around supplier qualification, transportation mode selection, and inventory policy. These outputs established the baseline for all subsequent phases.

Phase I - Supply Chain Strategy, Infrastructure, and Forecasting

Phase I presents our high-level design and management plan for the supply chain supporting the Emergency Ready Lock, a critical smart home device that prioritizes safety. The goal is to build a supply chain that aligns with Honeywell's differentiation priorities, including reliability, availability, and rapid response; allowing the product to meet high service expectations while reducing disruptions and quality risk across sourcing, assembly, and distribution. This is supported quantitatively by our demand estimation and forecasting results. Using an 8.5% CAGR and 8% market share, we project demand to grow from approximately 1,224,314 units in Year 1 to 1,328,381 units in Year 2. Seasonality further concentrates demand in Q2 and Q3 (30% each), with Q1 as the lowest (15%). Our static forecasting results also show a consistent negative bias and tracking signal trending from -1 to -8, indicating systematic over-forecasting that will need adjustment in later phases.

Phase II - Inventory Management & Initial Financial Modeling

Phase II continues to evaluate the supply chain; adding financial viability of the Emergency Ready Lock. Beginning with a demand estimation that applies quarterly seasonal weights to project sales growing from 16,800 units in 2024 to 42,917 units in Year 2. To project long term demand, five forecasting models were evaluated against standard error metrics (MSE, MAD, MAPE, and Tracking Signal). Winter's Method was selected for its optimal handling of trend, seasonality, and estimated demand projections. Inventory management for the finished product and three critical subsystems was subsequently optimized using Economic Order Quantity and a 95% Cycle Service Level, establishing a 17-day replenishment cycle for the finished lock. Integrating these operational requirements into a financial model revealed a $299 unit price, $115.28 unit production cost, and baseline capital/marketing expenditures at a 7% discount rate. This yielded a basecase Net Present Value of $25,466,000, confirming the product's strong financial feasibility.

Phase III - Operational Strategy and Financial Viability Analysis

Phase III focused on the quantitative modeling of market demand, supply chain logistics, and financial viability for the Emergency Ready Lock. The team executed a dual-method demand estimation to project a base-case demand of 28,878 units, 43,188 units, 53,555 units, and 64,254 units for Years 2026 through 2029, respectively. To optimize long-term planning, five forecasting models were tested, with Winter's method yielding the highest precision (MAPE 0.10) by accounting for a 35% Q4 seasonal surge. The team engineered a tiered inventory strategy for critical subsystems, calculating optimal lot sizes and re-order points that achieved a 94% cycle service level. These operational parameters were integrated into a comprehensive financial model, resulting in a base-case Net Present Value of $25,466,000. Sensitivity analysis confirmed project resilience, demonstrating that the investment remains profitable even under a 30% reduction in sales volume or significant R&D budget overruns.

Project Proposal: Honeywell Pre-Processing

1. Goals and Objectives

a. Project Goals:

i. To design and manage the supply chain for a Smart Home Security System in Honeywell.
⇒ This includes designing the physical network, selecting subsystem suppliers, and optimizing inventory and distribution strategies that closely align with Honeywell's own competitive strategy.

b. Project Proposal Objectives:

i. Establish a week-by-week project plan that assigns responsibilities, establishes a schedule for team meetings, and defines deliverables.
ii. Review and update the final product concept from the previous 172P lab, ensuring all feedback is reflected in our final product.
iii. Develop a clear and updated explanation of the final product concept, including identification of major subsystems and supporting design rationale.
iv. Construct a preliminary supply chain infrastructure based on the product's FAST diagram, progressing from subsystems to stage and network representations.
v. Analyze how the product's differentiation strategy (reliability, redundancy, and availability) impacts supply chain design decisions.
vi. Structure the proposal to reflect the plan provided in the assignment, with all steps clearly labeled.

c. Elaboration:

- The product differentiates on reliability (MEMS sensors) and redundancy (hardwire + battery). As such, the supply chain cannot be cheap and slow; it must ensure the components that are sourced for our product are high-quality (MEMS sensors are delicate) and that they have undergone rigorous testing.
- Since our product is directly in competition with Ring and SimpliSafe, our supply chain needs to support a "Differentiated Strategy", meaning that our product must have a high availability at major retailers for home improvement.
- Lead times and transportation modes must be selected to minimize handling risk and component damage, even if this results in higher unit costs compared to low-cost, offshore alternatives.
- Given the safety-critical nature of a home security system, inbound quality assurance and supplier qualification processes must be more stringent than those used for commodity consumer electronics.
- Inventory policies must balance high service levels with component sensitivity, ensuring sufficient safety stock while avoiding excessive storage time that could increase failure rates or degradation risk.

2. Final Conceptual Design Overview

This section outlines the proposed approach for developing and evaluating the product concept through a structured design flow process. We begin with a concept design description to establish the system's overall purpose, functional objectives, and design constraints. To further develop this concept, a morphological matrix is used to systematically explore alternative solution principles for each key function. From this matrix, the most suitable solution principles are selected and described in detail, with each mapped to a corresponding sub-system to clearly define how functional requirements are realized within the product architecture.

With the concept fully defined, we evaluate its strengths and weaknesses to assess feasibility, performance trade-offs, and potential areas for improvement. A conceptual product sketch is then developed to visually integrate the sub-systems into a cohesive configuration. To validate the functional logic of the proposed design, a FAST diagram is employed to illustrate the relationships between functions and ensure alignment with the original concept description.

We next present use cases to demonstrate how users are expected to interact with the product across representative operating scenarios. Building on these use cases, the proposal extends to a supply chain analysis, beginning with a stage representation that outlines the sequential flow of activities across the supply chain. Finally, a network representation captures the relationships and flows between supply chain entities, including materials and information. Together, these elements define a cohesive framework for evaluating the proposed product across design, user interaction, and supply chain dimensions.

Guided by the design methodology described above, the following section introduces the proposed product concept and defines its key functions and intended performance.

Emergency Ready Lock Concept

The **Emergency Ready Lock (Concept 6)** is a security system designed to prevent successful forced entry by detecting and escalating unauthorized intrusion attempts before a door or window is physically breached. Rather than responding only after access has been compromised, the system prioritizes early detection of abnormal mechanical activity and rapid escalation to alert occupants and, when necessary, external authorities.

Unlike conventional smart locks that emphasize access control and user convenience, the Emergency Ready Lock is designed around early intrusion detection, reliability, and emergency response, which were identified as high-priority criteria in our concept evaluation. Integrated vibration sensors detect high-energy tampering events such as hammering or drilling, allowing the system to respond before the locking mechanism is compromised. A stepper motor-driven locking mechanism provides precise and reliable actuation, while an integrated siren delivers immediate on-site deterrence.

To address reliability and failure mode risks, the system includes a rechargeable backup battery to ensure continued operation during power outages. A companion mobile application supports remote notifications and escalation capabilities; however, it functions as a supplementary interface rather than the primary safety mechanism, ensuring that core security functions remain operational independent of network connectivity.

To translate the conceptual product description into concrete design options, the following section presents a morphological matrix that explores alternative solution principles for each key function.

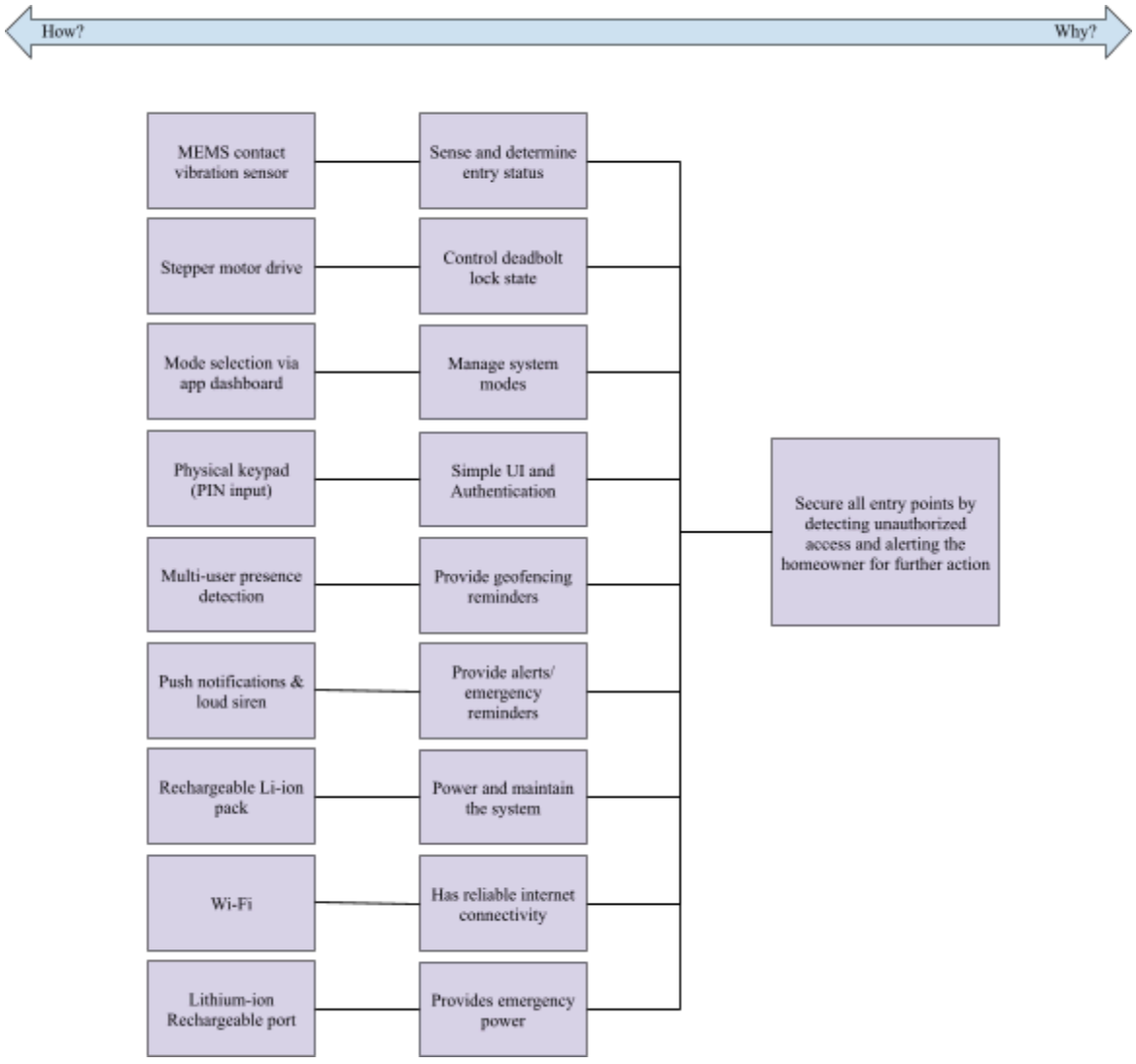
Morphological Matrix

We achieved the above design concept using a *morphological matrix*. This process allowed us to dissect our product "idea" into essential functions (*sub-functions*), like sensing, power, and user interface, and explore the various technical solutions (*solution principles*) for each. By identifying different alternatives, we were able to create a product concept with the most effective components to create a high-performance home security system.

Sub-Functions	SP1	SP2	SP3	SP4
Sense and determine entry status	Reed switch sensor (simple open/close detection)	Hall-effect sensor + small magnet	Optical interrupter sensor	MEMS contact vibration sensor (detects opening as a contact break)
Control deadbolt lock state	DC motor + gear train actuator	Stepper motor drive	Solenoid bolt actuator	Manual override key cylinder + motor assist
Manage system modes	Software logic toggling Home/Away/Disarmed	Mode selection via app dashboard	Auto-arming based on multi-user geofencing	Mode selection via lock keypad
Simple UI and Authentication	Physical keypad (PIN input)	Capacitive touchpad on the lock face	App-based authentication (PIN/biometric)	Voice-assistant integration (Alexa/Google/Siri)
Provide geofencing reminders	Basic GPS geofence on a single user	Multi-user presence detection (last user leaving triggers a reminder)	Wi-Fi/Bluetooth presence override (detect if someone is still home)	Location-based push notification via app
Provide alerts/emergency reminders	Push notifications (door opened while armed) & loud siren	SMS or email alerts	"Call law enforcement" button in app	Audible lock alarm (local siren on tamper/forced opening)
Power and maintain the system	AA/AAA replaceable battery pack	Rechargeable Li-ion pack	Hard-wire optional low-voltage power	Solar trickle charger (for exterior units)
Has reliable internet connectivity	Wi-Fi	Mobile Data	/	/
Provides Emergency Power	Power Terminals	Lithium-ion Rechargeable Port	Solar Power	/

FAST diagram

To validate that our selected sub-systems fulfill the requirements of our product, we developed a FAST diagram. This logic chain moves from the *primary function* (Secure all entry points by detecting unauthorized access and alerting the homeowner for further action), to *secondary functions*, like sensing the state of the entry point, controlling the deadbolt lock state, etc., and finally the *sub-system* that performs that function (MEMS contact vibration sensor, stepper motor drive, etc.). This way, we ensure that every component chosen has a logical and direct link to the user's safety.



Sub-Systems

From the matrix and FAST diagram above, the following sub-systems were selected, based on their ability to meet our core requirements and customer needs, like speed, reliability, and torque. The following sub-systems represent the best choices for our device. Evaluating these also allowed us to identify where the product excels (*strengths*) and where technical constraints (*weaknesses*) must be managed. We also decided to do a comparison regarding another concept to demonstrate why these sub-systems created a superior product compared to some of the other possible sub-system options.

Sub-System	Concept 6 (Highest Scoring)	Concept 1 (Lowest Scoring)	Comparison	Verified Potential Suppliers
Sense and determine entry status	MEMS Contact Sensor	Reed Switch Sensor	The **MEMS contact sensor** allows the security system to detect physical changes without needing any manual intervention. The **reed switch sensor** only activates when an inside user manually flips it into the on position.	Honeywell, Omron, Siemens
Control deadbolt lock state	Stepper Motor Drive	DC Motor + Gear Train Actuator	The **stepper motor** will provide the security system with the ability to apply high torque between two positions, which is ideal for the open and closed states of the deadbolt. This will also provide us with precise positioning. A **DC Motor + Gear Train Actuator** would require additional sensors for positioning and would provide weaker performance when the on-board battery is low.	Nidec, Oriental Motors, Moons
Manage System Mode	Mobile selection Via Application Dashboard	Software Logic Toggling (Home/Away/ Disarmed)	The **mobile application dashboard** gives users more options on top of the basic software toggling. This app would give users flexible mobility, allowing them to turn off & turn on the security system. The app also gives users a quick setting to alert law enforcement/paramedics if help is needed. The **software logic toggling** is restrictive and only gives users the option to either disarm or activate the security system via a keypad that is set up in a spot of the user's choice within	Apple(IOS), Google (Android)

			their home.	
Simple UI and Authentication	Physical Keypad	Physical Keypad	Both concepts utilize the same sub-system, the **physical keypad.** This function would be a simple UI that would be attached to the entrance of the house, allowing users to gain access via a numerical combination of their choosing.	JN White, ALMAX, Telamco, Techprint
Provide Geofencing reminders	Multi-user Presence Detection	GPS singular user	The **multi-user presence detection** uses a GPS that can be toggled within the app for user preference, which allows it to detect when the last user leaves the home, automatically activating the security system. This system allows for multiple users' locations to be accessed based on their permissions given via the mobile application. The **GPS singular user** roughly carries the same idea, with the only difference being that it can only track up to one user's location.	Broadcom, RealTek, Qualcomm
Provide alerts/emergency reminders	Push Notifications /Loud Siren	Push Notifications /Loud Siren	The **loud siren** with a high-decibel output, coupled with **push notifications** are used as an initial deterrent for intruders, coupled with an instant alert to the homeowner on the activity occurring, sent directly to their mobile device for prompt action.	Bose, Sony, Harman(speakers), Honeywell, Floyd Bell (sirens)
Power and maintain the system	Rechargeable Li-ion pack	Double/Triple A Replaceable Battery Pack	The **rechargeable Li-ion pack** has a higher energy density, allowing it to infuse more power into smaller units. It has an increased longevity and performs well in colder environments, all while providing consistent power. The **replaceable AAA battery pack** is convenient and affordable, but needs frequent replacements and has performance drops that steadily decline over time.	Panasonic, Samsung, BYD
Reliable Internet Connectivity	Wi-Fi	Wi-Fi	**Wi-fi** allows for real-time monitoring, supports app integration, and connects all the components to a hub where every	Broadcom, RealTek, Qualcomm

			aspect can be accessed through a 2.4 GHz network.	
Provides Emergency Power	Rechargeable Port	Power Terminals	A **rechargeable port** acts as a user-friendly, standardized source of power that has a medium power usage that is palatable to the typical user. Meanwhile, a **power terminal** tends to provide a high-current, unregulated power supply that requires specialized hardware and careful, supervised use.	Molex, Hirose, Amphenol

Concept 6 Strengths

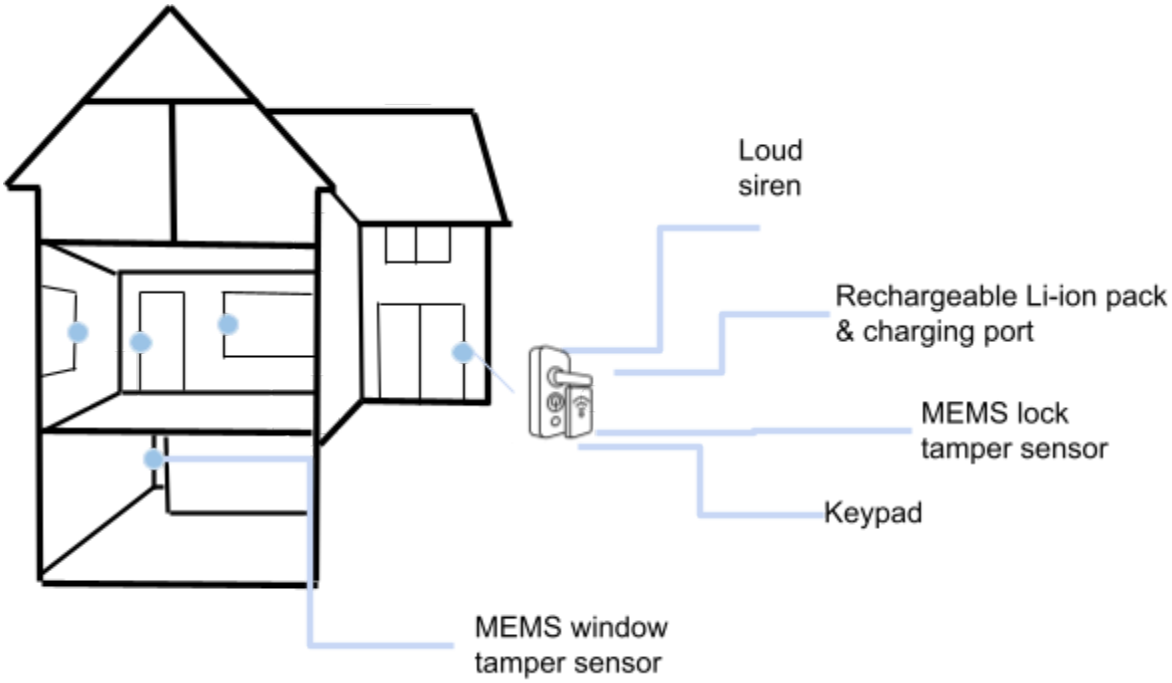
The security system's primary strength is the passive monitoring of active physical defense. The MEMS vibration sensor, when detecting a potential intruder, can trigger the stepper motor to apply torque to the deadbolt and sound the security alarm. This makes intrusion more challenging than traditional security locks. This converts the security system from passive to active, providing differentiation among competitors. This active response is only possible with the built-in lithium-ion battery, providing consistent and high power capabilities. The battery can be charged using a charging port, and during a power loss can also be powered. The integration of keypad authentication allows users to diminish vulnerabilities of losing a physical security key. This results in a highly powerful security system.

Concept 6 Weaknesses

The security system could potentially have remote connection issues during a power outage since it relies on Wi-Fi for remote connection. This means if the Wi-Fi does not work, users would be unable to turn on & off the system, and geofencing will not work correctly. The MEMS sensor would also require calibration to minimize false alarms. Without calibration, the sensor could be falsely triggered, leading to a quick battery drain. While these weaknesses exist, competitors also remain vulnerable to similar security issues.

Product Sketch

To visualize how these sub-systems integrate into a physical form, we created the following sketch that shows where the placement of the keypad, doorlock, and sensors are in the house.



Use Cases

The following scenarios demonstrate how the sub-systems work together in real-world situations. These use cases highlight how the logic programmed into the system handles every action, from daily automation to emergencies.

Primary Actor: Homeowner / Authorized User(s)

Goal: To ensure the home remains securely locked and capable of detecting and escalating intrusion attempts while occupants are away.

1. Automated Locking - Away Mode Scenario:
    1. The user sets the mode to “Away” via the mobile application.
    2. The system confirms that no additional authorized users are present in the home based on application geofencing.
    3. After confirmation, the stepper motor is enabled to extend the deadbolt and engage the lock, securing the home.
    4. The system confirms a successful locking event occurred and sends a confirmation notification to the homeowner.
    5. The home remains in a monitoring state. Utilizing the MEMS sensor, the sensor will actively sense for tampering or forced entry attempts.

2. Forced Entry/ Intruder Scenario:
    1. An unauthorized individual attempts to pry or tamper with the door.
    2. The MEMS sensor detects abnormal vibration patterns consistent with forced entry.
    3. The system immediately triggers the local siren to deter the intruder, then enables the stepper motor to apply torque on the deadbolt.
    4. A high-priority tamper alert is sent to the homeowner’s mobile device.
    5. The system remains locked and continues monitoring until the threat subsides or further escalation occurs through the application.

3. Authorized Entry Scenario:
    1. The homeowner returns but is unable to unlock the door via the keypad.
    2. The homeowner enters a valid PIN using the physical keypad.
    3. The system authenticates the input and disarms the alarm.
    4. The stepper motor retracts the deadbolt, allowing authorized entry.

4. Low Power Scenario:
    1. The system detects that the battery charge has fallen below a defined threshold.
    2. A low-battery alert is sent to the homeowner via the mobile application.
    3. The system continues operating in a power-conserving mode until recharged.

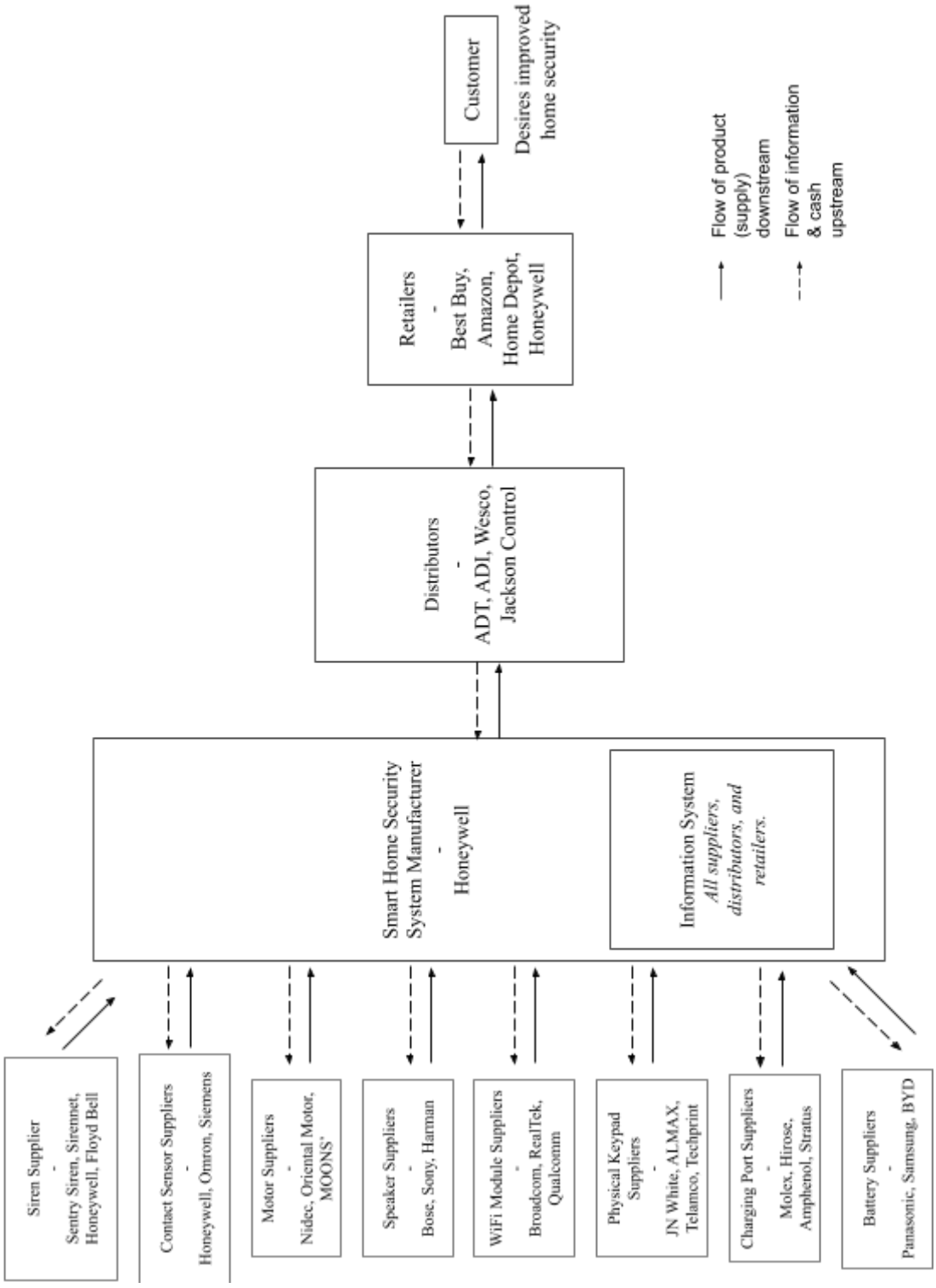
4. Preliminary Supply Chain Infrastructure

The following stage view illustrates the proposed supply chain structure for the Emergency Ready Lock, which is structured to support the nine core subsystems that we devised, moving from component sourcing to system integration.

Our upstream stage begins with the Tier 1 suppliers that provide the specific hardware, such as the sensor and siren, among many other subsystems; these feed directly into Honeywell's manufacturing operations. This is an "all-encompassing" stage, which includes manufacturing, but also assembly and testing operations.

The downstream flow utilizes Honeywell's existing distribution network to provide their product available to key retail partners (e.g., Best Buy, Amazon), which ensures high product availability that is required for a security device.

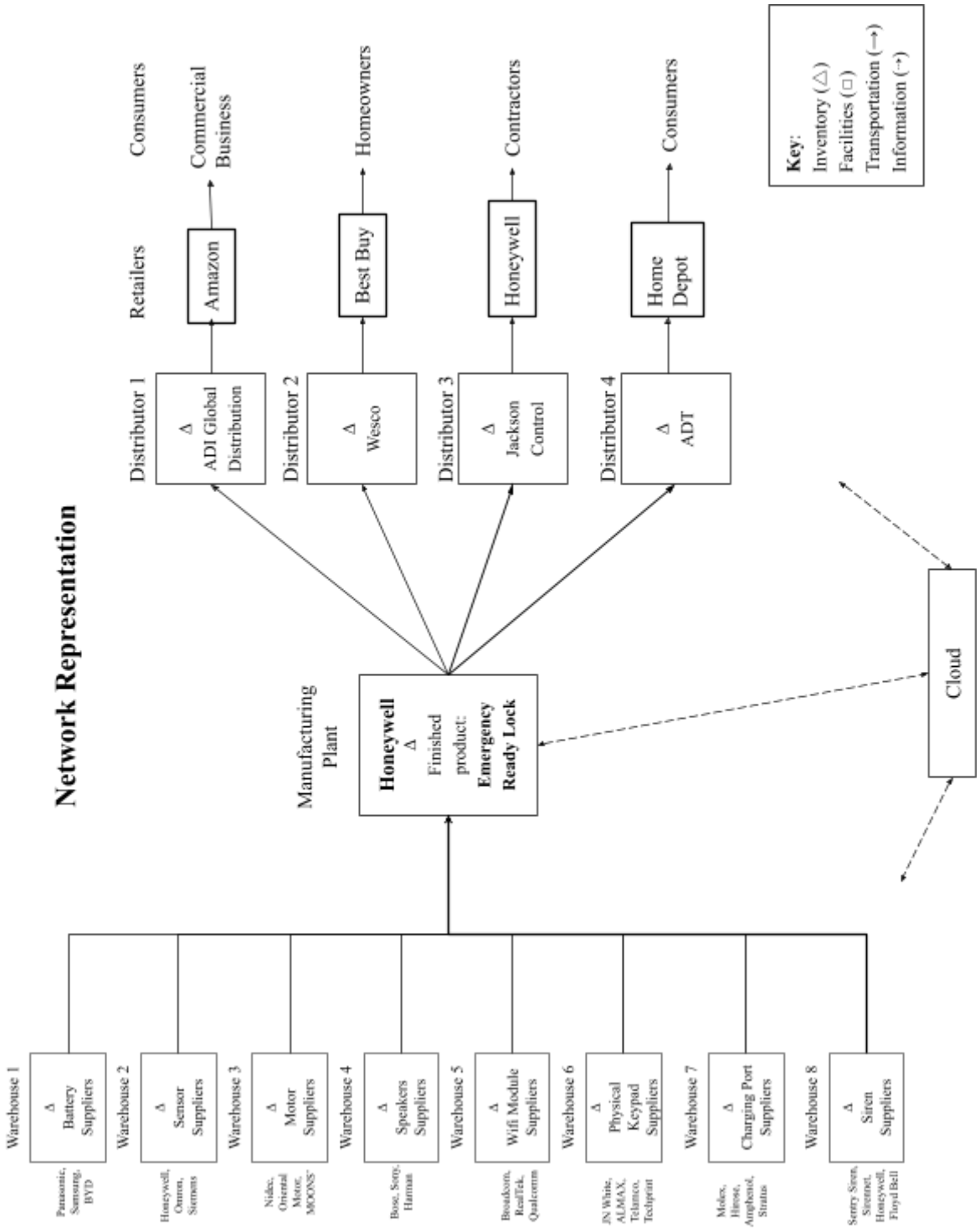
Stage Representation



The Stage View shows the linear flow of goods and information across the Emergency Ready Lock supply chain. Upstream, raw components are sourced from eight Tier-1 suppliers including siren, contact sensor, motor, speaker, WiFi module, physical keypad, charging port, and battery suppliers, and then delivered to Honeywell's centralized manufacturing plant. At this stage, components go through a strict incoming inspection before final assembly, calibration, and system-level testing are performed in-house. This ensures that safety-critical components such as the MEMS sensor and stepper motor meet Honeywell's quality standards before the product moves downstream. Once assembled and tested, the finished Emergency Ready Lock flows downstream from Honeywell's manufacturing plant to distributors including ADT, ADI, Wesco, and Jackson Control, who act as availability buffers to rapidly replenish retail partners. From the distributors, the product then flows to retailers including Best Buy, Amazon, Home Depot, and Honeywell's own direct channel, where it is made available to the end customer. Information flows in the opposite direction, moving upstream from customers to retailers, then to distributors, and finally back to Honeywell, providing demand signals, sales data, and inventory levels that help inform production planning and replenishment decisions.

Note: Our product includes 9 sub-systems; however, the Stage View and Network representation of the supply chain only includes 8 suppliers. We chose not to include the “Manage System Mode” sub-system as a supplier because this sub-system focuses on controlling the Smart Home Security System using software. The supply chain diagrams are representative of physical sub-system suppliers.

Network Representation



The Network Representation diagram maps out the supply chain network of the Honeywell Smart Home Security System. It illustrates how materials transform into finished products and then reach the consumer as the product that we created.

The network representation builds on the stage view by showing the specific structural relationships and flow paths between all supply chain entities. Goods flow outward from eight supplier warehouses, each storing a dedicated component category, into Honeywell's central manufacturing plant where the Emergency Ready Lock is assembled into its finished form. From the manufacturing plant, the finished product flows to four distributors: ADI Global Distribution, Wesco, Jackson Control, and ADT. Each distributor serves a specific retail channel where ADI Global supplies Amazon, Wesco supplies Best Buy, Jackson Control supplies Honeywell's direct channel, and ADT supplies Home Depot, ensuring broad market availability across both consumer and contractor segments. The end customers including homeowners, commercial businesses, and contractors receive the product through these retail channels. Information flows upstream through the same network, with sales and inventory data passing from retailers back through distributors to Honeywell, enabling responsive replenishment and production adjustments. Additionally, the product's built-in Wi-Fi connectivity creates a direct information link between deployed units and Honeywell's cloud infrastructure, providing real-time device data including battery status and signal strength to support continuous product improvement.

Phase 1: Supply Chain Strategy, Infrastructure, and Forecasting

1. Phase 1 Summary

Phase 1 develops the supply chain design for the Honeywell Emergency Ready Lock by linking the product concept to a stage view and network representation. A differentiated hybrid model is selected, keeping final assembly and testing in-house while sourcing key components from specialized suppliers. Based on customer needs and high implied demand uncertainty, the supply chain strategy prioritizes responsiveness over efficiency. The demand is then estimated and forecasted to support planning and ensure the design aligns with expected demand patterns.

Key Findings and Decisions (Steps 1 - 5):

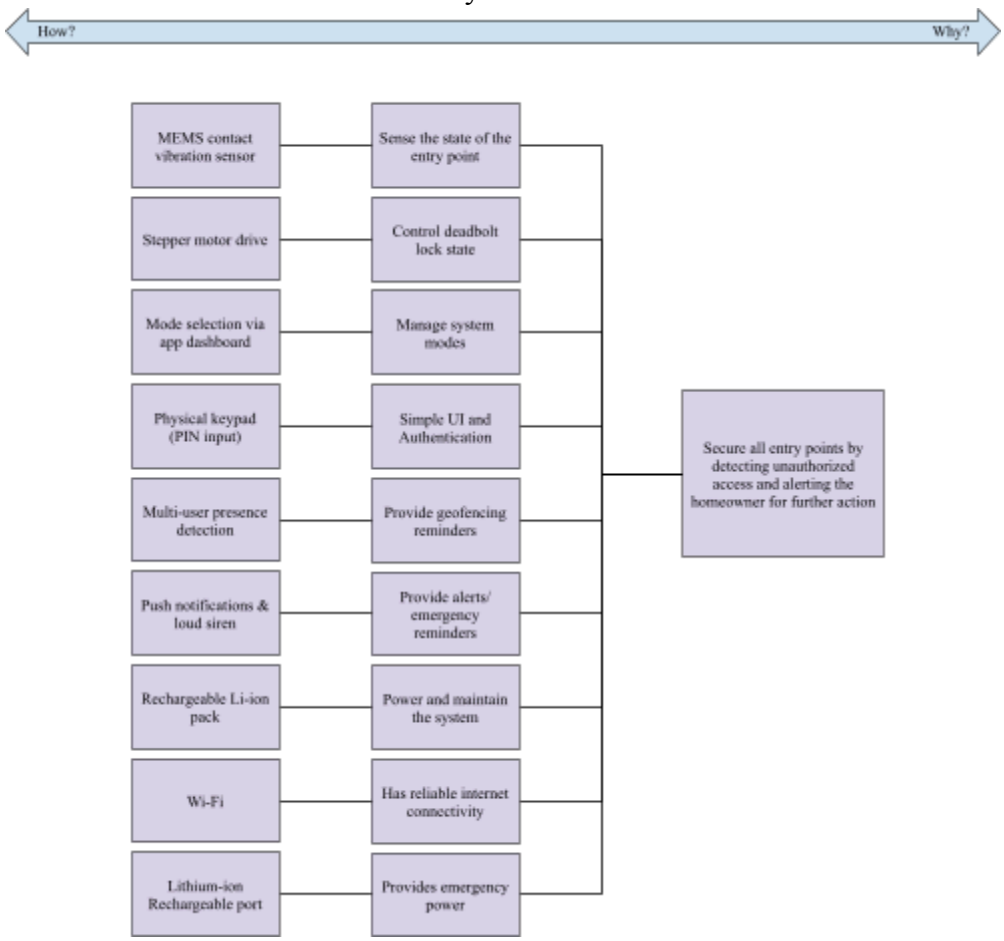
- **Business Mode:** A differentiated hybrid model best suits Honeywell for our product; they would retain the final integration, calibration/QA, firmware installation, and system-level testing in-house, while sourcing specialized components (MEMS sensors, stepper motors, Li-ion batteries, Wi-Fi modules) from Tier-1 suppliers. This borrows elements from Apple's quality-controlled integration and Dell's selective outsourcing.
- **Supply Chain Structure:** Eight upstream suppliers feed a centralized Honeywell assembly plant. Downstream, four distributors (ADT, ADI ,Wesco, Jackson Control) act as availability buffers to major retailers (Best Buy, Amazon, Home Depot) and professional installer channels.
- **Strategic Fit:** Our customer needs analyses showed high implied demand uncertainty due to variable order quantities, short lead-time expectations, multiple distribution channels, and a high required service level. The chosen strategy prioritizes responsiveness, placing the product within the zone of strategic fit.
- **Demand Estimates:** Using an 8.5% CAGR and Honeywell's 8% market share, Year 1 annual demand is estimated at about 1.2 million units and Year 2 at about 1.3 million units, distributed across quarters using seasonal indices (Q1: 15%, Q2: 30%, Q3: 30%, Q4: 25%). Our static forecasting applied to Year 3 - 4 (Q9 - Q16) resulted in forecasts with a MAPE of approximately 33%, with a consistently negative bias and tracking signal, which indicates serious under-forecasting that will be addressed in subsequent phases.

2. Concept Overview

The Emergency Ready Lock is a security system designed to prioritize rapid alerts, tamper detection, and fast emergency escalation. Unlike standard smart locks, it utilizes vibration sensors to detect forced entry attempts, such as hammering or drilling. The system features a stepper motor for lock control and a loud local siren to deter intruders. It also mitigates power failure risks by offering a hard-wired connection with a battery backup. Because of these performance requirements, the supply chain must prioritize reliability, rapid response, and component quality, which directly informs the hybrid sourcing and responsive strategy developed in the following sections. With the product architecture defined, we now translate these requirements into a structured supply chain representation.

FAST Diagram

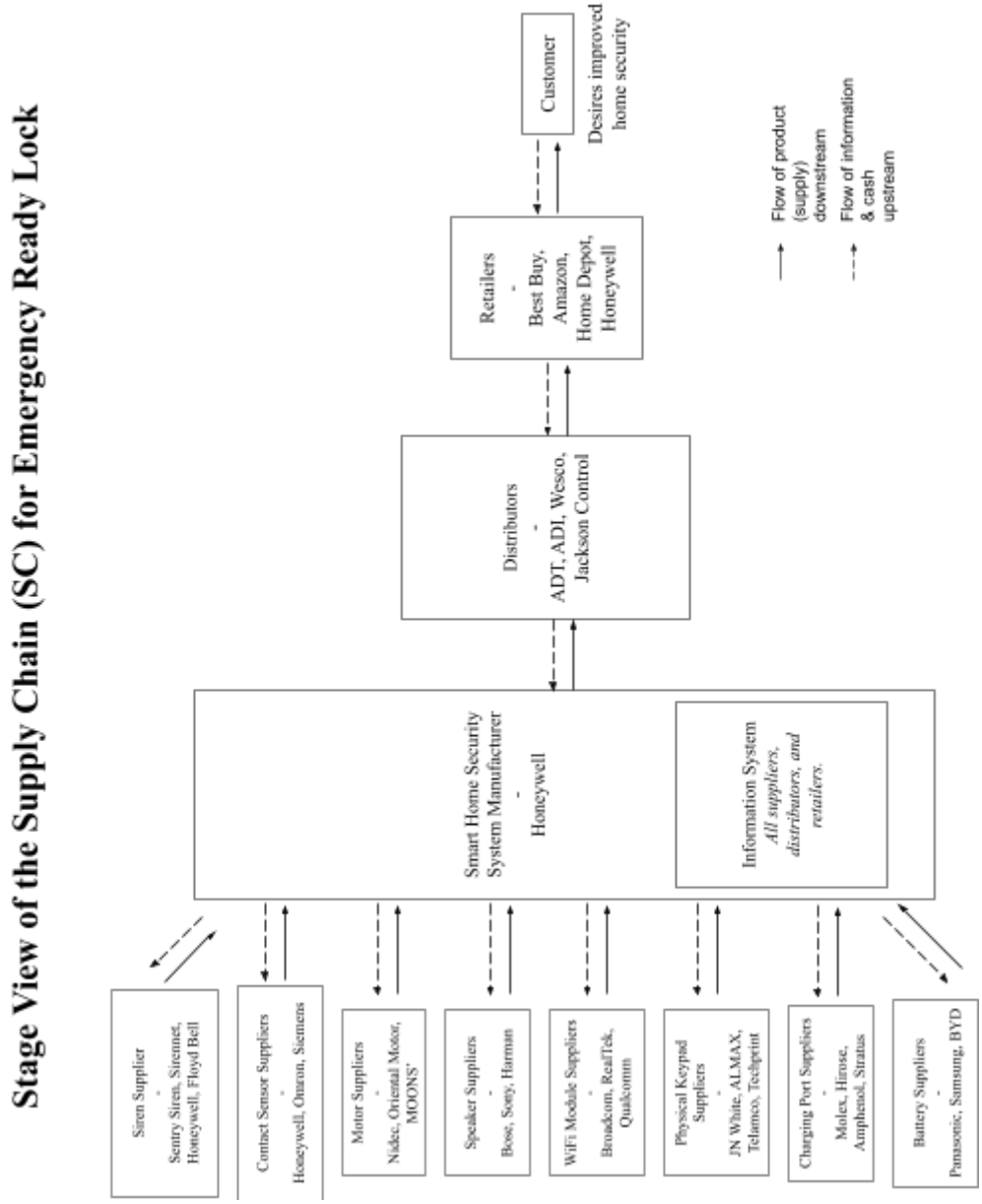
The selected sub-systems fulfill the description of our product. The FAST diagram chain moves from the primary function to the secondary function, to the sub-system that performs that function. Every component chosen has a logical and direct link to the user's safety.



Supply Chain Representation

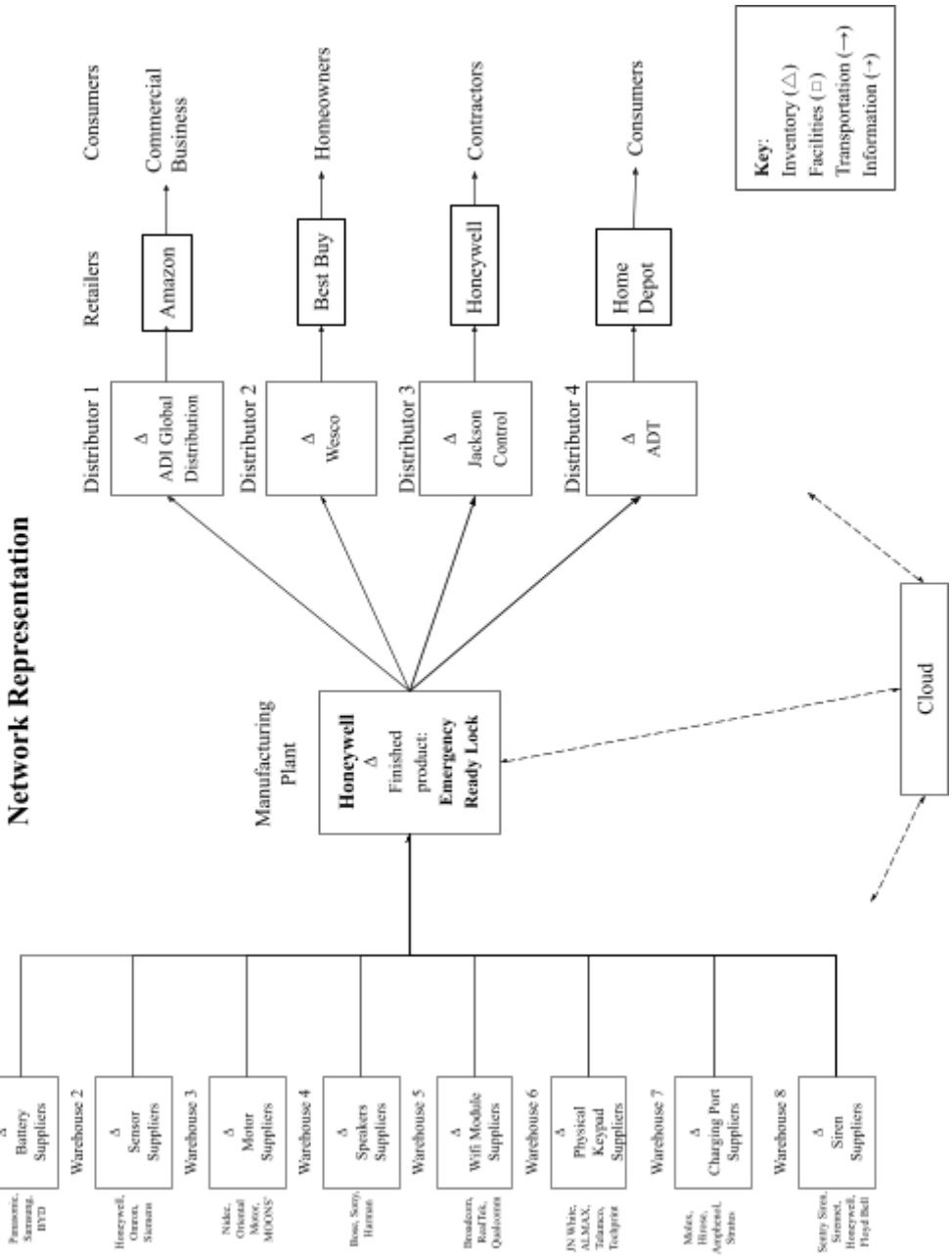
Our product includes 9 sub-systems; however, the Stage View and Network representation of the supply chain only includes 8 suppliers. For the purpose of our supply chain modeling, we decided to focus on the 8 physical hardware subsystems.

Stage View



The goal for our model is to highlight the linear flow designed with reliability and market availability in mind. Above, we source high-quality components from trusted suppliers, followed by a strict inspection process, and then ending with in-house assembly. Below, we use our established distribution network to ensure availability at major retail partners.

Network Representation



Our network uses a hybrid sourcing strategy, and reduces supplier-related risk while maintaining flexibility, which is critical given the high IDU and the need for consistent component performance across all production batches. Building on this structure, we next define the business model that governs sourcing, production, and distribution decisions.

3. Business Model for Honeywell's Smart Home Security System Supply Chain

The stage and network representations that we created above help define the structure of the supply chain, such as who supplies what, and how the product flows to the customer. Pivoting from that, the following section aims to explain how Honeywell manages these relationships. This section defines the business model that dictates sourcing decisions, integration boundaries, and distribution strategy, while also revisiting the stage and network diagrams from above to ensure that they are consistent with this model.

Comparison Table

Dimensions	Honeywell	Ecobee	Ring
Core Value Proposition	Safety-critical, reliability-driven security with early intrusion detection and redundancy	Energy efficiency, sustainability, and smart climate control	Consumer-friendly, camera-centric home security
Primary Revenue Model	Hardware sales through retail + potential premium service add-ons (alerts, integrations)	Hardware sales + energy services partnerships	Hardware sales + subscription monitoring & cloud services
Manufacturing Approach	Hybrid: in-house integration & testing + Tier-1 specialized suppliers	Mostly outsourced manufacturing	Largely outsourced, Amazon-optimized
Supplier Selection Criteria	Quality, reliability, certification, handling sensitivity	Cost + energy efficiency	Cost, scale, speed
Inventory Strategy	Higher safety stock to ensure availability and reliability	Smaller, more barebones inventory	Push toward fast turnover
Distribution Channels	Big-box retail (Home Depot, Best Buy), online	Direct-to-consumer + retail	Direct-to-consumer + Amazon

The comparison table is structured to emphasize supply-chain-relevant differences in business models, enabling a clear linkage between competitive strategy and operational design choices. Honeywell's approach most closely resembles a hybrid of Apple's tightly controlled integration model and Dell's hybrid sourcing strategy, where system architecture, testing, and final assembly are retained in-house. At the same time, specialized components are sourced from qualified Tier-1 suppliers. Apple's supply chain demonstrates that premium, reliability-driven products justify higher unit costs in exchange for tighter quality control and supplier

discipline, a principle that directly informs Honeywell's treatment of MEMS sensors, actuators, and power systems. Similarly, Dell's model illustrates how selective outsourcing can preserve flexibility and scalability without sacrificing system reliability, reinforcing Honeywell's decision to avoid full vertical integration. By contrasting this hybrid model with Ring's cost-optimized consumer electronics supply chain, the table justifies why Honeywell's supply chain decisions prioritize reliability, availability, and risk mitigation over lowest-cost sourcing for a safety-critical security system. This decision is further supported by the product's demand characteristics, including over 1.2 million projected units annually and high service-level expectations, which require tighter control over quality and faster response times than a fully outsourced model would allow.

Proposed Smart Home Security System Supply Chain Business Model

Honeywell should utilize a Differentiated Hybrid Supply Chain Model that balances in-house controls with specialized external suppliers.

Key Characteristics:

I. In-House Responsibilities:
   i. Final system integration
   ii. Calibration and quality assurance of MEMS sensors
   iii. Firmware installation and system-level testing

II. Externally Sourced Components:
   i. MEMS vibration sensors
   ii. Stepper motors
   iii. Lithium-ion battery packs
   iv. Wireless connectivity modules

The proposed differentiated hybrid supply chain model is justified by the product's need for high reliability, redundancy, and early intrusion detection, which are central to the Emergency Ready Lock concept. Fully outsourcing manufacturing would increase risk in calibration, testing, and integration of MEMS sensors and electromechanical subsystems, while full vertical integration would be inefficient and unnecessary. A hybrid approach allows Honeywell to retain control over final assembly, quality assurance, and system-level testing while leveraging specialized Tier-1 suppliers for high-precision components. This structure aligns with Honeywell's historical strengths as a systems integrator rather than a commodity electronics manufacturer. As a result, the supply chain directly supports the product's differentiated strategy by prioritizing performance consistency and availability over unit cost minimization.

Implications for Stage and Network Diagram

i. Stage Diagram
   1. Final assembly and testing remain centralized at Honeywell
   2. Supplier stages become more selective, emphasizing certified Tier-1 partners
   3. Quality assurance is elevated from a sub-step to a core stage activity

ii. Network Diagram
   1. Fewer, higher-quality suppliers per subsystem (risk reduction)
   2. Stronger, more direct links between Honeywell and distributors
   3. Distributors act as availability buffers, not just logistics nodes

The proposed business model follows the stage and network diagrams to reflect greater centralization of integration and testing activities at Honeywell. Since the system is safety-critical and includes sensitive subsystems, quality assurance cannot be treated as a peripheral activity and must be kept within Honeywell's manufacturing stage. The network diagram's emphasis on fewer, highly qualified suppliers reduces disruption risk and ensures consistent component performance across production batches. Distributors are positioned as availability buffers to support high service levels at major retailers, which aligns with the product's need for rapid replacement and consumer trust. These changes are justified because the differentiated business model requires tighter coordination, higher safety inventory, and stronger upstream and downstream control than a cost-driven consumer electronics supply chain.

This business model directly aligns with the stage and network representations by placing final assembly, testing, and quality control at Honeywell's centralized facility, while upstream suppliers provide specialized components. The distribution channels shown in the network diagram reflect this structure, where finished products move from centralized production to retailers and customers. This ensures consistency between sourcing decisions, production flow, and delivery to end users.

4. Supply Chain for Honeywell's Smart Home Security System

To accurately perform the implied demand uncertainty analysis, we must establish customer needs. To ensure our supply chain can withstand unpredictability, the following customer needs were identified, and their potential impacts:

Customer Needs for IDU

Customer Need		Impact
Range of Quantity	● The number of exterior doors and windows varies widely across homes. ● On average, most homes have 2 exterior entrances and 14 windows.	HIGH, INCREASES IDU
Lead time	● Customers view the security system as critical security infrastructure. ● They require the system within days after placing an order and within hours to replace malfunctioning sensors to ensure security integrity.	HIGH, INCREASES IDU
Variety of products	● The Emergency Ready Lock and Window/Door sensors are the two core components of the system. ● These components can be sold as a kit with many sensors or sold individually.	MEDIUM, INCREASES IDU
Number of channels	● Customers should expect our products to be installable on their own or by professionals. ● A complex amount of channels is required to support online marketplace, direct to consumers, and security system installer demand.	MEDIUM, INCREASES IDU
Rate of innovation	● Hardware innovation will be minimal, with security systems having a multi-year life cycle. ● Provides a stable and functional product, with minimal obsolescence.	LOW, DECREASES IDU
Required service level	● The security system requires a high service level since it is a critical infrastructure.	HIGH, INCREASES IDU

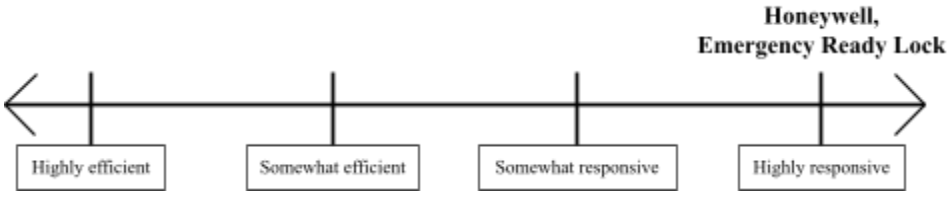
Based on the impact of the customer needs identified, the implied demand uncertainty has a strong lean toward HIGH IDU.

Implied Demand Uncertainty Spectrum:



Honeywell will focus on the Emergency Ready Lock through market-wide segments, with a differentiated hybrid supply chain model. This is possible through the variety of channels the product will support and the criticality of the hardware. This will prioritize responsiveness over efficiency in the supply chain.

Efficiency & Responsiveness Spectrum:



Supply Chain Drivers

Facilities

- Manufacturing: Centralized at Honeywell's manufacturing plant, which is where components will be delivered for final assembly.
- Warehousing: The security system will rely on upstream vendors to store sub-system components before they reach the manufacturing plant.
- Support: Reliance on local security system installers such as ADT, ADI, Wesco, and Jackson Control will provide local installers and specialists to customers.
- Retail: The security system & components will be available at Home Depot, Best Buy, and Amazon.

Inventory

- Push/Pull Strategy: The product will be pushed to security system installers and retail stores. From here, customers can create a pull in the supply chain to meet their needs.
- Responsiveness: Our supply chain will be highly responsive to meet critical customer needs, available within hours.

Transportation

- Flow: Sub systems will be delivered to Honeywell manufacturing, the finished product will be delivered to security system installers & retailers, who then provide it to the customer. This creates 4 primary stages in the supply chain.
- Speed: The product must be available everywhere with rapid replenishment, which is a major differentiation factor.

Information

- Downstream: Products will flow toward customers.
- Upstream: Information will flow toward Honeywell, providing supply and demand metrics to better forecast future demand.
- User connectivity: The product will have built-in Wi-Fi, allowing Honey will to collect data on these devices, including signal strength, battery life, and other metrics to improve future products.

Chosen Supply Chain Strategy

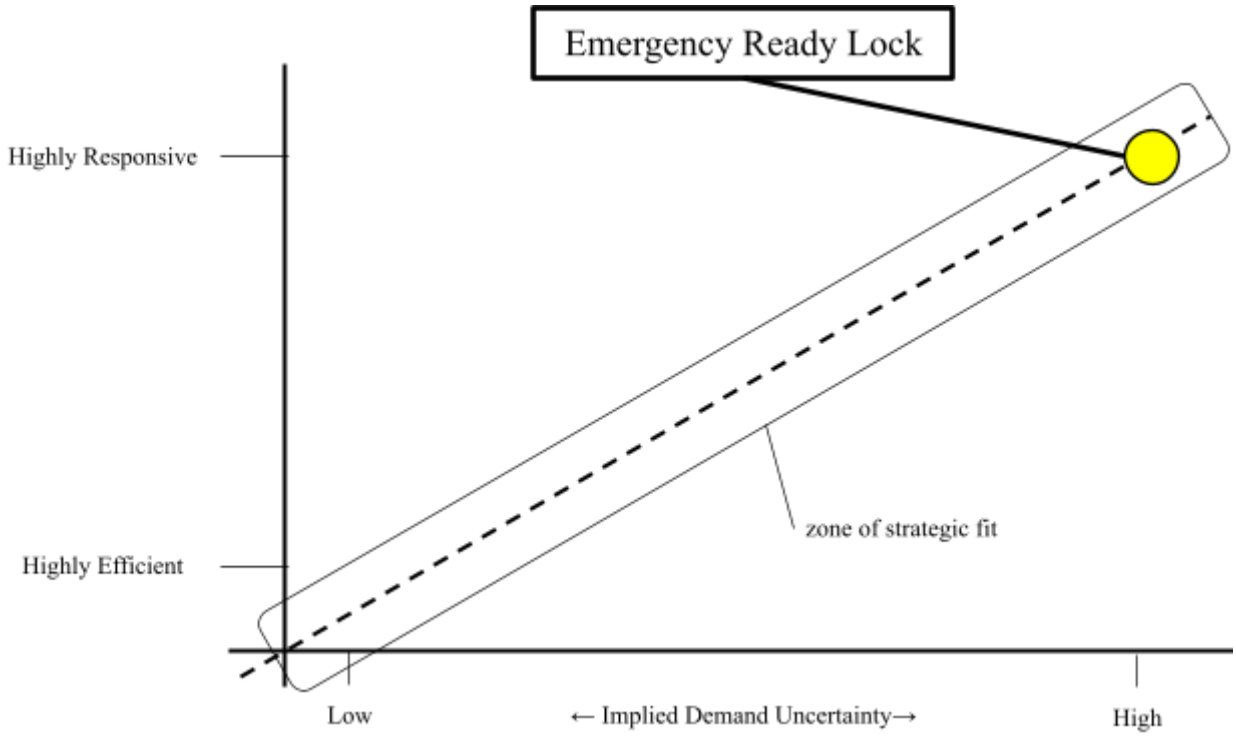
The IDU analysis that we performed above identified that the Emergency Ready Lock faces high implied demand uncertainty, mostly due to variable order quantities, short lead-time expectations, a high required service level, and multiple distribution channels. Of the six customer needs that we evaluated, these five push IDU upward, while only the low rate of hardware innovation provides a counterbalance.

Given this high IDU, the supply chain must prioritize responsiveness in the form of speed, flexibility, and reliable service levels. This is the best decision for the supply chain, as cost efficiency would prove to be an unreliable and ineffective strategy. This can be achieved through higher safety stock at distributor locations, rapid replenishment cycles from the centralized Honeywell plant, and multi-channel availability (retail, online, and professional installer networks). The resulting position on the efficiency-responsiveness spectrum falls toward the “highly responsive” mark which places the product within the zone of strategic fit.

This supply chain strategy aligns with the competitive strategy of differentiation by prioritizing reliability, product quality, and rapid response. Since the product competes on performance and service rather than cost, a responsive supply chain ensures high availability and consistent quality, supporting the overall competitive positioning.

Strategic Fit

Our product must be inside the “Zone of Strategic Fit” to ensure customer needs are met. This is verified by graphing our placement of efficiency & responsiveness against our placement of implied demand uncertainty.



After graphing our efficiency & responsiveness against our placement of implied demand uncertainty, we have identified that the product is within the “Zone of Strategic Fit,” which means our supply chain model aligns with customer needs. This aligns well with the critical needs our product is serving and showcases a highly responsive supply chain.

5. Demand Estimation Process (Years 1 and 2, Q1-Q8)

Why Static?
Static forecasting is the appropriate method for the Honeywell Emergency Ready Lock based on the following factors:

1. New product with Limited Historical Data
   Qualitative forecasting methods are most appropriate when little historical data is available. As a new product entering Year 1, the emergency-ready lock has no pre-launch demand history.

2. Zone of Strategic Fit
   Our supply chain prioritizes responsiveness over efficiency. Static forecasting provides responsiveness over efficiency. Static forecasting provides clear insight into the level, trend, and seasonality

Estimating Demand Approach

According to SafeHome.org, the Home Security market estimates that 13 million new homes will get a security system in 2024. Honeywell currently holds 8% of the market share. So, the baseline for Honeywell in 2024 is approximately 1,040,000 units (13M × 8%).

However, our product was designed in the year 2025 (Y0). So, to forecast into Y1 and Y2 (2026 and 2027), we will apply a Compound Annual Growth Rate formula (CAGR).

Step 1: Start with a baseline of 13,000,000 units in 2024. To find the next year, we will apply the growth rate of 8.5% (0.085):

Future Value = Present Value × (1 + Growth Rate)

2025 Projection: 13,000,000 × 1.085 ≈ 14,105,000
2026 (Year 1) Projection: 14,105,000 × 1.085 ≈ 15,303,925
2027 (Year 2) Projection: 15,303,925 × 1.085 ≈ 16,604,758

Step 2: Calculating Honeywell's Annual Demand
Once we have the total market, we will multiply it by the 8% market share to find Honeywell's share:
Annual Demand = Total Industry Units × 0.08

2026 (Year 1): 15,303,925 × 0.08 ≈ 1,224,314 units
2027 (Year 2): 16,604,758 × 0.08 ≈ 1,328,381 units

Year	Total Industry Units (Estimated)	Honeywell Market Share (8%)	Annual Unit Demand
2024	13,000,000	8%	1,040,000

2025	14,105,000	8%	1,128,400
2026 (Year 1)	15,303,925	8%	1,224,314
2027 (Year 2)	16,604,758	8%	1,328,381

Step 3: Since the product was designed in 2025 (Year 0), we would need to transition from these annual numbers to actionable quarterly data

The estimated demand is not flat. It follows a seasonal trend and is influenced by holiday spikes and moving seasons. So we will apply a seasonal index to distribute the annual totals.

Seasonal weighting:

- Quarter 2 and Quarter 3 (High → 30% each): High volume of new home purchases and renovations
- Quarter 4 (Peak → 25%): Holiday season/spike, as Black Friday and Cyber Monday increase sales
- Quarter 1 (Low → 15%): Post-holiday slump and colder weather, so fewer moves

Based on these seasonal factors and our estimated annual demand, we will calculate the quarterly demands.

*Seasonal factors: 15% + 30% + 30% + 25% = 100%

Year / Annual demand	Quarter	Seasonal Factor	Estimated Demand (Annual demand × Seasonal factor)
Year 1: 1,224,314	Q1	15%	183,647
	Q2	30%	367,294
	Q3	30%	367,294
	Q4	25%	306,079
Year 2: 1,328,381	Q1	15%	199,257
	Q2	30%	398,514
	Q3	30%	398,514
	Q4	25%	332,095

Using these estimates as a foundation, we extend the analysis into forecasting future demand and evaluating model performance.

6. Demand Forecasting Process (Years 3 and 4, Q9-Q16)

Static Forecasting Implementation

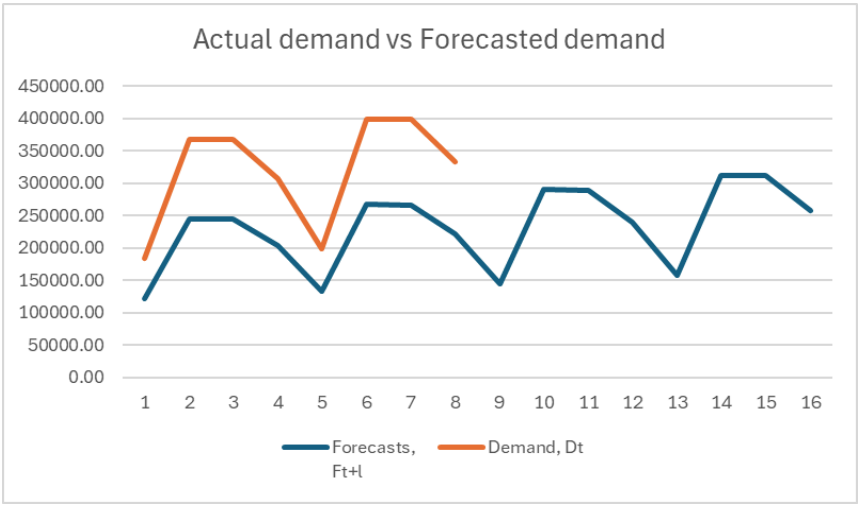
Year	Quarter	Period, t	Demand, D_t	Deseasonalized Demand, $D_t^{-'}$	Regressed deseasonalized demand, D_t^-	Seasonal Factor, S_i	Average Seasonal factor, S_i^-	Forecasts, F_{t+l}
1	1	1	183,647		293525.5	0.63	0.4151	121832.12
1	2	2	367,294		300550.0	1.22	0.8116	243912.29
1	3	3	367,294	308029.75	307574.5	1.19	0.7938	244149.95
1	4	4	306,079	313883.50	314599.0	0.97	0.6473	203648.45
2	1	5	199,257	321688.50	321623.4	0.62	0.4151	133494.57
2	2	6	398,514	328843.00	328647.9	1.21	0.8116	266715.23
2	3	7	398,514		335672.4	1.19	0.7938	266453.82
2	4	8	332,095		342696.9	0.97	0.6473	221836.98
3	1	9			349721.3		0.4151	145157.02
3	2	10			356745.8		0.8116	289518.17
3	3	11			363770.3		0.7938	288757.69
3	4	12			370794.8		0.6473	240025.52
4	1	13			377819.2		0.4151	156819.47
4	2	14			384843.7		0.8116	312321.10
4	3	15			391868.2		0.7938	311061.56
4	4	16			398892.7		0.6473	258214.05

Errors:

Forecast Error	Mean Square Error	Absolute Error or Deviation	Mean Absolute Deviation	Percent Error	Absolute Percent Error	Mean Absolute Percent Error	Bias	Tracking Signal
-61814.88	3821079192.39	61814.88	61814.88	-33.66	33.66	33.66	-61814.88	-1.00
-123381.71	9522062927.06	123381.71	92598.29	-33.59	33.59	33.63	-185196.59	-2.00
-123144.05	11402860730.49	123144.05	102780.21	-33.53	33.53	33.59	-308340.64	-3.00
-102430.55	11175149917.07	102430.55	102692.80	-33.47	33.47	33.56	-410771.19	-4.00
-65762.43	9805059330.71	65762.43	95306.72	-33.00	33.00	33.45	-476533.61	-5.00
-131798.77	11066035533.48	131798.77	101388.73	-33.07	33.07	33.39	-608332.39	-6.00
-132060.18	11976586243.11	132060.18	105770.37	-33.14	33.14	33.35	-740392.57	-7.00
-110258.02	11999116702.68	110258.02	106331.32	-33.20	33.20	33.33	-850650.58	-8.00

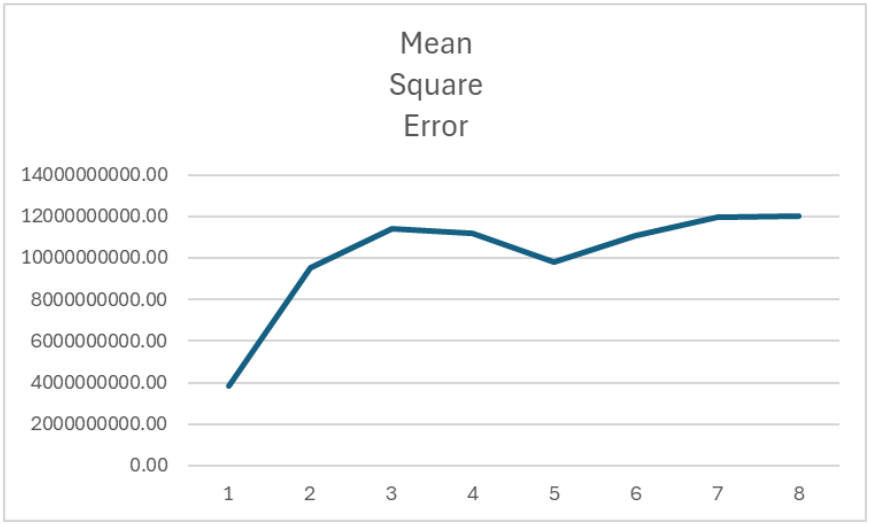
Error Metrics graphs:

Demand vs. Forecasting



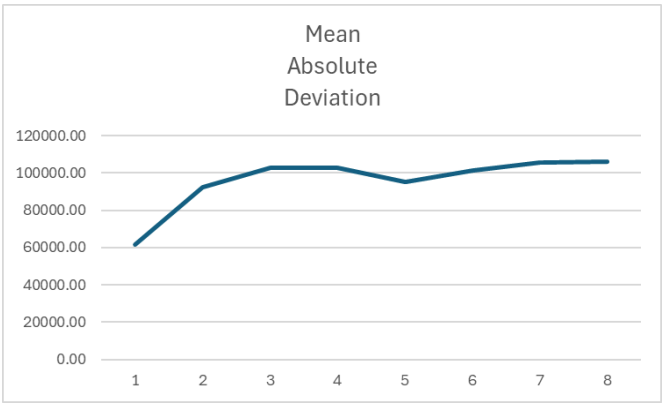
This graph compares actual demand with the forecasted values over time. The forecasts range from about 121,832 to 311,061 units, while actual demand ranges from about 183,647 to 398,514 units. The model captures the seasonal pattern, but consistently underestimates demand in most periods. The gap between the lines shows a clear and repeated difference in magnitude.

MSE vs. Period



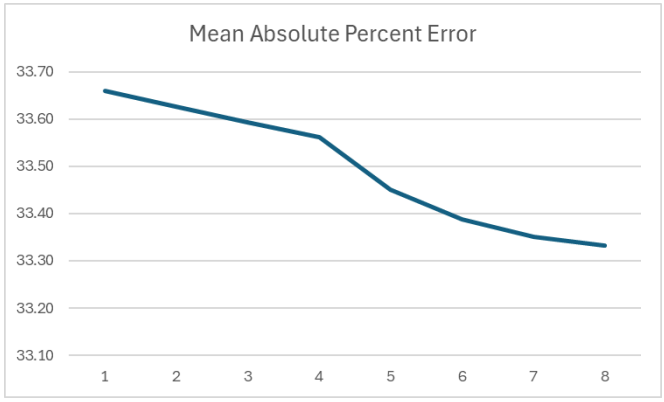
This graph shows how the squared error changes over time. The values range from about 3.82 billion to 11.99 billion. The error increases early on and then stabilizes at a high level. This shows that the model consistently produces large errors.

MAD vs. Period



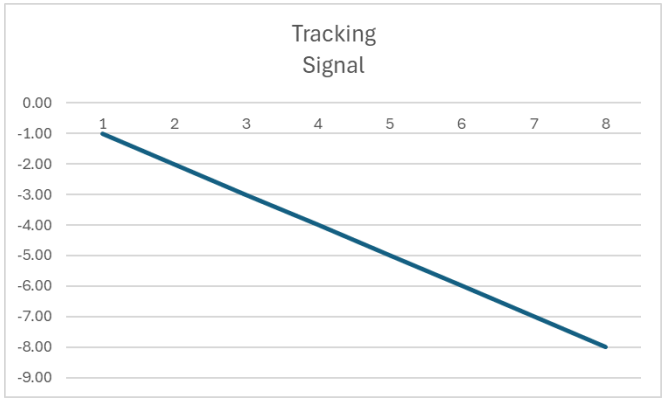
This graph shows the average size of the errors. The MAD increases from about 61,814 to around 106,331 units over time. This indicates that the model's errors are consistently large and do not significantly improve.

MAPE vs. Period



This graph shows the percentage error over time. The values stay between about 33.66 percent and 33.33 percent. There is a slight decrease, but the error remains high. This suggests the model has moderate accuracy at best.

Period vs Tracking Signal



This graph shows the tracking signal values. It decreases from -1.00 to -8.00 across periods. This confirms a strong and increasing negative bias.

Bringing together these analyses, we can now evaluate how well the supply chain design supports both demand and operational requirements.

7. Conclusion

Our Phase 1 report creates the foundation for the supply chain design for our product, the Emergency Ready Lock. We've taken our FAST diagrams and turned them into a Stage View, as well as a Network Representation that reflects our approach towards a differentiated hybrid business model. The model is simple: Honeywell keeps in-house control of final integration and system testing while sourcing special components from trusted suppliers. This model is directly supported by our competitive strategy of relying on reliability, redundancy, and abundant availability. Our IDU analysis helped confirm that we need a highly responsive supply chain, one with short lead times, variable quantities, and critical service levels. To further our points regarding how our strategy would affect us quantitatively, we estimated Year 1 and Year 2 demand using CAGR-based market projections as well as a Static forecasting approach to create a 16 quarter forecast. Overall, these elements help create a reasonable and defensible supply chain foundation, one that aligns product design, business model, operational strategy, and demand planning.

Phase 2: Inventory Management & Initial Financial Modeling

1. Phase 2 Summary

This section presents a supply chain analysis for the Honeywell Emergency Ready Lock, a smart home security system. The supply chain is structured around specialized suppliers providing key components to Honeywell's manufacturing and assembly facility.

Demand Estimation. Demand was estimated using the U.S. home security market, projected at 1.5 million new installs in 2024. With Honeywell holding an 8% market share, the baseline annual demand is 120,000 units, with an 8.5% growth rate applied through 2027. A Product Penetration Factor (PPF) was modeled using a logistic S curve, with a launch year PPF of 14% and a long run ceiling of 35%, producing annual demand estimates of 16,800 units in 2024, 24,738 in 2025, 33,904 in Year 1, and 42,917 in Year 2. Quarterly seasonal weights of 15% (Q1), 30% (Q2), 20% (Q3), and 35% (Q4) were then applied, yielding estimates from 5,086 units in Year 1 Q1 to 15,021 units in Year 2 Q4.

Demand Forecasting. Five methods were tested against the eight quarters of estimated demand including Static, Moving Average, Simple Exponential Smoothing, Holt's Method, and Winter's Method, and evaluated using MSE, MAD, MAPE, and Tracking Signal. Winter's Method was selected because it is the only method that captures both trend and seasonality simultaneously. Using smoothing constants of 0.3 (Level), 0.2 (Trend), and 0.3 (Seasonal Factor), it achieved the lowest MAPE of all methods tested. Projected annual demand grows from 28,878 units in Year 1 to 64,254 units in Year 4, a 122% increase.

Inventory Management. Cycle and safety inventory were analyzed for three subsystems including the Contact Sensor ($21.99/unit), Speaker ($3.52/unit), and Motor ($9.60/unit), as well as the finished product ($115.28/unit), using EOQ and a 95% Cycle Service Level ($z = 1.65$). Contact Sensor lot sizes grow from 2,562 to 3,822 units with a 22 to 32 day replenishment cycle; Speaker lot sizes peak above 9,500 units by 2029; Motor lot sizes scale from 3,878 to 5,785 units; and the finished product uses a lean approach with an 11 to 17 day replenishment cycle and average flow time under 9 days. By Year 4, annual safety inventory holding costs reach $58,434 for Contact Sensors, $13,208 for Speakers, $36,020 for Motors, and approximately $216,275 for the finished product. To reduce the Bullwhip Effect, Honeywell should update lot sizes semi annually.

Financial Modeling. The NPV model was built using a unit price of $299.00, unit production cost of $115.28, development cost of $60,480, ramp up cost of $6,048, annual marketing and support cost of $840,000, and a 7% discount factor, with sales volumes drawn from the Winter's Method forecasts. The base case NPV is $25,466,000, indicating the product is financially viable.

2. Demand Estimation for Years 1 and 2

Step 1: Establish Purpose

This section provides a demand forecast for the proposed Honeywell product across the first four years following launch. The objective is not to predict exact sales volume, but to establish a planning range that can support phased decisions related to production, sourcing, inventory, and launch readiness.

To improve reliability, demand was estimated using two complementary approaches:

- A top-down portfolio-based forecast, which estimates the share of Honeywell's annual category demand captured by the product over time
- An independent market-share build-up method to cross check the forecast

Quarterly demand was then derived from the annual forecast using a seasonal allocation model to support near-term operational planning.

Step 2: Define Key Variables

Variable	Definition	Unit
t	Years since launch	years
N_t	Relevant new homes in year t	homes
A_t	Security-system adoption rate	%
S_h	Honeywell's share of category demand	%
H_t	Honeywell's annual category demand	units
PPF_t	Product Penetration factor in year t	%
PPF_{max}	Long-run maximum product penetration	%
k	Adoption rate constant in the S-curve	-
A	S-curve anchor constant	-
D_t	Annual demand for the proposed product	units
SF_q	Seasonal factor for quarter q	%

Step 3: Estimation Approach

Demand for the product was modeled as a function of three main components:

- Total category demand available to Honeywell
- Product penetration within Honeywell's portfolio
- Seasonalities across the calendar year

The annual forecast structure is shown below:

$$D_t = H_t \times PPF_t$$

Then, to validate the previous estimate, we utilized the following cross-check procedure:

$$D_t = N_t \times A_t \times S_h \times PPF_t$$

This method separates housing activity, category adoption, brand share, and product-specific adoption rather than combining them into a single growth assumption, which proves useful for a second check.

Step 4: Planning Assumptions

To create our base-case demand estimates, we used the following planning assumptions:

Assumption	**Value**	**Rationale**
U.S homes installing security systems in baseline year, 2024	1,500,000	Initial market estimate
Honeywell category share	8%	Used to estimate the annual Honeywell demand
Honeywell baseline annual category demand	120,000 units	$1,500,000 \times 8\%$
Annual category growth rate	8.5%	Planning a growth scenario for the category demand
Launch year product penetration	14%	Baseline share adjusted for launch support and adoption friction
Long-run product penetration ceiling	35%	Assumes one product in the niche does not dominate the full portfolio
Adoption rate constant (k)	0.6	Reflects gradual but accelerating product adoption

These assumptions are simply treated as planning inputs, not as fixed outcomes. Actual demand will depend on market conditions, launch execution, customer adoption, channel support, and competitive response.

Step 5: Annual Category Demand Estimation

Honeywell's annual category demand was projected from the 2024 baseline of 120,000 units using the planning growth assumption of 8.5%:

$$H_t = H_{t-1}(1 + g),\ where\ g = 8.5\%$$

Year	Honeywell Annual Category Demand
2024	120,000
2025	130,200
2026	141,267
2027	153,275

Step 6: Product Penetration Model

Since the proposed product is newly introduced, its share of Honeywell's category demand is expected to increase gradually rather than begin at steady-state adoption. To reflect this ramp, product penetration was modeled using a logistic S-curve

$$PPF(t) = \frac{PPF_{max}}{1+Ae^{-kt}}$$

Parameter Selection		
Parameter	**Value**	**Interpretation**
PPF_{max}	0.35	Long-run ceiling for product penetration
k	0.60	Moderate adoption speed
$PPF(0)$	0.14	Launch-year product penetration
A	1.50	Anchor constant matching the chosen launch-year penetration

The launch-year penetration of 14% was derived from three considerations:

- an equal-share baseline across Honeywell's nine products in the category, or about 11%

- a launch support uplift factor of 1.6×
- an adoption-friction adjustment of 0.8×

$PPF(0) \approx 0.11 \times 1.6 \times 0.8 \approx 0.14$

So the final product penetration function:

$$PPF(t) = \frac{0.35}{1+1.5e^{-0.6t}}$$

Product Penetration by Year		
Year	t	PPF
2024	0	14%
2025	1	19%
2026	2	24%
2027	3	28%

Step 7: Base-Case Annual Demand Estimation

Annual product demand was estimated by applying the product penetration factor to Honeywell's annual category demand.

$$D_t = H_t \times PPF_t$$

Year	Honeywell Annual Category Demand	PPF	Estimated Annual Product Demand
2024	120,000	14%	16,800
2025	130,200	19%	24,738
2026	141,267	24%	33,904
2027	153,275	28%	42,917

This forecast represents the base case and should be used as a primary estimate for the early volume assumptions.

Step 8: Quarterly Demand Allocation

The estimated demand is not flat. It follows a seasonal trend and is influenced by holiday spikes and moving seasons, depending on the quarter (3-month periods). So we will apply a seasonal index to distribute the annual totals.

Seasonal Weighting:

- Quarter 1 [January to March]: (Low → 15%): Post-holiday slump and colder weather, so fewer moves
- Quarter 2 [April to June] (High → 30%): More homes are left unprotected when families go on vacation
- Quarter 3 [July to September] (Low → 20%): Lower demand for security systems post summer (vacations)
- Quarter 4 [October to December] (High → 35%): Holiday season/spike, as Black Friday and Cyber Monday increase sales

Based on these seasonal factors and our estimated annual demand, we will calculate the quarterly demands.

*Seasonal factors: 15% + 30% + 20% + 35% = 100%

Estimated Demands

Year / Annual Demand	Quarter	Seasonal Factor	Estimated Demand (Annual Demand × Seasonal Factor)
Year 1:	Q1	15%	5,086
	Q2	30%	10,171
	Q3	20%	6,781
	Q4	35%	11,866
Year 2:	Q1	15%	6,438
	Q2	30%	12,875
	Q3	20%	8,583
	Q4	35%	15,021

Step 9: Forecast and Planning Implications

The demand forecast indicates that annual product demand is expected to grow from approximately *16,800 units at launch to about 42,900 units by Year 3,* based on the portfolio-based model applying a logistic penetration curve to Honeywell's projected category demand. To cross check, we implemented an independent market-share validation method and using a relevant annual market of 1.45 to 1.48 million homes and an external product share ramp of *1.7%, 2.3%, 2.9%, and 3.5%,* which produced comparable demand estimates of *24,650, 33,580, 42,630, and 51,800 units,* confirming that the base-case forecast falls within a reasonable planning range. Together, these results support phased production and sourcing planning while indicating that capacity and inventory strategies should be prepared for demand levels approaching *40,000–50,000 units annually* as the product reaches broader market adoption.

3. Demand Forecasting for Years 3 and 4

Step 1: Forecasting Process and Implementation

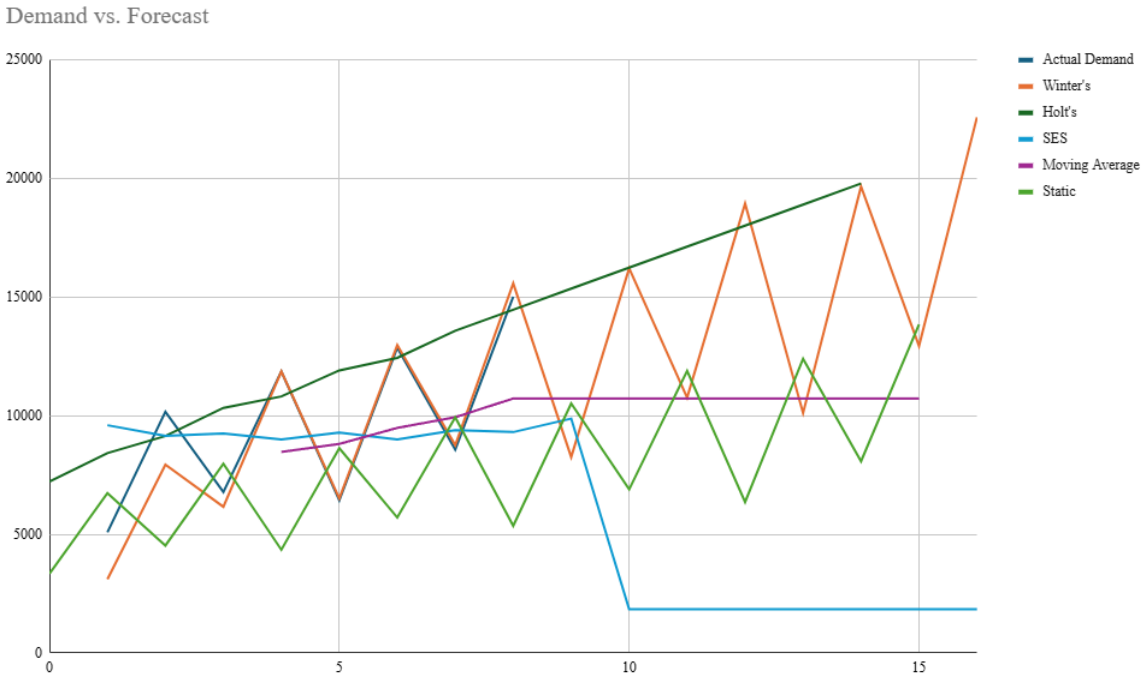
Using the demand estimation values from above, we proceeded with the demand forecasting with the five methods: Static, Moving Average, Simple Exponential Smoothing, Holt's Method, and Winter's Method. To calculate the relevant values per method, we used the following calculations below:

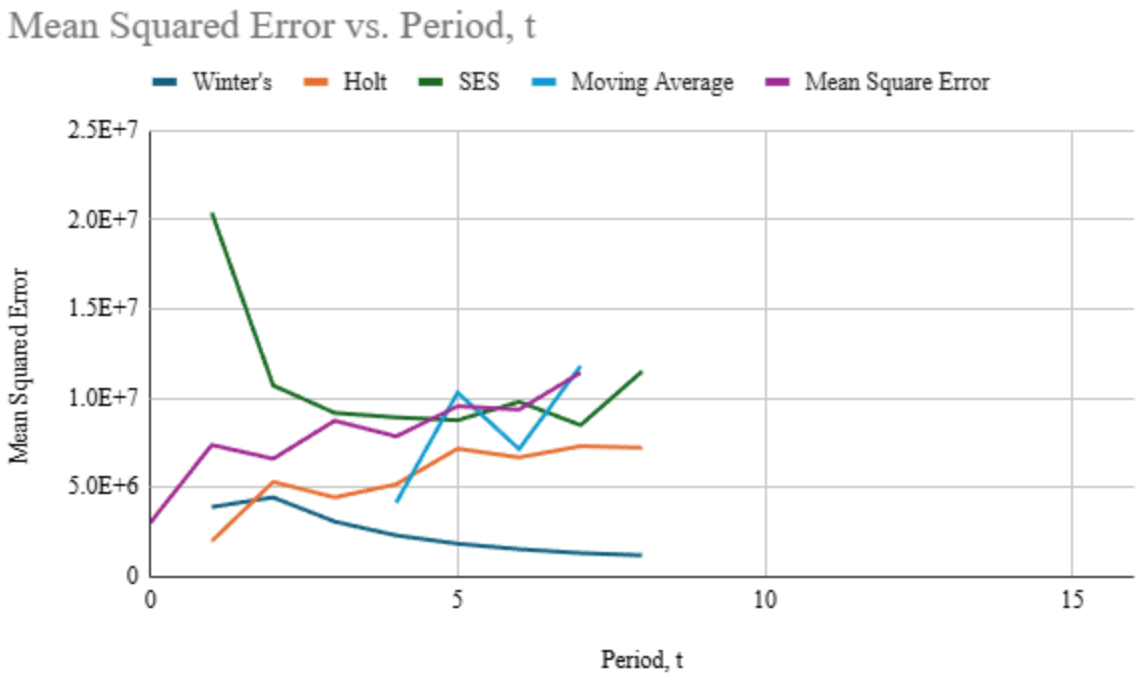
Value Type	Formula	Method(s)
Deseasonalized Demand	$\frac{Actual\ Demand}{Average\ Seasonal\ Factor}$	Static Only
Regressed Deseasonalized Demand	Intercept of Demand Data + (Current Demand * X-Var)	Static Only
Seasonal Factor	**Static:** $\frac{Actual\ Demand}{Deseasonalized\ Demand}$ **Winter:** Seasonality x $\frac{Current\ Demand}{Current\ Level}$ + (1- Seasonality) * Previous Seasonal Factor	Static and Winter
Average Seasonal Factor	Average of Seasonal Factors for the corresponding period across all cycles.	Static
Forecast	**Static:** Regressed Deseasonalized Demand x Average Seasonal Factor **Moving Average:** Sum of actual demands from previous n periods **SES:** Previous Forecast + Level * (Previous Actual Demand - Previous Forecast) **Holt:** Current Level + Current Trend **Winter:** (Current Level + Current Trend) x Previous Seasonal Factor	All Methods
Forecast Error	Actual Demand - Forecast	All Methods
Mean Square Error	Running Average of Forecast Error after Squaring	All Methods

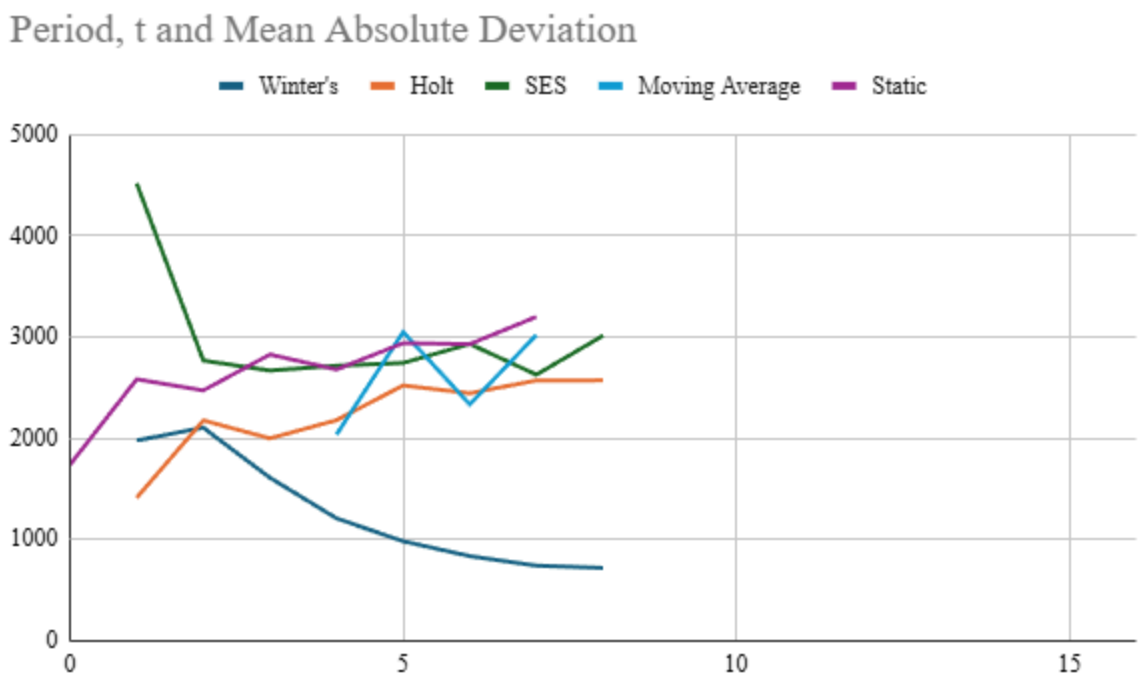
Absolute Error	Absolute Value of Forecast Error	All Methods
Mean Absolute Deviation	$\frac{Sum\ of\ Absolute\ Errors}{Number\ of\ Periods}$	All Methods
Percent Error	$\frac{Forecast\ Error}{Actual\ Demand}$	All Methods
Absolute Percent Error	Absolute Value of Percent Error	All Methods
Mean Absolute Percent Error	$\frac{Sum\ of\ Absolute\ Percent\ Errors}{Number\ of\ Periods} * 100$	All Methods
Bias	Forecast Error + Previous Bias	All Methods
Tracking Signal	$\frac{Bias}{MAD}$	All Methods

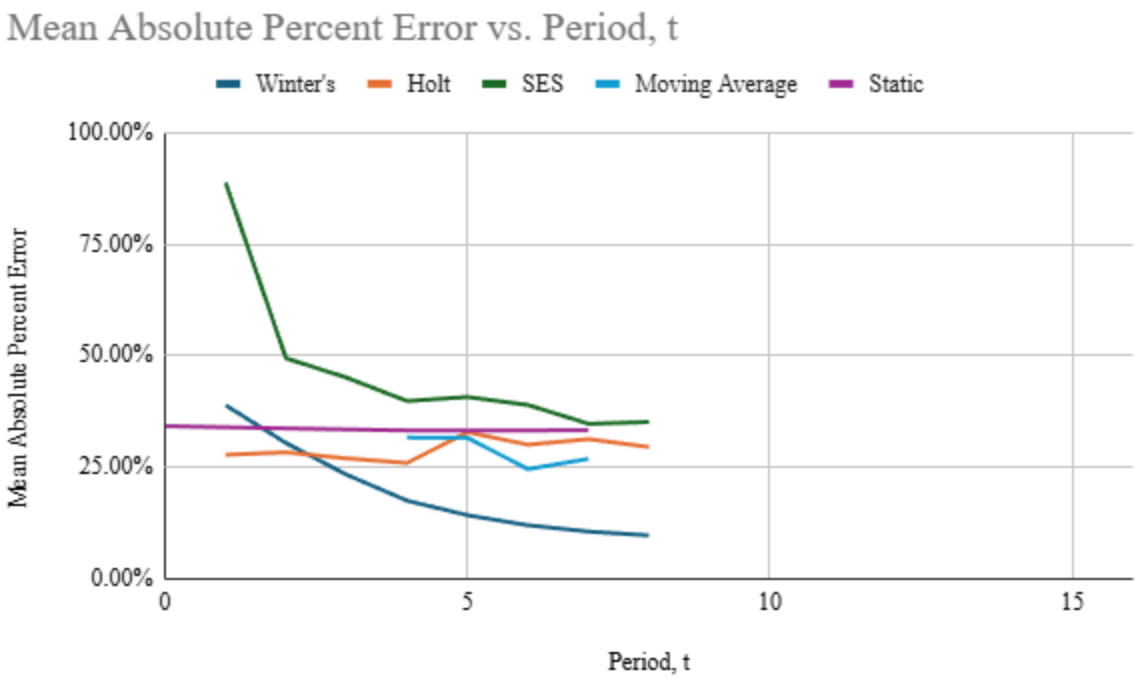
Step 2: Forecasting and Error Results

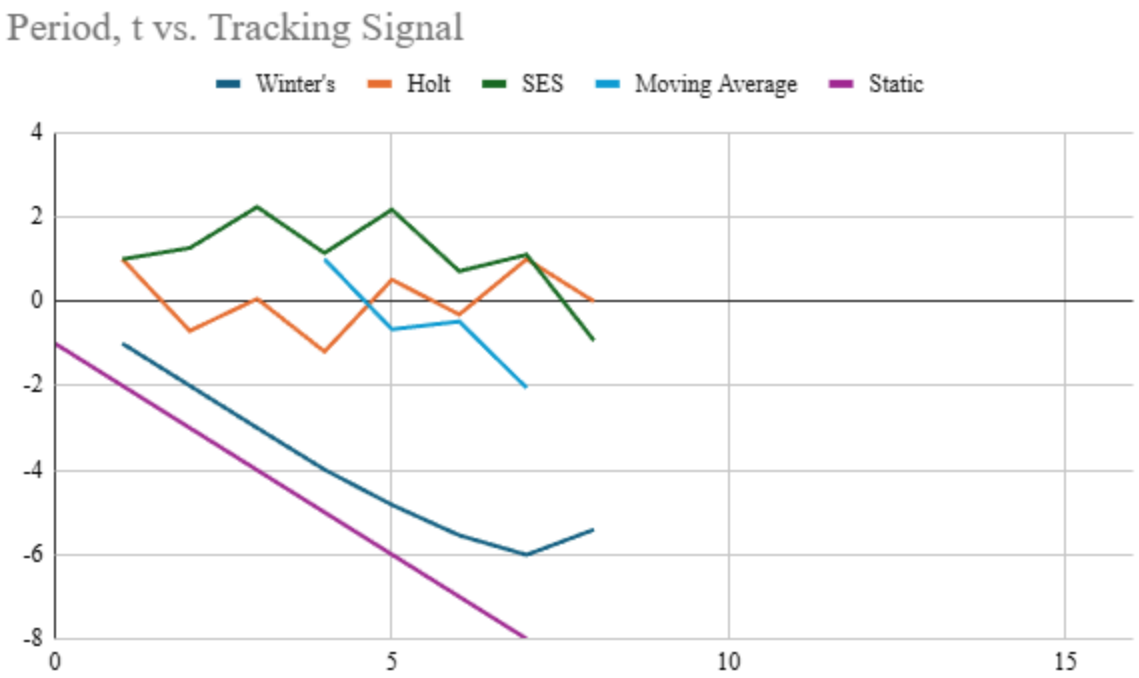
To identify the best forecasting model for our data, we looked closely at error metrics for all forecasting methods and graphed them. The graphs will provide a visual representation of errors for all methods used, allowing us to determine the best forecasting method. Each error of the methods were consolidated into one graph for better readability.

Forecasting and Error Results	
Demand vs. Forecasting	 This graph compares actual demand with several forecasting models across multiple time periods. The actual demand line fluctuates significantly, showing strong seasonal increases and decreases. Each forecasting method responds differently, with some models following the overall upward trend while others smooth out the fluctuations. Methods like Holt's and Winters capture the growth pattern more closely, while simpler models such as the moving average and SES produce smoother but less responsive forecasts. Overall, the graph highlights how different forecasting techniques vary in their ability to track changing demand patterns.

MSE vs. Period	 This graph compares the mean squared error (MSE) of the forecasting methods over time. Winter's method consistently shows the lowest MSE and continues to decline as more periods are observed, indicating that it produces the most accurate forecasts overall. Holt's method performs moderately well with relatively stable error values, while the moving average shows more variability across periods. SES begins with very high error levels but improves as more data becomes incorporated into the forecast.

MAD vs. Period	 This graph shows the mean absolute deviation of each forecasting method across time periods. Winter's method demonstrates a clear downward trend, indicating that forecast errors decrease as the model adjusts to the data. Holt's method remains relatively stable with moderate error levels, while the moving average fluctuates more noticeably. SES begins with the largest deviation but gradually improves over time as the smoothing process stabilizes.

MAPE vs. Period	 This chart measures the percentage error of each forecasting technique relative to the actual demand. Winter's method again shows the lowest and most consistent MAPE values, indicating strong forecasting accuracy. Holt's and the moving average models maintain moderate percentage errors across periods. SES starts with extremely high percentage error in early periods but steadily declines as the model adapts to the demand pattern.

Period vs. Tracking Signal	 The tracking signal graph evaluates whether forecasts are systematically biased over time. A tracking signal close to zero suggests that forecasts are balanced and not consistently over or under-predicting demand. Holt's and SES remain relatively close to zero, indicating more stable forecasting performance. Winter's method trends more negatively across periods, suggesting that it may slightly overestimate demand in later periods, while the static model shows the largest bias over time.

Step 3: Error Metrics Summary table

Forecasting Method	Forecast Error	Absolute Error	Mean Squared Error	Mean Absolute Deviation	Percent Error	Absolute Percent Error	Mean Absolute Percent Error	Bias	Tracking Signal
Static	-5087.88	11435413.93	5087.88	3200.11	-33.87	0.34	0.33	-25600.91	-8.00
Moving Average	-5080.50	5080.50	11819823.56	3021.63	-0.34	0.34	0.27	-6196.50	-2.05
SES	-5708.76	5708.76	11511054.23	3013.78	-0.38	0.38	0.35	-2804.95	-0.93
Holt's	-2577.83	2577.83	7236967.30	2572.74	-0.17	0.17	0.30	-9.55	0.00
Winter's	565.14	565.14	1206990.50	720.73	0.04	0.04	0.10	-3899.32	-5.41

Using these formulas for their respective methods, we calculated the relevant values for all five methods, where we summarized our final values in the table below. The columns represent the error type for Period 8, while the rows on the left represent the relevant forecasting method. To clearly compare the performance of each forecasting method, the key error metrics are summarized below.

Compact Summary of Forecasting Model Performance (MAPE & Tracking Signal)

Forecasting Method	Mean Absolute Percent Error	Tracking Signal
Static	0.33	-8.00
Moving Average	0.27	-2.05
SES	0.35	-0.93
Holt's	0.30	0.00
Winter's	0.10	-5.41

This table clearly shows that Winter's Method achieves the lowest MAPE, indicating the highest accuracy, while Holt's Method shows the least bias with a tracking signal closest to zero. Based on these results, Winter's Method is selected as the most appropriate forecasting model due to its superior accuracy and ability to capture both trend and seasonality.

MAPE and tracking signal were used to evaluate which forecasting model performed the best. MAPE measures the average percent difference between the forecasted demand and the actual demand, which shows how accurate the forecasts are. A lower MAPE means the forecast is closer to the real demand and the model is more accurate. In our results, Winter's method had the lowest mean absolute percent error at about 0.10. The other models had higher values, such as 0.33 for the static model, 0.27 for the moving average model, 0.35 for simple exponential smoothing, and 0.30 for Holt's method. Because Winter's method produced the lowest MAPE, it generated forecasts that were closest to the actual demand overall.

The tracking signal was used to determine whether the forecasting models showed consistent bias over time. A tracking signal close to zero means the model is not consistently overestimating or underestimating demand. In our results, Holt's method had a tracking signal of 0, while simple exponential smoothing had −0.93 and moving average had −2.05, which all indicate relatively balanced forecasts. The static model had a tracking signal of −8, which shows strong bias and poor performance. Winter's method had a tracking signal of about −5.41, which suggests some overestimation in later periods. However, because it had the lowest MAPE and captured both trend and seasonality in the demand pattern, it still produced the most accurate forecasts overall and was selected as the best forecasting method.

Step 4: Conclusions and Findings

To achieve the data as shown on the graph, we experimented with different constants for the Level, Trend, and Seasonal Factor, before arriving at the most optimal constants of 0.3, 0.2, and 0.3, respectively. **Winter's Method** is the forecasting method that we decided would be the best possible option.

The reasoning for this is that, although the forecast starts out below the demand, we see that eventually they are roughly equivalent throughout the rest of the timeframe. This indicates that, for our specific use case and scenario, Winter's Method is a highly effective and accurate forecasting method that outperforms all of the other forecasting methods as seen above.

4. Cycle and Safety Inventory Management

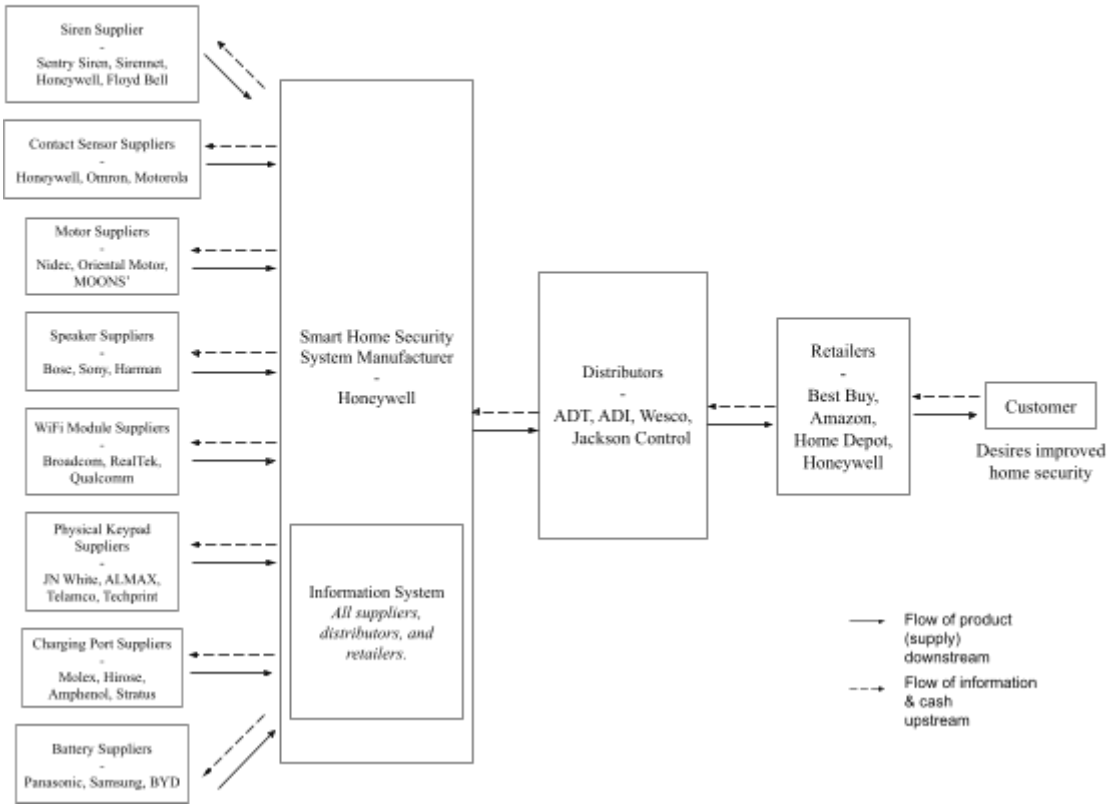
To calculate the Cycle and Safety Inventory for our product's supply chain, specifically for at least three key subsystems of the product and the finished product, we first need to start with the network representation of our product's supply chain and explain the flow from the various suppliers of subsystems through the manufacturer, distributors, retailers, and finally to the customers. Then, we introduce the purpose of Cycle and Safety Inventory within our supply chain, which will help us set up and perform a complete set of cycle and safety inventory calculations for the three subsystems and our finished product.

Step 1: Analyze Supply Chain Network Diagrams

Stage View Representation

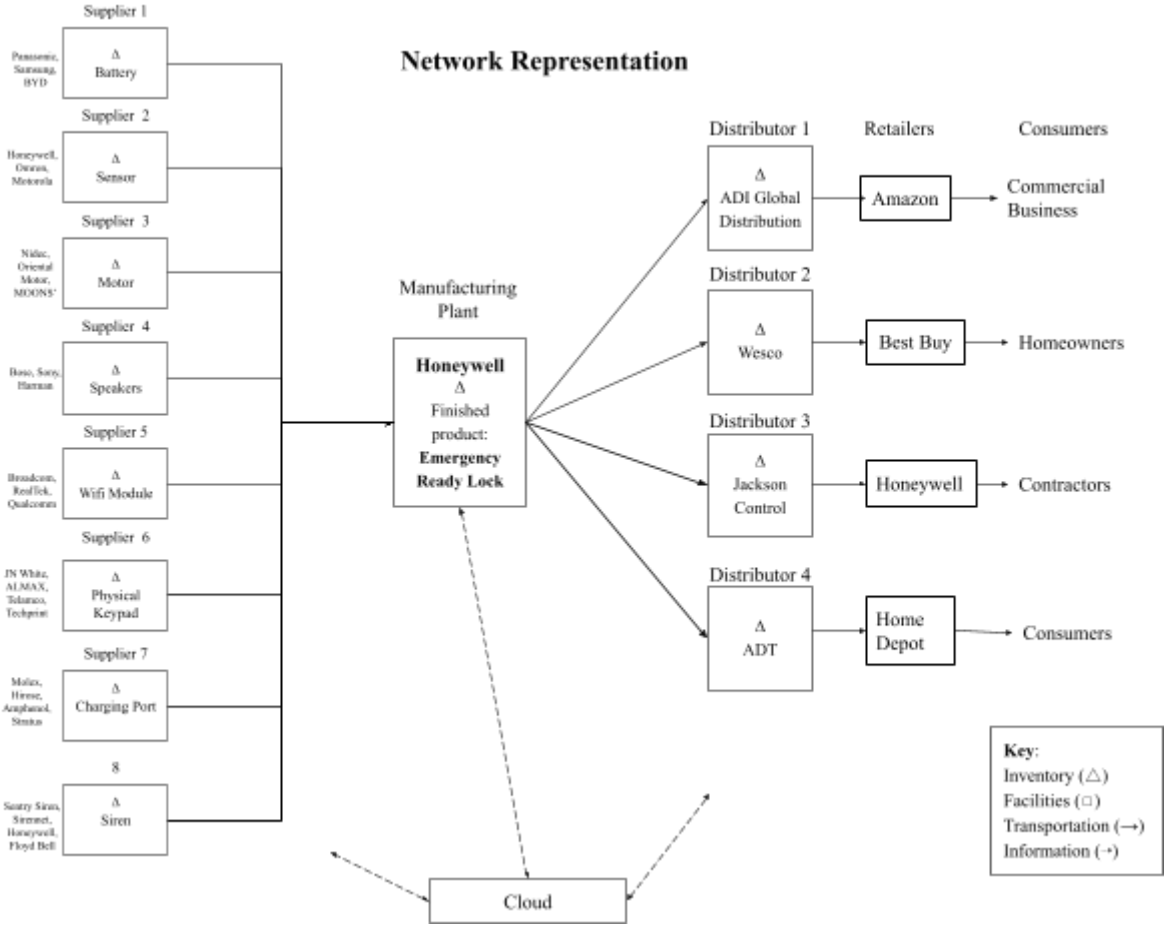
The stage view provides a simplified representation of the same supply chain, organized by the supply chain stage. On the left, eight supplier categories provide individual components to the Honeywell Manufacturing Plant at the center. The finished Emergency Ready Lock then moves through distributors and retailers before reaching the end customer. This diagram makes clear the stages through which both components and finished products travel, and serves as the foundation for identifying which specific stages and subsystems will be analyzed in the cycle inventory calculations that follow

Stage View of the Supply Chain (SC) for Emergency Ready Lock

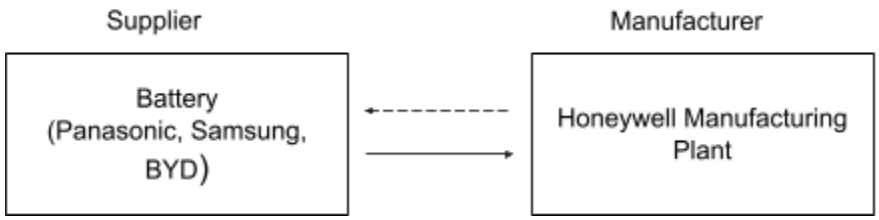


Network Representation

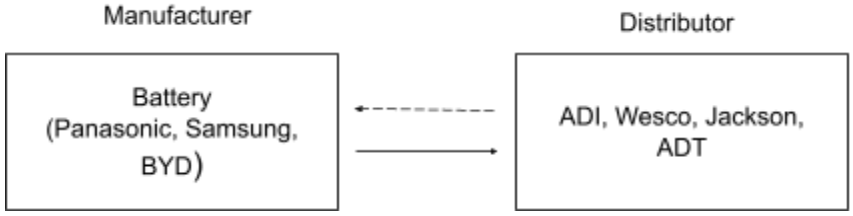
The network representation shows the full supply chain for the Honeywell Emergency Ready Lock. Eight supplier categories (Battery, Sensor, Motor, Speaker, WiFi Module, Physical Keypad, Charging Port, and Siren) feed directly into the Honeywell Manufacturing Plant, where the finished product is assembled. From there, the finished product flows downstream through four distributors (ADI Global Distribution, Wesco, Jackson Control, and ADT) to retailers (Amazon, Best Buy, Honeywell, Home Depot) and finally to end customers, including commercial businesses, homeowners, contractors, and consumers. Information flows upstream through a centralized cloud system connecting all stages of the network.



Component flow (Battery Supply → Manufacturing Plant):



Finished Product Flow (Manufacturing Plant → Distributor):



Step 2: Subsystem Selection

Based on our FAST and Network representation, the Emergency Ready Lock consists of multiple hardware subsystems sourced from specialized T1 suppliers and integrated at a centralized Honeywell manufacturing facility. For our cycle and safety inventory analysis, the two key subsystems we selected were: **speakers and contact sensors.** We chose these for our analysis as they are critical to the product's safety and intrusion detection function, sourced externally from specialized suppliers, replenished repeatedly over the product's lifecycle, and represent the different costs, volume, and variability profiles present in our system overall. The cycle and safety inventory for the completed Emergency Ready Lock is analyzed to ensure alignment between component replenishment and final product availability.

Step 3: Inventory Calculations

Step 3.a: To calculate the Cycle and Safety Inventory, we would first need the forecasted demand data for the years 2026 to 2029. These were obtained from our best forecasting results, which were Winter's method.

Annual Demand for 2026 (Winter forecasting):

Year	Forecasts, Ft+l	Total Yearly Demand, D
2026	3,107	28,878
	7,938	
	6,117	
	11,716	
2027	6,412	43,188
	12,756	
	8,630	
	15,390	
2028	8,178	53,555
	16,029	
	10,639	
	18,709	
2029	9,961	64,254
	19,345	
	12,732	
	22,216	

Step 3.b: Then, based on internet research, we estimated the unit costs, holding costs, and shipping costs per order for the two chosen subsystems and our finished product. The complete product cost was estimated in the NPV section of the report.

Costs	Contact Sensor	Speaker	Complete Product
Unit Cost, C	$21.99	$3.52	$115.28
Holding Cost, h	20% = 0.2	20% = 0.2	20% = 0.2
Shipping Cost Per Order, S	$500	$500	$750

These forecasted demands and estimated costs were then used to calculate Cycle and Safety Inventory for the two subsystems and our finished product.

Step 3.c: Cycle Inventory for 3 subsystems and finished product

To determine Cycle Inventory and Cycle Inventory Holding Cost, the following formulas were used:

Calculation	Formula
Annual Demand, D Unit cost, C Holding cost, h Shipping cost, S	Determined above
Optimal Lot Size	$Q_L^* = \sqrt{\frac{2DS}{hC}}$
Number of shipments per year	$n = D / Q_L$
Cycle inventory	$Q_L / 2$
Cycle inventory holding cost	$CI = (Q_L / 2) \times (hC)$
Replenishment cycle time	$T = 365 / n$
Average flow time	T/2

Cycle Inventory Results:

Year	Annual Demand, D	Sub system / Finished product	Unit cost, C	Holding cost, h	Shipping cost, S	Optimal Lot Size, Q_L^*	# of shipments/year, $n = D/Q_L$	Cycle Inventory, $Q_L/2$	Cycle inventory holding cost, CI = (QL / 2) × (hC)	Replenishment cycle time, T = 365/n	Average flow time, T/2
Year 1 - 2026	28878	Contact Sensor	21.99	0.2	500	2562.45	11.27	1281.23	5634.83	32.39	16.19
	28878	Speaker	3.52	0.2	500	6404.68	4.51	3202.34	2254.45	80.95	40.48
	28878	Finished Product	115.28	0.2	750	1370.68	21.07	685.34	15801.24	17.32	8.66
Year 2 - 2027	43188	Contact Sensor	21.99	0.2	500	3133.67	13.78	1566.84	6890.95	26.48	13.24
	43188	Speaker	3.52	0.2	500	7832.41	5.51	3916.20	2757.01	66.19	33.10
	43188	Finished Product	115.28	0.2	750	1676.24	25.76	838.12	19323.65	14.17	7.08
Year 3 - 2028	53555	Contact Sensor	21.99	0.2	500	3489.57	15.35	1744.79	7673.57	23.78	11.89
	53555	Speaker	3.52	0.2	500	8721.95	6.14	4360.98	3070.13	59.44	29.72
	53555	Finished Product	115.28	0.2	750	1866.61	28.69	933.31	21518.28	12.72	6.36
Year 4 - 2029	64254	Contact Sensor	21.99	0.2	500	3822.28	16.81	1911.14	8405.19	21.71	10.86
	64254	Speaker	3.52	0.2	500	9553.53	6.73	4776.76	3362.84	54.27	27.13
	64254	Finished Product	115.28	0.2	750	2044.58	31.43	1022.29	23569.90	11.61	5.81

Step 3.d: Safety Inventory for 2 subsystems and finished product

In order to determine Safety (Stock) Inventory and Annual Safety Inventory Holding Cost, the following formulas were used:

Calculation	Formula
Annual Demand, D Unit cost, C Holding cost, h Shipping cost, S	Determined above
Mean weekly demand	$D_W = D_{annual} / 52$
Standard deviation of weekly demand	σ_W (2026) =STDEV(3107,7938,6117,11716) σ_W (2027) =STDEV(6412,12756,8630,15390) σ_W (2028) =STDEV(8178,16029,10639,18709) σ_W (2029) =STDEV(9961,19345,12732,22216) *using Excel formula* =STDEV(forecasted quarterly demands for each year)
Supplier lead time (weeks), L	Contact sensor, L = 2 Speaker, L = 4 Finished product, L = 1 *Lead times were curated based on information regarding our 3rd party sourcing (real lead times based on website info).*
Mean value of demand during the supplier lead time	$D_L = (D_L)_m = LD_W$
Standard deviation of demand during the supplier lead time	$\sigma_L = \sqrt{L}\ \sigma_W$
$(CSL)_{desired}$	95% = 0.95 ⇒ 1.65
Safety stock	$ss = \sqrt{L}\ \sigma_W\ F^{-1}[(CSL)_{desired}]$
Re-order point	$ROP = LD_W + ss$
Annual safety inventory holding cost	(ss) × (hC)

Safety Inventory results:

Year	Annual Demand, D	Sub system / Finished product	Unit cost, C	Holding cost, h	Shipping cost, S	Mean weekly demand, Dw	Std dev of weekly demand, σw	Supplier lead time (weeks), L	Mean value of D_L, $(D_L)_m$ = LDw	Std dev of D_L, σ_{D_L} = √L σw	(CSL)desired = 95% = 0.95	Safety stock, ss = √L σw F-1[(CSL)desired]	Reorder point, ROP = LDw + ss	Annual safety inventory holding cost, (ss)(hC)
Year 1 - 2026	28878	Contact Sensor	21.99	0.2	500	555.35	3599.21	2	1110.69	5090.05	1.65	8398.58	9509.28	36936.97
	28878	Speaker	3.52	0.2	500	555.35	3599.21	4	2221.38	7198.42	1.65	11877.39	14098.77	8361.68
	28878	Finished Product	115.28	0.2	750	555.35	3599.21	1	555.35	3599.21	1.65	5938.69	6494.04	136922.55
Year 2 - 2027	43188	Contact Sensor	21.99	0.2	500	830.54	4035.57	2	1661.08	5707.16	1.65	9416.81	11077.88	41415.12
	43188	Speaker	3.52	0.2	500	830.54	4035.57	4	3322.15	8071.14	1.65	13317.38	16639.53	9375.43
	43188	Finished Product	115.28	0.2	750	830.54	4035.57	1	830.54	4035.57	1.65	6658.69	7489.23	153522.72
Year 3 - 2028	53555	Contact Sensor	21.99	0.2	500	1029.90	4830.08	2	2059.81	6830.76	1.65	11270.76	13330.57	49568.81
	53555	Speaker	3.52	0.2	500	1029.90	4830.08	4	4119.62	9660.16	1.65	15939.26	20058.88	11221.24
	53555	Finished Product	115.28	0.2	750	1029.90	4830.08	1	1029.90	4830.08	1.65	7969.63	8999.54	183747.84
Year 4 - 2029	64254	Contact Sensor	21.99	0.2	500	1235.65	5685.09	2	2471.31	8039.94	1.65	13265.90	15737.20	58343.42
	64254	Speaker	3.52	0.2	500	1235.65	5685.09	4	4942.62	11370.19	1.65	18760.81	23703.43	13207.61
	64254	Finished Product	115.28	0.2	750	1235.65	5685.09	1	1235.65	5685.09	1.65	9380.41	10616.06	216274.64

Step 3.e: Total Inventory Holding Cost (Annual)

In order to determine the Total Inventory Holding Cost, the following formulas were used:

Calculation	Formula
Annual Demand, D Unit cost, C Holding cost, h Shipping cost, S	Determined above
Cycle inventory	$Q_L / 2$
Safety stock	$ss = \sqrt{L}\,\sigma_W\,F^{-1}[(CSL)_{desired}]$
Total inventory holding cost	$((Q_L / 2) + ss) \times (hC)$

Total Inventory Holding Cost results:

Year	Annual Demand, D	Sub system / Finished product	Unit cost, C	Holding cost, h	Cycle Inventory, QL/2	Safety stock, ss = √L σW F-1[(CSL)desired]	Total inventory holding cost = (QL / 2 + ss) (hC)
Year 1 - 2026	28878	Contact Sensor	21.99	0.2	14439.00	8398.58	100439.69
	28878	Speaker	3.52	0.2	14439.00	11877.39	18526.74
	28878	Finished Product	115.28	0.2	14439.00	5938.69	469828.13
Year 2 - 2027	43188	Contact Sensor	21.99	0.2	21594.00	9416.81	136385.53
	43188	Speaker	3.52	0.2	21594.00	13317.38	24577.61
	43188	Finished Product	115.28	0.2	21594.00	6658.69	651393.98
Year 3 - 2028	53555	Contact Sensor	21.99	0.2	26777.50	11270.76	167336.26
	53555	Speaker	3.52	0.2	26777.50	15939.26	30072.60
	53555	Finished Product	115.28	0.2	26777.50	7969.63	801129.88
Year 4 - 2029	64254	Contact Sensor	21.99	0.2	32127.00	13265.90	199637.96
	64254	Speaker	3.52	0.2	32127.00	18760.81	35825.02
	64254	Finished Product	115.28	0.2	32127.00	9380.41	956994.75

Step 4: Results & Interpretations

Annual Demand

The demand for the Emergency Ready Lock is projected to grow immensely, rising from 28,878 units in 2206 to 64,254 units by 2029. Since this demand is calculated using Winter's method, it accurately accounts for the seasonal fluctuations that are prevalent in the home security market. Since there is almost a 122% increase in volume, we anticipate the necessity for a highly scalable supply chain so as to prevent shortages during peak purchasing periods. In conjunction with mitigating the Bullwhip Effect across its global T1 supplier network, Honeywell needs to proactively update its optimal lot size every six months.

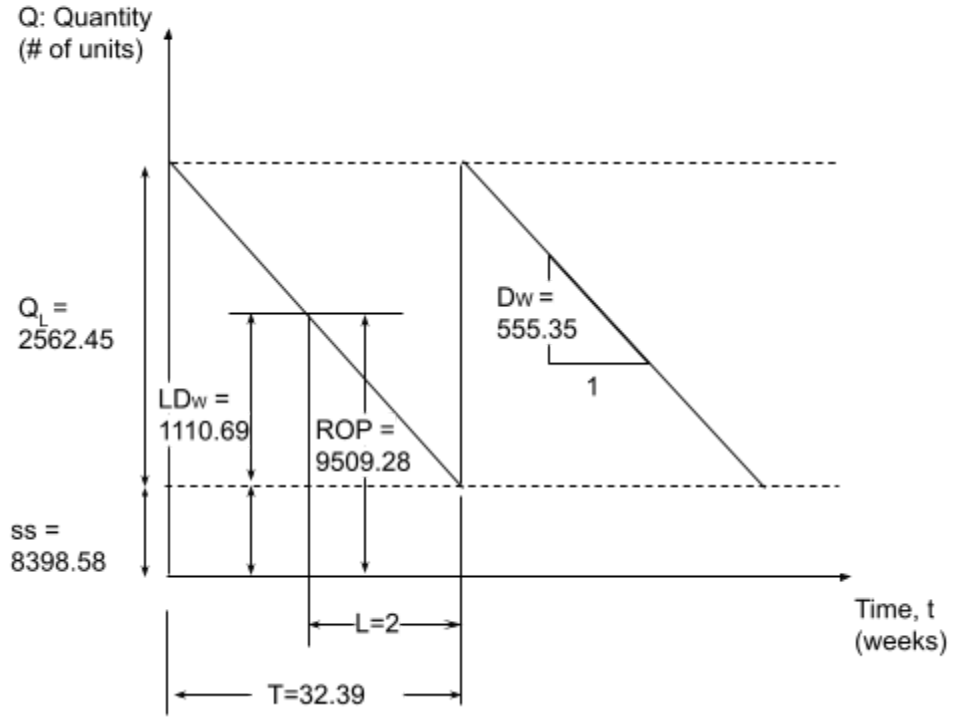
Cycle Inventory Analysis for Contact Sensor

The Contact Sensor unit carries a moderate unit cost of $21.99, as it is a critical component utilized for the lock's primary intrusion detection function. Our analysis shows that the optimal lot size should increase from 2,562 units to 3,822 units over the 4 years to match the estimated rising demand. By maintaining a replenishment cycle of around 22-32 days, Honeywell ensures a high level of agility for these high-value components, and the increased frequency of replenishment in the strategy minimizes the risk of having defective or obsolete hardware, while keeping holding costs balanced.

Safety Inventory Analysis for Contact Sensor

With a two-week supplier lead time and a target 95% CSL, safety stock grows from 8,399 to 13,266 units over the four years, driving annual holding costs from $36,939 to $58,434. The relatively short lead time keeps safety stock manageable, but Honeywell should monitor demand variability closely as volume scales.

Cycle and Safety Inventory for Contact Sensor (Year 2026)

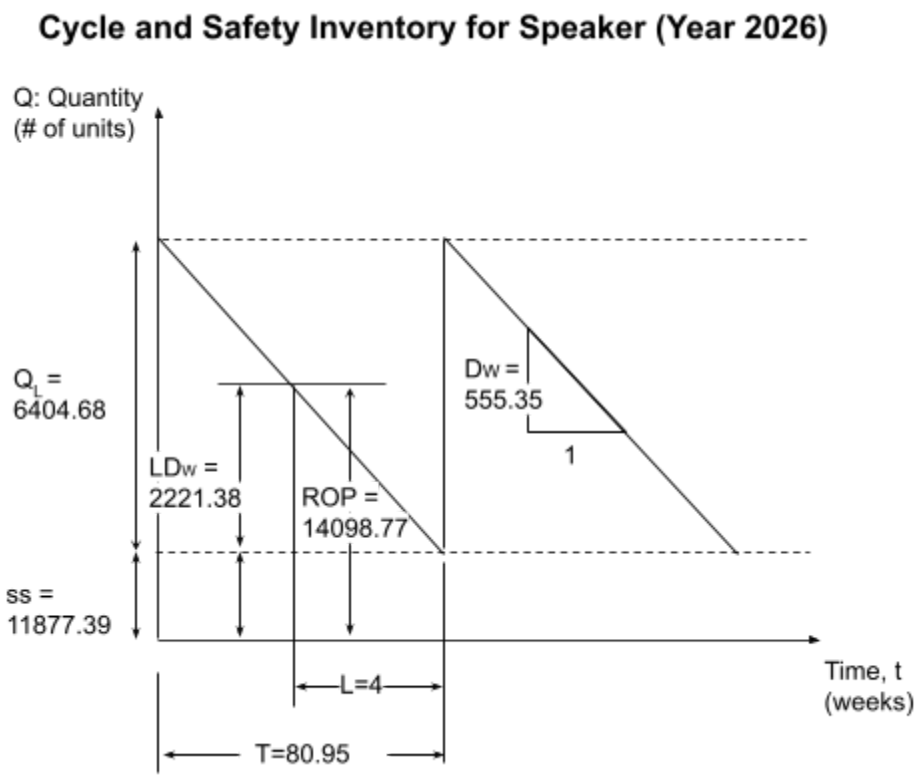


Cycle Inventory Analysis for Speaker

The Speaker unit represents a bulk-ordered component that has the lowest unit cost in the system at $3.52.Since the shipping cost per order is fixed at $500, the EOQ model generates the largest lot size, peaking at over 9,500 units by 2029. Through this, we get the longest average flow time of roughly 40 days, indicating that Honeywell is intentionally leveraging storage to minimize transportation expenses. Since these mechanical components have low technological obsolescence, holding a larger cycle inventory here is an effective strategy to lower the total cost of goods sold

Safety Inventory Analysis for Speaker

The speaker's 4-week lead time doubles demand exposure during replenishment, pushing safety stock from 11,877 to 18,761 units by year 4. Despite the low unit cost of $3.52, annual holding cost still reached $13,208, justifying close coordination with suppliers to potentially reduce lead time and buffer requirements.



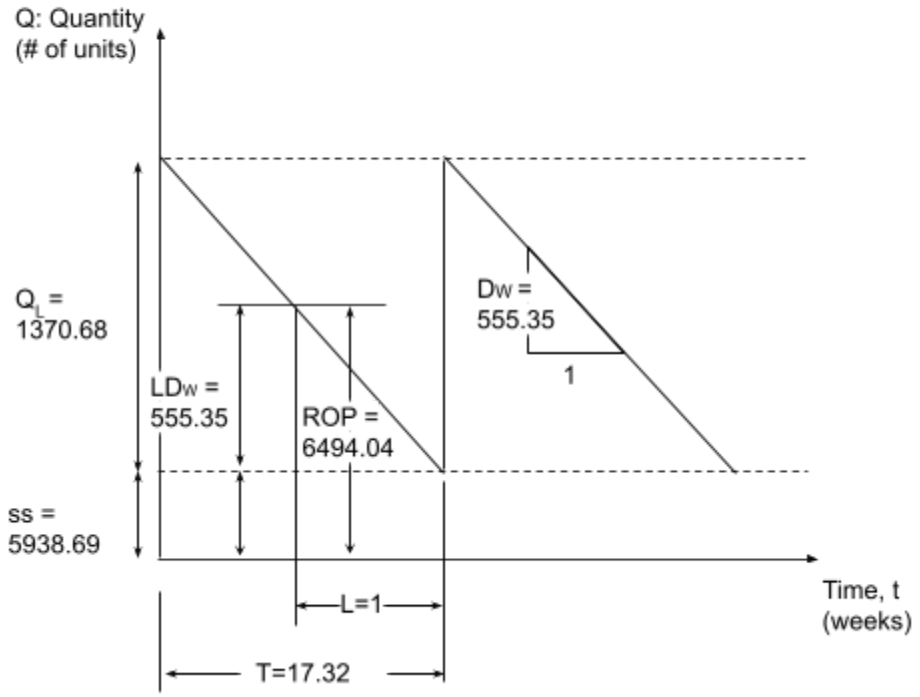
Cycle Inventory Analysis for Finished Product Interpretation

The Emergency Ready Lock is the most expensive item in the chain at $115.28, which demands a "Lean" inventory approach to avoid tying up excessive capital. To minimize high holding costs, the system utilizes the smallest relative lot sizes and a rapid replenishment cycle of just 11 to 17 days. This results in a very low average flow time of under 9 days, meaning the completed locks move through the Honeywell plant and out to distributors almost immediately. This strategy optimizes cash flow and ensures that the highest-value inventory is always moving towards the customer.

Safety Inventory Analysis for Finished Product Interpretation

The finished product benefits from a short 1-week lead time, keeping safety inventory stock relatively low at 5,393 to 9,380 units. However, at $115.28 per unit, holding costs are the highest in the supply chain, reaching ~$216, 275 annually by year 4, reinforcing the need for tight demand forecasting accuracy to avoid excess finished goods in inventory.

Cycle and Safety Inventory for Finished Product (Year 2026)



5. Financial Modeling

A financial model will allow Honeywell to determine the feasibility of the project. The calculation focuses on calculating the net present value (NPV) and considers sales volume per year, developmental cost, unit price, unit product cost, ramp-up cost, marketing cost, and annual discount factor. A negative NPV means the project is not feasible, and a positive NPV means that the project will be feasible. A simple step-wise process is the following:

Step 1: Set up the NPV Excel sheet.
Step 2: State NPV parameter estimations with rationale.
Step 3: Explain NPV Base Case Results

Step 1: NPV Process

We calculated the necessary values to set up the base case for *Net Present Value Estimations;* then referenced the "Excel Tutorial" written by R.Polany and S.Desa to establish the NPV Base Case. After following the tutorial's steps and inputting our own estimations of development cost, unit price, etc., we were able to complete the calculations for the NPV analysis.

Step 2: NPV Input Estimation

The following table showcases how we were able to estimate each input value in the net present value calculation:

Net Present Values Estimations		
Sales & Product Volume (units/quarter)	106,825 units	The unit quantity for each quarter was imported from the forecasting model. For example, "Year 1, Projection Q1" in the financial modeling spreadsheet directly corresponds to Year 1 Q1 in our forecasting model (Winter's).
Development Cost (total $)	$60,480	The development cost was determined using an estimation of estimated cost based on the team's time commitment. Assuming a 9-hour commitment for 15 weeks by a team of 8 provides the following estimation: $= 9\ hours\ *\ 15\ weeks\ *\ 8\ members\ *\ \$56/hr$ $= \$60,480$ Where $56/hr is the average hourly wage of a research & development engineer according to ZipRecruiter (see appendix for additional information).
Unit price ($/unit)	$299	The unit price was determined by comparing competing product prices and identifying a competitive price. The following competing products were compared in MSRP: ● SimpliSafe Security ($250-$750, $440 is most comparable)

		● Ring Alarm Pro ($430) ● EcoBee ($460) With this information, we establish our customer's willingness to pay up to $460 for similar competing products. Placing our product at $299 would considerably undercut competing products, with only SimpliSafe offering a more basic security system at this price range.
Unit Productions Cost ($/unit)	$115.28	The unit product cost was systematically determined by researching the cost of major subsystems identified in previous FAST diagrams, then providing an estimation of missing components. A comprehensive table for this result is found in the next section.
Ramp-up cost (total $)	$6,048	According to Tacto Management, Ramp-up cost is typically 5%-15% of annual cost. For our model, we utilized a 10% ramp up cost, which is between the typical 5%-15%, and applied this toward the current development cost. $= \$60,480 * 10\%$ $= \$6,048$
Marketing & support cost ($/year)	$840,000	According to Proto Marketing, marketing & support cost is typically 5%-15% of revenue. Our financial model without marketing & support determined an NPV of $28,000,000; or an NPV of 5,600,000 per year. Considering that NPV is profit and not actual revenue, we will consider using the NPV value with a higher marketing & support cost percentage to account for the difference between profit and revenue. $= \$5,600,000 * 0.15$ $= \$840\,000$
Annual Discount Factor (%)	7%	Our financial model will utilize a 7% discount factor which aligns with another discount factor Honeywell previously used when estimating the cost of an environmental clean-up project. This also aligns closely with the 10% estimation For Entrepreneurs recommends using for public companies.

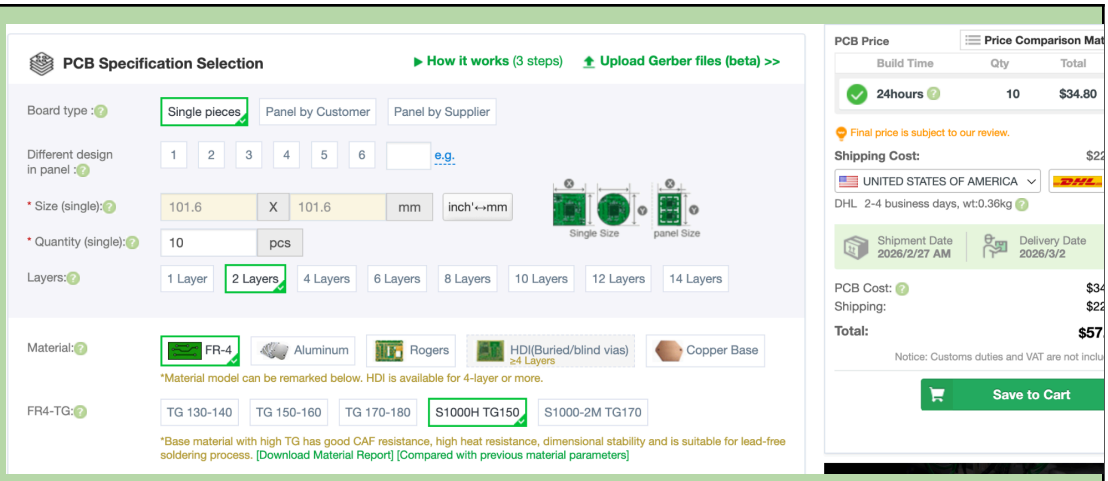
Determining Unit Price

The unit price was determined by identifying the cost for each subsystem in the product. This is assuming third-party components are utilized without bulk discounts. Actual cost will be less when using Honeywell-made components, and bulk discounts are applied. We are unable to obtain the actual component cost from Honeywell.

Components	Primary Source	Alternative Source
Battery * 4 batteries in series are needed to reach sufficient voltage to power the stepper motor.	 Panasonic BK-200AAB9B Unit cost: $4.29	 Panasonic HHR-210AAC4B Unit cost: $4.29
Contact Sensor	 ControlByWeb GRI-28C Unit Cost: $21.99	 Comet America SA200A Unit Cost: $19.20

Motor	 MIKROE 1530 Unit Cost: $9.60	 Seeed Technology 108990003 Unit Cost: $4.50
Speaker	 PUI Audio 668-1398-ND Unit cost: $3.52	 2056-W170-8OHM-ND Unit cost: $3.82
Wireless connectivity (Wifi + Bluetooth)	 Raspberry PI 2648-SC0020-ND Unit cost: $15.00	 SparkFun Electronics: 1568-11114-ND Unit cost: $11.25

Keypad	 Adafruit Industries 1528-1136-ND Unit cost: $3.95	 Knitter-Switch 3746-YFTKNI12B19-19 Unit cost: $14.44
Charging Port	 Adafruit Industries 1528-6071-ND Unit cost: $2.50	 Bulgin 708-2061-ND Unit cost: $23.16
Siren	 Floyd Bell 2339-M-80-ND Unit cost: $10.80	 Floyd Bell 2339-PM-80-ND Unit cost: $12.33

PCB	 Estimated Unit cost: $5.76 at 10 qty **Per PCB Way, assuming a standard 4x4-inch double-layered PCB.*
Enclosure Will be manufactured in-house using strong metals.	 Unit Cost: $20-$30
Product Estimated Product Cost: $115.28	

Step 3: NPV Base Case Results

The net present value calculation revealed a unit cost of $115.28 and an NPV of $25,466,000. This means over 4 years, the product will generate an estimated baseline of 25 million dollars in net present value. As mentioned above, it is also assumed that the unit cost was determined using third-party components, assuming no bulk purchasing discount; it is strongly recommended to use first-party components where possible to further increase NVP. This showcases that the product is profitable under third-party and first-party component sourcing. Honeywell should pursue developing the product.

NPV Spreadsheet, input tab:

Scenario input parameters	
Sales & Product Volume (units/year)	Values imported from forecast
Development Cost (total $)	60,480.00
Unit price ($/unit)	299.00
Unit Productions Cost ($/unit)	115.28
Ramp-up cost (total S)	6,048.00
Marketing & suppost cost ($/year)	840,000.00
Annual Discount Factor (%)	7.00

Data imported from Winter's method

Year	Quarter	Forecast
1	1	3,107
1	2	7,938
1	3	6,160
1	4	11,873
2	1	6,518
2	2	12,978
2	3	8,761
2	4	15,586
3	1	8,261
3	2	16,216
3	3	10,767
3	4	18,944
4	1	10,119
4	2	19,669
4	3	12,945
4	4	22,591

NPV Spreadsheet, basecase tab:

BaseCase																				
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection			
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Development Cost	-15.12	-15.12	-15.12	-15.12																
Ramp-up cost				-3.02	-3.02															
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26
Production volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit product cost					-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12
Sales revenue					928.98	2,373.48	1,841.95	3,550.00	1,948.96	3,880.32	2,619.63	4,660.26	2,469.99	4,848.48	3,219.41	5,664.35	3,025.62	5,881.10	3,870.41	6,754.63
Forecasted Sales Volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Period Cash Flow	-15	-15	-15	-18	358	1,248	922	1,971	988	2,174	1,400	2,653	1,308	2,769	1,768	3,270	1,649	3,404	2,168	3,940
Unit Price	-15	-15	-15	-17	334	1,145	831	1,746	860	1,860	1,177	2,192	1,062	2,210	1,387	2,521	1,249	2,534	1,587	2,834
Project NVP	25,466																			

Phase 3: Operational Strategy and Financial Viability Analysis

1. Phase 3 Summary

This report presents a supply chain analysis for the Honeywell Emergency Ready Lock, a smart home security system. The supply chain is structured around specialized suppliers providing key components to Honeywell's manufacturing and assembly facility.

Demand Estimation. Demand was estimated using the U.S. home security market, projected at 1.5 million new installs in 2024. With Honeywell holding an 8% market share, the baseline annual demand is 120,000 units, with an 8.5% growth rate applied through 2027. A Product Penetration Factor (PPF) was modeled using a logistic S curve, with a launch year PPF of 14% and a long run ceiling of 35%, producing annual demand estimates of 16,800 units in 2024, 24,738 in 2025, 33,904 in Year 1, and 42,917 in Year 2. Quarterly seasonal weights of 15% (Q1), 30% (Q2), 20% (Q3), and 35% (Q4) were then applied, yielding estimates from 5,086 units in Year 1 Q1 to 15,021 units in Year 2 Q4.

Demand Forecasting. Five methods were tested against the eight quarters of estimated demand including Static, Moving Average, Simple Exponential Smoothing, Holt's Method, and Winter's Method, and evaluated using MSE, MAD, MAPE, and Tracking Signal. Winter's Method was selected because it is the only method that captures both trend and seasonality simultaneously. Using smoothing constants of 0.3 (Level), 0.2 (Trend), and 0.3 (Seasonal Factor), it achieved the lowest MAPE of all methods tested. Projected annual demand grows from 28,878 units in Year 1 to 64,254 units in Year 4, a 122% increase.

Inventory Management. Cycle and safety inventory were analyzed for three subsystems including the Contact Sensor ($21.99/unit), Speaker ($3.52/unit), and Motor ($9.60/unit), as well as the finished product ($115.28/unit), using EOQ and a 95% Cycle Service Level (z = 1.65). Contact Sensor lot sizes grow from 2,562 to 3,822 units with a 22 to 32 day replenishment cycle; Speaker lot sizes peak above 9,500 units by 2029; Motor lot sizes scale from 3,878 to 5,785 units; and the finished product uses a lean approach with an 11 to 17 day replenishment cycle and average flow time under 9 days. By Year 4, annual safety inventory holding costs reach $58,434 for Contact Sensors, $13,208 for Speakers, $36,020 for Motors, and approximately $216,275 for the finished product. To reduce the Bullwhip Effect, Honeywell should update lot sizes semi annually.

Financial Modeling. The NPV model was built using a unit price of $299.00, unit production cost of $115.28, development cost of $60,480, ramp up cost of $6,048, annual marketing and support cost of $840,000, and a 7% discount factor, with sales volumes drawn from the Winter's Method forecasts. The base case NPV is $25,466,000, indicating the product is financially viable. Sensitivity analysis shows that a plus or minus 30% change in sales volume shifts NPV between $11,653,000 and $39,279,000; development cost changes of plus or minus 30% shift NPV by less than 0.1%; and unit cost changes of plus or minus 30% shift NPV between $19,403,000 and $30,395,000. The product remains profitable across all scenarios tested.

2. Demand Estimation for Years 1 and 2

Step 1: Establish Purpose

This section provides a demand forecast for the proposed Honeywell product across the first four years following launch. The objective is not to predict exact sales volume, but to establish a planning range that can support phased decisions related to production, sourcing, inventory, and launch readiness.

To improve reliability, demand was estimated using two complementary approaches:

- A top-down portfolio-based forecast, which estimates the share of Honeywell's annual category demand captured by the product over time
- An independent market-share build-up method to cross check the forecast

Quarterly demand was then derived from the annual forecast using a seasonal allocation model to support near-term operational planning.

Step 2: Define Key Variables

Variable	**Definition**	**Unit**
t	Years since launch	years
N_t	Relevant new homes in year t	homes
A_t	Security-system adoption rate	%
S_h	Honeywell's share of category demand	%
H_t	Honeywell's annual category demand	units
PPF_t	Product Penetration factor in year t	%
PPF_{max}	Long-run maximum product penetration	%
k	Adoption rate constant in the S-curve	-
A	S-curve anchor constant	-
D_t	Annual demand for the proposed product	units
SF_q	Seasonal factor for quarter q	%

Step 3: Estimation Approach

Demand for the product was modeled as a function of three main components:

- Total category demand available to Honeywell
- Product penetration within Honeywell's portfolio
- Seasonalities across the calendar year

The annual forecast structure is shown below:

$$D_t = H_t \times PPF_t$$

Then, to validate the previous estimate, we utilized the following cross-check procedure:

$$D_t = N_t \times A_t \times S_h \times PPF_t$$

This method separates housing activity, category adoption, brand share, and product-specific adoption rather than combining them into a single growth assumption, which proves useful for a second check.

Step 4: Planning Assumptions

To create our base-case demand estimates, we used the following planning assumptions:

Assumption	**Value**	**Rationale**
U.S homes installing security systems in baseline year, 2024	1,500,000	Initial market estimate
Honeywell category share	8%	Used to estimate the annual Honeywell demand
Honeywell baseline annual category demand	120,000 units	$1,500,000 \times 8\%$
Annual category growth rate	8.5%	Planning a growth scenario for the category demand
Launch year product penetration	14%	Baseline share adjusted for launch support and adoption friction
Long-run product penetration ceiling	35%	Assumes one product in the niche does not dominate the full portfolio
Adoption rate constant (k)	0.6	Reflects gradual but accelerating product adoption

These assumptions are simply treated as planning inputs, not as fixed outcomes. Actual demand will depend on market conditions, launch execution, customer adoption, channel support, and competitive response.

Step 5: Annual Category Demand Estimation

Honeywell's annual category demand was projected from the 2024 baseline of 120,000 units using the planning growth assumption of 8.5%:

$$H_t = H_{t-1}(1 + g),\ where\ g = 8.5\%$$

Year	Honeywell Annual Category Demand
2024	120,000
2025	130,200
2026	141,267
2027	153,275

Step 6: Product Penetration Model

Since the proposed product is newly introduced, its share of Honeywell's category demand is expected to increase gradually rather than begin at steady-state adoption. To reflect this ramp, product penetration was modeled using a logistic S-curve

$$PPF(t) = \frac{PPF_{max}}{1+Ae^{-kt}}$$

Parameter Selection		
Parameter	**Value**	**Interpretation**
PPF_{max}	0.35	Long-run ceiling for product penetration
k	0.60	Moderate adoption speed
$PPF(0)$	0.14	Launch-year product penetration
A	1.50	Anchor constant matching the chosen launch-year penetration

The launch-year penetration of 14% was derived from three considerations:

- an equal-share baseline across Honeywell's nine products in the category, or about 11%
- a launch support uplift factor of 1.6×
- an adoption-friction adjustment of 0.8×

$PPF(0) \approx 0.11 \times 1.6 \times 0.8 \approx 0.14$

So the final product penetration function:

$$PPF(t) = \frac{0.35}{1+1.5e^{-0.6t}}$$

Product Penetration by Year		
Year	t	PPF
2024	0	14%
2025	1	19%
2026	2	24%
2027	3	28%

Step 7: Base-Case Annual Demand Estimation

Annual product demand was estimated by applying the product penetration factor to Honeywell's annual category demand.

$$D_t = H_t \times PPF_t$$

Year	**Honeywell Annual Category Demand**	**PPF**	**Estimated Annual Product Demand**
2024	120,000	14%	16,800
2025	130,200	19%	24,738
2026	141,267	24%	33,904
2027	153,275	28%	42,917

This forecast represents the base case and should be used as a primary estimate for the early volume assumptions.

Step 8: Quarterly Demand Allocation

The estimated demand is not flat. It follows a seasonal trend and is influenced by holiday spikes and moving seasons, depending on the quarter (3-month periods). So we will apply a seasonal index to distribute the annual totals.

Seasonal Weighting:

- Quarter 1 [January to March]: (Low → 15%): Post-holiday slump and colder weather, so fewer moves
- Quarter 2 [April to June] (High → 30%): More homes are left unprotected when families go on vacation
- Quarter 3 [July to September] (Low → 20%): Lower demand for security systems post summer (vacations)
- Quarter 4 [October to December] (High → 35%): Holiday season/spike, as Black Friday and Cyber Monday increase sales

Based on these seasonal factors and our estimated annual demand, we will calculate the quarterly demands.

*Seasonal factors: 15% + 30% + 20% + 35% = 100%

Estimated Demands

Year / Annual Demand	Quarter	Seasonal Factor	Estimated Demand (Annual Demand × Seasonal Factor)
Year 1:	Q1	15%	5,086
	Q2	30%	10,171
	Q3	20%	6,781
	Q4	35%	11,866
Year 2:	Q1	15%	6,438
	Q2	30%	12,875
	Q3	20%	8,583
	Q4	35%	15,021

Step 9: Cross-Check Demand Framework

To validate the portfolio-based demand forecast, an independent market-share build-up method was developed. Unlike the primary method, which estimates demand as a share of Honeywell's annual category demand, this second method starts from the external relevant market and applies an expected product share trajectory based on launch maturity.

Model Structure

Variable	Definition
D_t	Annual demand for the proposed product in year t

R_t	Relevant annual market opportunity
S_t	Expected external market share captured by the product in year t

$$D_t = R_t \times S_t$$

The relevant annual market was held in the range of 1.45 to 1.48 million units, reflecting recent U.S. housing activity and a stable security-oriented market opportunity rather than aggressive construction growth. Recent Census data shows that U.S. housing authorizations declined from 1,478,000 in 2024 to 1,425,200 in 2025, indicating that housing-linked demand should be modeled conservatively

At the same time, broader smart-home and smart-home-security categories remain growth-positive. Grand View Research estimates 23.4% CAGR for the U.S. smart home market from 2025 to 2030 and 15.2% CAGR for the global smart home security market over the same period, supporting continued category adoption even under flatter housing conditions.

A gradual external market-share ramp was then applied to reflect launch constraints in Year 1 and stronger channel adoption over time.

Year	**Relevant Annual Market**	**Expected Product Share**	**Independent Forecast Demand**	**Forecast from the Previous Method**
2024	1,450,000	1.7%	24,650	16,800
2025	1,460,000	2.3%	33,580	24,738
2026	1,470,000	2.9%	42,630	33,904
2027	1,480,000	3.5%	51,800	42,917

To validate the portfolio-based forecast, an independent market-share build-up method was used to estimate demand directly from the relevant housing market. The assumed product share trajectory of *1.7%, 2.3%, 2.9%, and 3.5%* reflects realistic launch dynamics, including limited initial distribution, lower early customer awareness, and gradual increases in channel acceptance and product adoption over time. Comparing this estimate with the primary forecast provides a reasonableness check to ensure the base-case demand projection remains within a realistic planning range.

This independent method produces results that are directionally consistent with the primary portfolio-based forecast. The close alignment between the two methods increases confidence that the base-case demand estimate is reasonable for phase planning purposes.

Step 10: Forecast and Planning Implications

The demand forecast indicates that annual product demand is expected to grow from approximately *16,800 units at launch to about 42,900 units by Year 3,* based on the portfolio-based model applying a logistic

penetration curve to Honeywell's projected category demand. To cross check, we implemented an independent market-share validation method and using a relevant annual market of 1.45 to 1.48 million homes and an external product share ramp of *1.7%, 2.3%, 2.9%, and 3.5%,* which produced comparable demand estimates of *24,650, 33,580, 42,630, and 51,800 units,* confirming that the base-case forecast falls within a reasonable planning range. Together, these results support phased production and sourcing planning while indicating that capacity and inventory strategies should be prepared for demand levels approaching *40,000–50,000 units annually* as the product reaches broader market adoption.

3. Demand Forecasting for Years 3 and 4

Step 1: Forecasting Process and Implementation

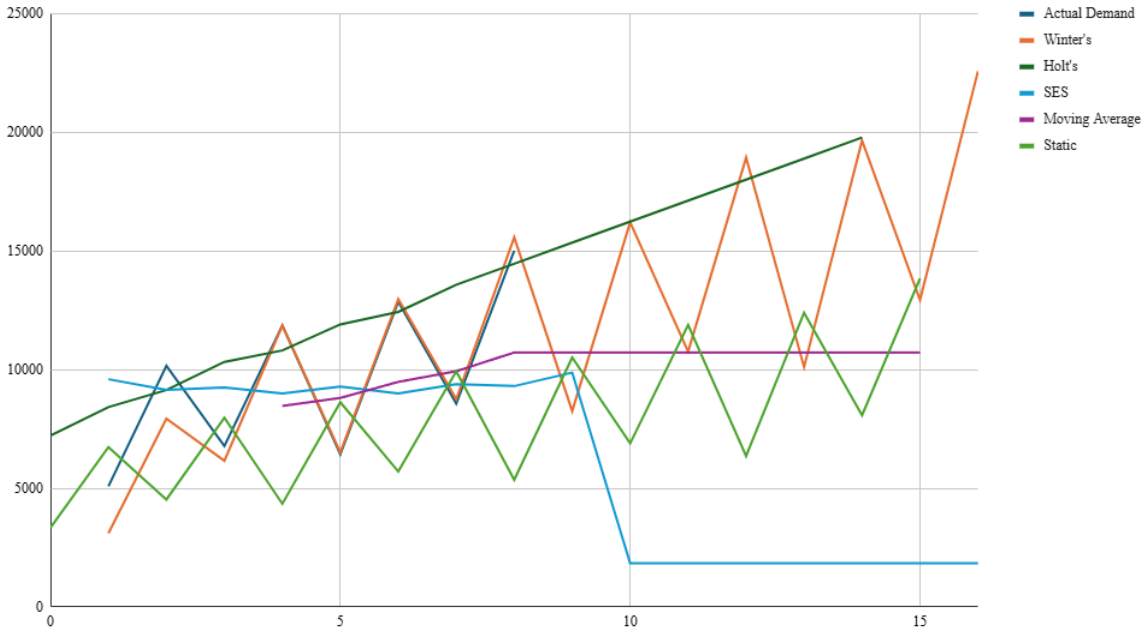
Using the demand estimation values from above, we proceeded with the demand forecasting with the five methods: Static, Moving Average, Simple Exponential Smoothing, Holt's Method, and Winter's Method. To calculate the relevant values per method, we used the following calculations below:

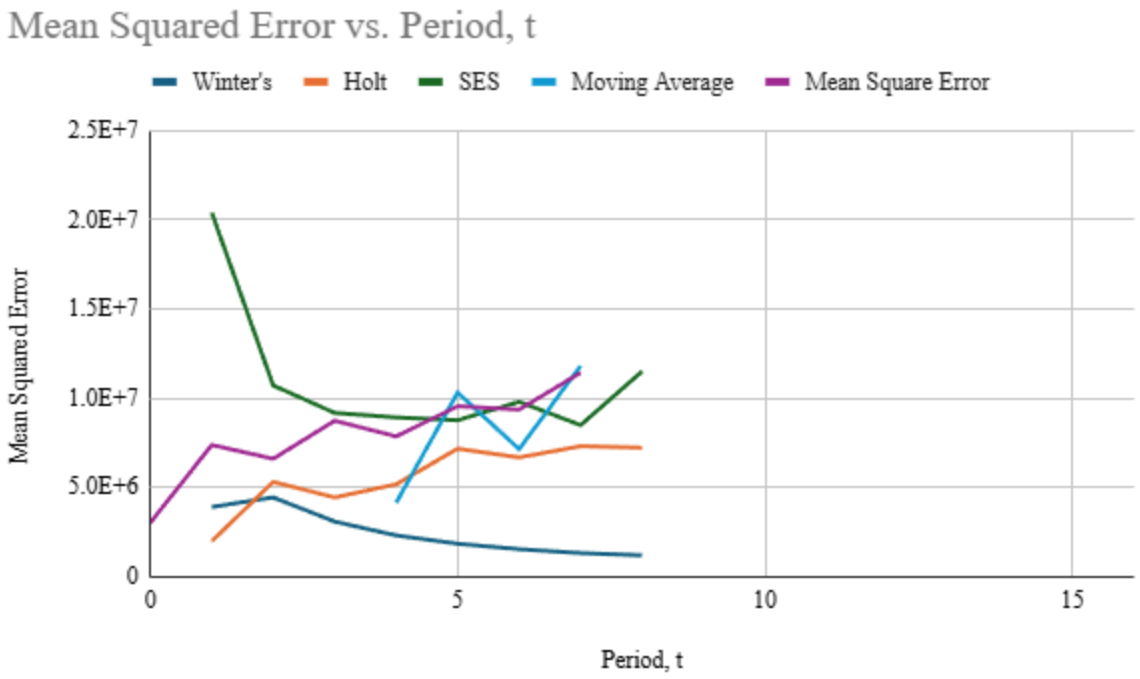
Value Type	**Formula**	**Method(s)**
Deseasonalized Demand	$\frac{Actual\ Demand}{Average\ Seasonal\ Factor}$	Static Only
Regressed Deseasonalized Demand	Intercept of Demand Data + (Current Demand * X-Var)	Static Only
Seasonal Factor	**Static:** $\frac{Actual\ Demand}{Deseasonalized\ Demand}$ **Winter:** Seasonality x $\frac{Current\ Demand}{Current\ Level}$ + (1- Seasonality) * Previous Seasonal Factor	Static and Winter
Average Seasonal Factor	Average of Seasonal Factors for the corresponding period across all cycles.	Static
Forecast	**Static:** Regressed Deseasonalized Demand x Average Seasonal Factor **Moving Average:** Sum of actual demands from previous n periods **SES:** Previous Forecast + Level * (Previous Actual Demand - Previous Forecast) **Holt:** Current Level + Current Trend **Winter:** (Current Level + Current Trend) x Previous Seasonal Factor	All Methods
Forecast Error	Actual Demand - Forecast	All Methods
Mean Square Error	Running Average of Forecast Error after Squaring	All Methods
Absolute Error	Absolute Value of Forecast Error	All Methods
Mean Absolute Deviation	$\frac{Sum\ of\ Absolute\ Errors}{Number\ of\ Periods}$	All Methods
Percent Error	$\frac{Forecast\ Error}{Actual\ Demand}$	All Methods

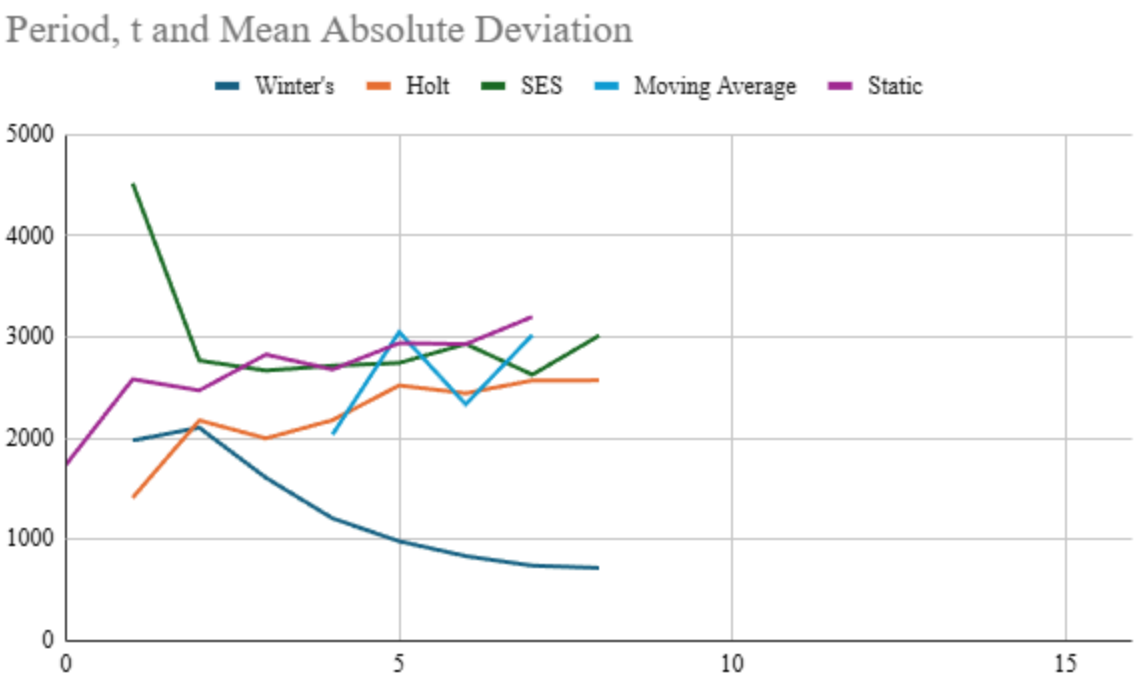
Absolute Percent Error	Absolute Value of Percent Error	All Methods
Mean Absolute Percent Error	$\frac{Sum\ of\ Absolute\ Percent\ Errors}{Number\ of\ Periods} * 100$	All Methods
Bias	Forecast Error + Previous Bias	All Methods
Tracking Signal	$\frac{Bias}{MAD}$	All Methods

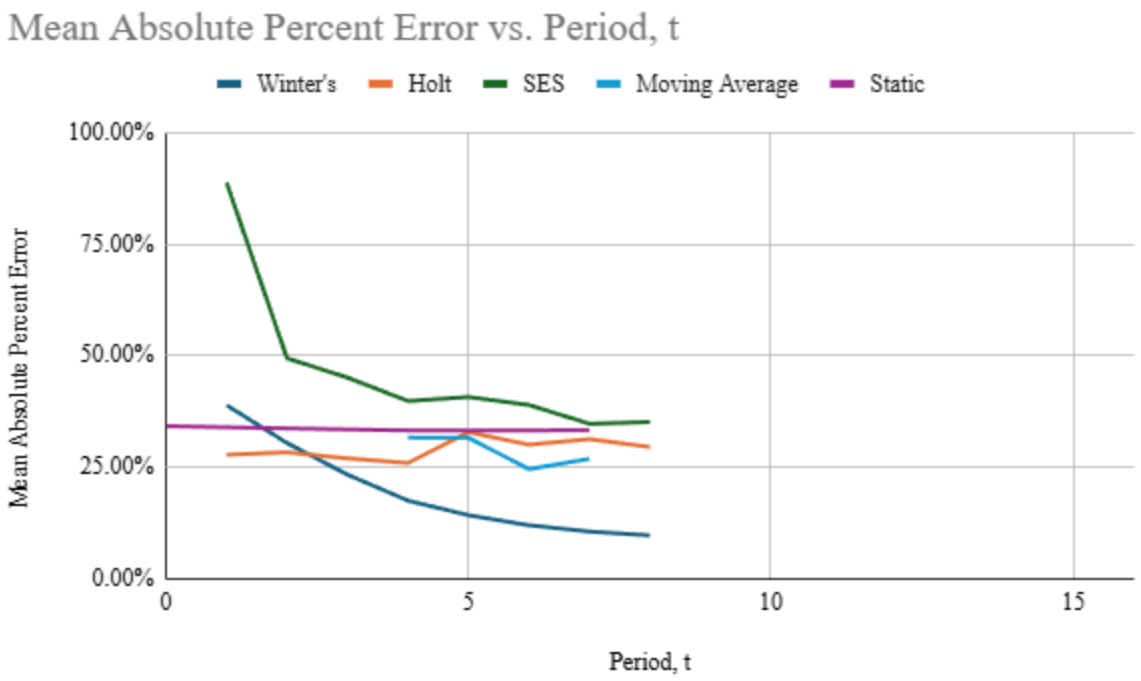
Step 2: Forecasting and Error Results

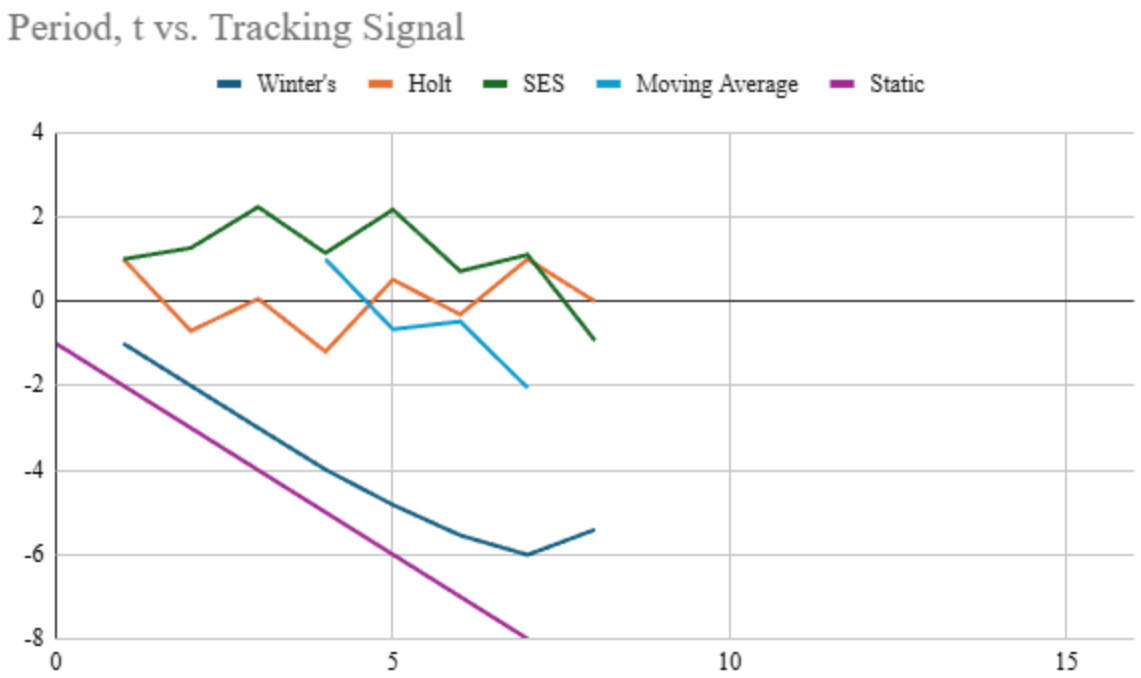
To identify the best forecasting model for our data, we looked closely at error metrics for all forecasting methods and graphed them. The graphs will provide a visual representation of errors for all methods used, allowing us to determine the best forecasting method. Each error of the methods were consolidated into one graph for better readability.

Forecasting and Error Results	
Demand vs. Forecasting	 This graph compares actual demand with several forecasting models across multiple time periods. The actual demand line fluctuates significantly, showing strong seasonal increases and decreases. Each forecasting method responds differently, with some models following the overall upward trend while others smooth out the fluctuations. Methods like Holt's and Winters capture the growth pattern more closely, while simpler models such as the moving average and SES produce smoother but less responsive forecasts. Overall, the graph highlights how different forecasting techniques vary in their ability to track changing demand patterns.

MSE vs. Period	 This graph compares the mean squared error (MSE) of the forecasting methods over time. Winter's method consistently shows the lowest MSE and continues to decline as more periods are observed, indicating that it produces the most accurate forecasts overall. Holt's method performs moderately well with relatively stable error values, while the moving average shows more variability across periods. SES begins with very high error levels but improves as more data becomes incorporated into the forecast.

MAD vs. Period	 This graph shows the mean absolute deviation of each forecasting method across time periods. Winter's method demonstrates a clear downward trend, indicating that forecast errors decrease as the model adjusts to the data. Holt's method remains relatively stable with moderate error levels, while the moving average fluctuates more noticeably. SES begins with the largest deviation but gradually improves over time as the smoothing process stabilizes.

MAPE vs. Period	 This chart measures the percentage error of each forecasting technique relative to the actual demand. Winter's method again shows the lowest and most consistent MAPE values, indicating strong forecasting accuracy. Holt's and the moving average models maintain moderate percentage errors across periods. SES starts with extremely high percentage error in early periods but steadily declines as the model adapts to the demand pattern.

Period vs. Tracking Signal	 The tracking signal graph evaluates whether forecasts are systematically biased over time. A tracking signal close to zero suggests that forecasts are balanced and not consistently over or under-predicting demand. Holt's and SES remain relatively close to zero, indicating more stable forecasting performance. Winter's method trends more negatively across periods, suggesting that it may slightly overestimate demand in later periods, while the static model shows the largest bias over time.

Step 3: Error Metrics Summary table

Forecasting Method	Forecast Error	Absolute Error	Mean Squared Error	Mean Absolute Deviation	Percent Error	Absolute Percent Error	Mean Absolute Percent Error	Bias	Tracking Signal
Static	-5087.88	11435413.93	5087.88	3200.11	-33.87	0.34	0.33	-25600.91	-8.00
Moving Average	-5080.50	5080.50	11819823.56	3021.63	-0.34	0.34	0.27	-6196.50	-2.05
SES	-5708.76	5708.76	11511054.23	3013.78	-0.38	0.38	0.35	-2804.95	-0.93
Holt's	-2577.83	2577.83	7236967.30	2572.74	-0.17	0.17	0.30	-9.55	0.00
Winter's	565.14	565.14	1206990.50	720.73	0.04	0.04	0.10	-3899.32	-5.41

Using these formulas for their respective methods, we calculated the relevant values for all five methods, where we summarized our final values in the table below. The columns represent the error type for Period 8, while the rows on the left represent the relevant forecasting method.

MAPE and Tracking Signal Analysis

Forecasting Method	Mean Absolute Percent Error	Tracking Signal
Static	0.33	-8.00
Moving Average	0.27	-2.05
SES	0.35	-0.93
Holt's	0.30	0.00
Winter's	0.10	-5.41

MAPE and tracking signal were used to evaluate which forecasting model performed the best. MAPE measures the average percent difference between the forecasted demand and the actual demand, which shows how accurate the forecasts are. A lower MAPE means the forecast is closer to the real demand and the model is more accurate. In our results, Winter's method had the lowest mean absolute percent error at about 0.10. The other models had higher values, such as 0.33 for the static model, 0.27 for the moving average model, 0.35 for simple exponential smoothing, and 0.30 for Holt's method. Because Winter's method produced the lowest MAPE, it generated forecasts that were closest to the actual demand overall.

The tracking signal was used to determine whether the forecasting models showed consistent bias over time. A tracking signal close to zero means the model is not consistently overestimating or underestimating demand. In our results, Holt's method had a tracking signal of 0, while simple exponential smoothing had −0.93 and moving average had −2.05, which all indicate relatively balanced forecasts. The static model had a tracking signal of −8, which shows strong bias and poor performance. Winter's method had a tracking signal of about −5.41, which suggests some overestimation in later periods. However, because it had the lowest MAPE and captured both trend and seasonality in the demand pattern, it still produced the most accurate forecasts overall and was selected as the best forecasting method.

Step 4: Conclusions and Findings

- To achieve the data as shown on the graph, we experimented with different constants for the Level, Trend, and Seasonal Factor, before arriving at the most optimal constants of 0.3, 0.2, and 0.3, respectively.
- Winter's Method is the forecasting method that we decided would be the best possible option.
    - The reasoning for this is that, although the forecast starts out below the demand, we see that eventually they are roughly equivalent throughout the rest of the timeframe.
    - This indicates that, for our specific use case and scenario, Winter's Method is a highly effective and accurate forecasting method that outperforms all of the other forecasting methods as seen above.

4. Cycle and Safety Inventory Management

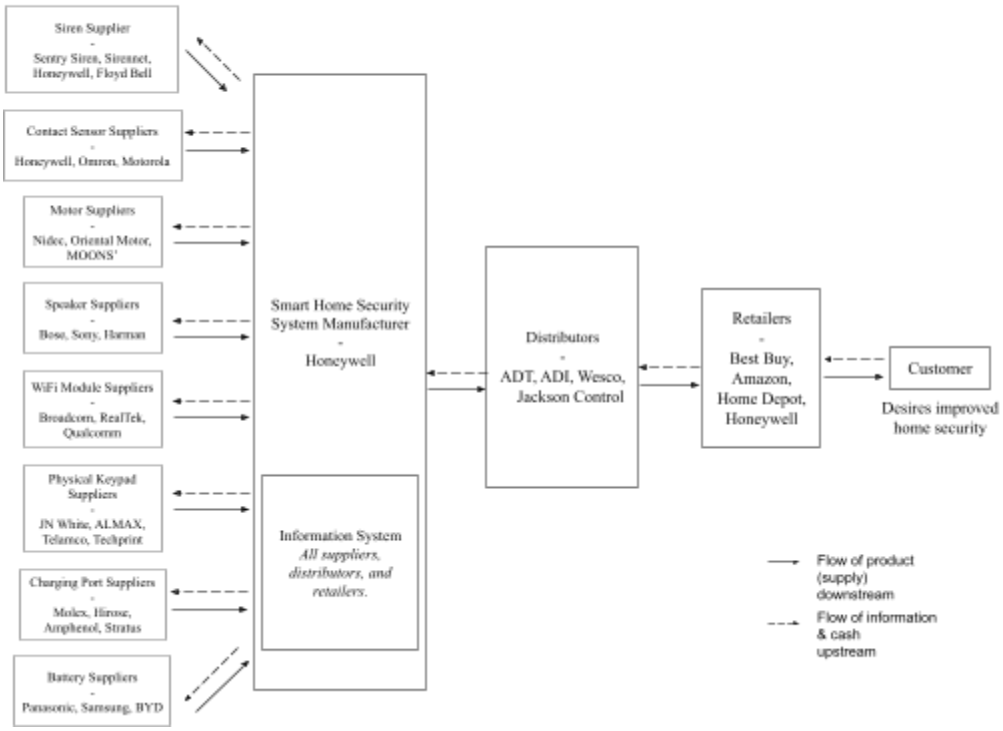
In order to calculate the Cycle and Safety Inventory for our product's supply chain, specifically for at least three key subsystems of the product and the finished product, we first need to start with the network representation of our product's supply chain and explain the flow from the various suppliers of subsystems through the manufacturer, distributors, retailers, and finally to the customers. Then, we introduce the purpose of Cycle and Safety Inventory within our supply chain, which will help us set up and perform a complete set of cycle and safety inventory calculations for the three subsystems and our finished product.

Step 1: Analyze Supply Chain Network Diagrams

Stage View Representation

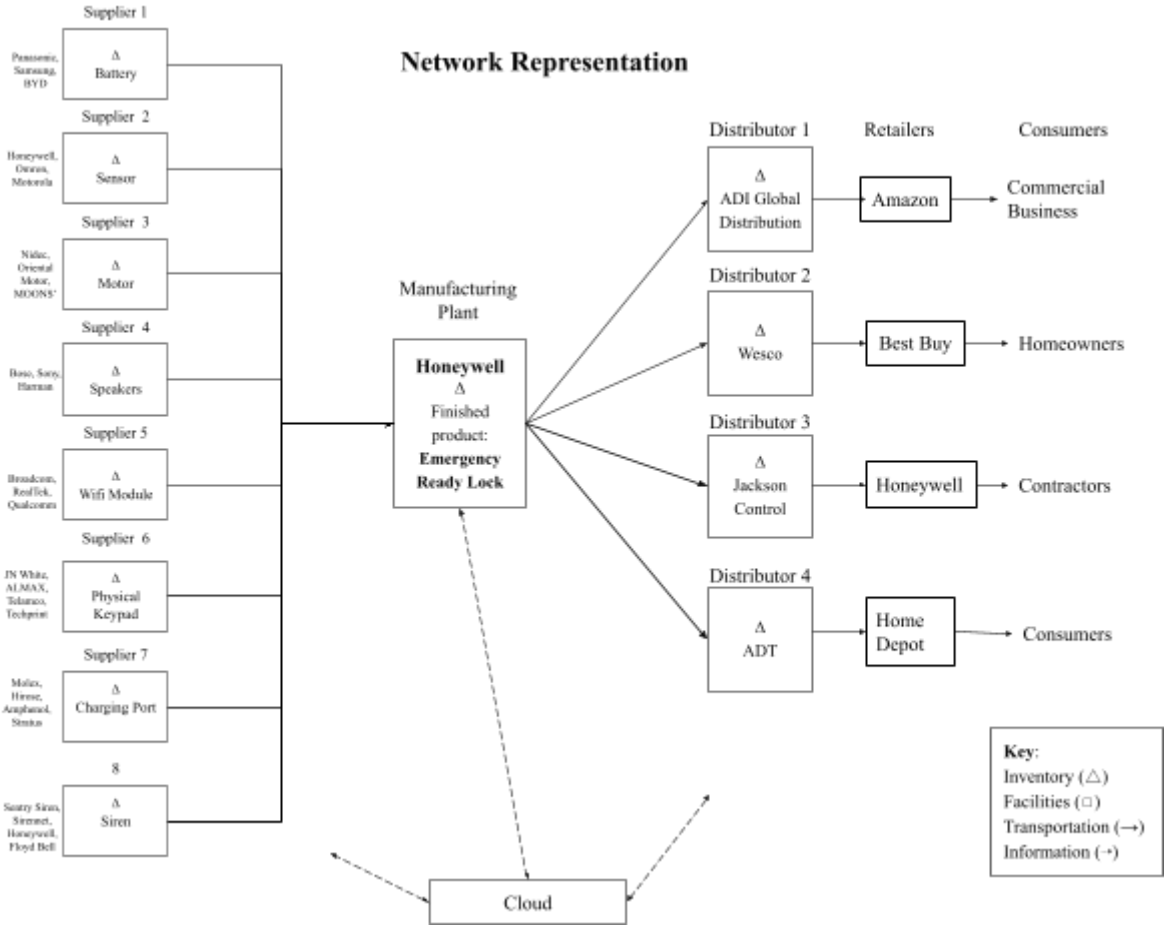
The stage view provides a simplified representation of the same supply chain, organized by the supply chain stage. On the left, eight supplier categories provide individual components to the Honeywell Manufacturing Plant at the center. The finished Emergency Ready Lock then moves through distributors and retailers before reaching the end customer. This diagram makes clear the stages through which both components and finished products travel, and serves as the foundation for identifying which specific stages and subsystems will be analyzed in the cycle inventory calculations that follow

Stage View of the Supply Chain (SC) for Emergency Ready Lock

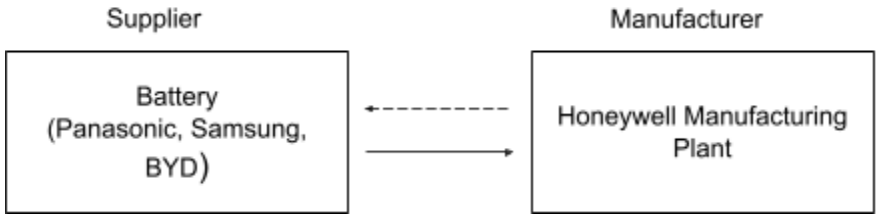


Network Representation

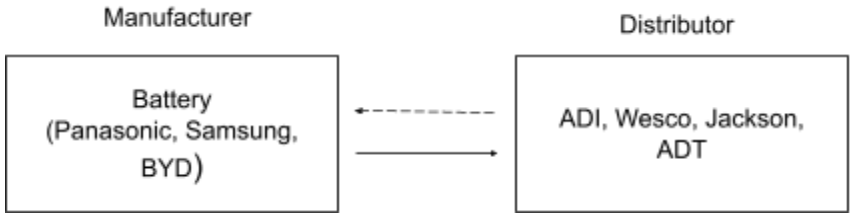
The network representation shows the full supply chain for the Honeywell Emergency Ready Lock. Eight supplier categories (Battery, Sensor, Motor, Speaker, WiFi Module, Physical Keypad, Charging Port, and Siren) feed directly into the Honeywell Manufacturing Plant, where the finished product is assembled. From there, the finished product flows downstream through four distributors (ADI Global Distribution, Wesco, Jackson Control, and ADT) to retailers (Amazon, Best Buy, Honeywell, Home Depot) and finally to end customers, including commercial businesses, homeowners, contractors, and consumers. Information flows upstream through a centralized cloud system connecting all stages of the network.



Component flow (Battery Supply → Manufacturing Plant):



Finished Product Flow (Manufacturing Plant → Distributor):



Step 2: Subsystem Selection

Based on our FAST and Network representation, the Emergency Ready Lock consists of multiple hardware subsystems sourced from specialized T1 suppliers and integrated at a centralized Honeywell manufacturing facility. For our cycle and safety inventory analysis, the three key subsystems we selected were: **speakers, contact sensor, and motor subsystems**. We chose these for our analysis as they are critical to the product's safety and intrusion detection function, sourced externally from specialized suppliers, replenished repeatedly over the product's lifecycle, and represent the different costs, volume, and variability profiles present in our system overall. The cycle and safety inventory for the completed Emergency Ready Lock is analyzed to ensure alignment between component replenishment and final product availability.

Step 3: Inventory Calculations

Step 3.a: To calculate the Cycle and Safety Inventory, we would first need the forecasted demand data for the years 2026 to 2029. These were obtained from our best forecasting results, which were Winter's method.

Annual Demand for 2026 (Winter forecasting):

Year	Forecasts, Ft+l	Total Yearly Demand, D
2026	3,107	28,878
	7,938	
	6,117	
	11,716	
2027	6,412	43,188
	12,756	
	8,630	
	15,390	
2028	8,178	53,555
	16,029	
	10,639	
	18,709	
2029	9,961	64,254
	19,345	
	12,732	
	22,216	

Step 3.b: Then based on internet research, we estimated the unit costs, holding costs, and shipping cost per order for the three chosen sub systems and our finished product.

Costs	Contact Sensor	Speaker	Motor	Complete Product
Unit Cost, C	$21.99	$3.52	$9.60	$115.28
Holding Cost, h	20% = 0.2	20% = 0.2	20% = 0.2	20% = 0.2
Shipping Cost Per Order, S	$500	$500	$500	$750

These forecasted demands and estimated costs were then used to calculate Cycle and Safety Inventory for the three subsystems and our finished product.

Step 3.c: Cycle Inventory for 3 subsystems and finished product

To determine Cycle Inventory and Cycle Inventory Holding Cost, the following formulas were used:

Calculation	Formula
Annual Demand, D Unit cost, C Holding cost, h Shipping cost, S	Determined above
Optimal Lot Size	$Q_L^* = \sqrt{\frac{2DS}{hC}}$
Number of shipments per year	$n = D / Q_L$
Cycle inventory	$Q_L / 2$
Cycle inventory holding cost	$CI = (Q_L / 2) \times (hC)$
Replenishment cycle time	$T = 365 / n$
Average flow time	T/2

Cycle Inventory Results:

Year	Annual Demand, D	Sub system / Finished product	Unit cost, C	Holding cost, h	Shipping cost, S	Optimal Lot Size, Q_L^*	# of shipments/year, n = D/Q_L	Cycle Inventory, $Q_L/2$	Cycle inventory holding cost, CI = (QL / 2) × (hC)	Replenishment cycle time, T = 365/n	Average flow time, T/2
Year 1 - 2026	28878	Contact Sensor	21.99	0.2	500	2562.45	11.27	1281.23	$5,634.83	32.39	16.19
	28878	Speaker	3.52	0.2	500	6404.68	4.51	3202.34	$2,254.45	80.95	40.48
	28878	Motor	9.6	0.2	500	3878.22	7.45	1939.11	$3,723.10	49.02	24.51
	28878	Finished Product	115.28	0.2	750	1370.68	21.07	685.34	$15,801.24	17.32	8.66
Year 2 - 2027	43188	Contact Sensor	21.99	0.2	500	3133.67	13.78	1566.84	$6,890.95	26.48	13.24
	43188	Speaker	3.52	0.2	500	7832.41	5.51	3916.20	$2,757.01	66.19	33.10
	43188	Motor	9.6	0.2	500	4742.76	9.11	2371.38	$4,553.05	40.08	20.04
	43188	Finished Product	115.28	0.2	750	1676.24	25.76	838.12	$19,323.65	14.17	7.08
Year 3 - 2028	53555	Contact Sensor	21.99	0.2	500	3489.57	15.35	1744.79	$7,673.57	23.78	11.89
	53555	Speaker	3.52	0.2	500	8721.95	6.14	4360.98	$3,070.13	59.44	29.72
	53555	Motor	9.6	0.2	500	5281.40	10.14	2640.70	$5,070.15	36.00	18.00
	53555	Finished Product	115.28	0.2	750	1866.61	28.69	933.31	$21,518.28	12.72	6.36
Year 4 - 2029	64254	Contact Sensor	21.99	0.2	500	3822.28	16.81	1911.14	$8,405.19	21.71	10.86
	64254	Speaker	3.52	0.2	500	9553.53	6.73	4776.76	$3,362.84	54.27	27.13
	64254	Motor	9.6	0.2	500	5784.95	11.11	2892.47	$5,553.55	32.86	16.43
	64254	Finished Product	115.28	0.2	750	2044.58	31.43	1022.29	$23,569.90	11.61	5.81

Step 3.d: Safety Inventory for 3 subsystems and finished product

In order to determine Safety (Stock) Inventory and Annual Safety Inventory Holding Cost, the following formulas were used:

Calculation	**Formula**
Annual Demand, D Unit cost, C Holding cost, h Shipping cost, S	Determined above
Mean weekly demand	$D_W = D_{annual} / 52$
Standard deviation of weekly demand	σ_W (2026) =STDEV(3107,7938,6117,11716) σ_W (2027) =STDEV(6412,12756,8630,15390) σ_W (2028) =STDEV(8178,16029,10639,18709) σ_W (2029) =STDEV(9961,19345,12732,22216) *using Excel formula* =STDEV(forecasted quarterly demands for each year)
Supplier lead time (weeks), L	Contact sensor, L = 2 Speaker, L = 4 Motor, L = 4 Finished product, L = 1 *Lead times were curated based on information regarding our 3rd party sourcing (real lead times based on website info).*
Mean value of demand during the supplier lead time	$D_L = (D_L)_m = LD_W$
Standard deviation of demand during the supplier lead time	$\sigma_L = \sqrt{L}\,\sigma_W$
$(CSL)_{desired}$	95% = 0.95 ⇒ 1.65
Safety stock	$ss = \sqrt{L}\,\sigma_W\,F^{-1}[(CSL)_{desired}]$
Re-order point	$ROP = LD_W + ss$
Annual safety inventory holding cost	(ss) × (hC)

Expected Shortage per Cycle, ESC	$\text{ESC} = -\ ss\left[1 - F\left(z = \frac{ss}{\sigma_L}\right)\right] + \sigma_L\left[f\left(z = \frac{ss}{\sigma_L}\right)\right]$
Fill Rate, fr	$\text{fr} = 1 - \frac{ESC}{Q_L}$

Safety Inventory results:

Year	Annual Demand, D	Sub system / Finished product	Unit cost, C	Holding cost, h	Shipping cost, S	Mean weekly demand, Dw	Std dev of weekly demand, σw	Supplier lead time (weeks), L	Mean value of DL, (DL)m = LDw	Std dev of DL, σL = √L σw	(CSL)desired = 95% = 0.95
Year 1 - 2026	28878	Contact Sensor	21.99	0.2	500	555.35	3599.21	2	1110.69	5090.05	1.65
	28878	Speaker	3.52	0.2	500	555.35	3599.21	4	2221.38	7198.42	1.65
	28878	Motor	9.6	0.2	500	555.35	3599.21	4	2221.38	7198.42	1.65
	28878	Finished Product	115.28	0.2	750	555.35	3599.21	1	555.35	3599.21	1.65
Year 2 - 2027	43188	Contact Sensor	21.99	0.2	500	830.54	4035.57	2	1661.08	5707.16	1.65
	43188	Speaker	3.52	0.2	500	830.54	4035.57	4	3322.15	8071.14	1.65
	43188	Motor	9.6	0.2	500	830.54	4035.57	4	3322.15	8071.14	1.65
	43188	Finished Product	115.28	0.2	750	830.54	4035.57	1	830.54	4035.57	1.65
Year 3 - 2028	53555	Contact Sensor	21.99	0.2	500	1029.90	4830.08	2	2059.81	6830.76	1.65
	53555	Speaker	3.52	0.2	500	1029.90	4830.08	4	4119.62	9660.16	1.65
	53555	Motor	9.6	0.2	500	1029.90	4830.08	4	4119.62	9660.16	1.65
	53555	Finished Product	115.28	0.2	750	1029.90	4830.08	1	1029.90	4830.08	1.65
Year 4 - 2029	64254	Contact Sensor	21.99	0.2	500	1235.65	5685.09	2	2471.31	8039.94	1.65
	64254	Speaker	3.52	0.2	500	1235.65	5685.09	4	4942.62	11370.19	1.65
	64254	Motor	9.6	0.2	500	1235.65	5685.09	4	4942.62	11370.19	1.65
	64254	Finished Product	115.28	0.2	750	1235.65	5685.09	1	1235.65	5685.09	1.65

Safety stock, ss = √L σw F-1[(CSL)desired]	Reorder point, ROP = LDw + ss	Annual safety inventory holding cost, (ss)(hC)	Optimal Lot Size, QL*	Expected Shortage per Cycle, ESC	Fill Rate
8398.58	9509.28	$36,936.97	2562.45	100.60	96.07%
11877.39	14098.77	$8,361.68	6404.68	142.28	97.78%
11877.39	14098.77	$22,804.59	3878.22	142.28	96.33%
5938.69	6494.04	$136,922.55	1370.68	71.14	94.81%
9416.81	11077.88	$41,415.12	3133.67	112.80	96.40%
13317.38	16639.53	$9,375.43	7832.41	159.53	97.96%
13317.38	16639.53	$25,569.36	4742.76	159.53	96.64%
6658.69	7489.23	$153,522.72	1676.24	79.76	95.24%
11270.76	13330.57	$49,568.81	3489.57	135.01	96.13%
15939.26	20058.88	$11,221.24	8721.95	190.93	97.81%
15939.26	20058.88	$30,603.39	5281.40	190.93	96.38%
7969.63	8999.54	$183,747.84	1866.61	95.47	94.89%
13265.90	15737.20	$58,343.42	3822.28	158.91	95.84%
18760.81	23703.43	$13,207.61	9553.53	224.73	97.65%
18760.81	23703.43	$36,020.76	5784.95	224.73	96.12%
9380.41	10616.06	$216,274.64	2044.58	112.37	94.50%

Step 3.e: Total Inventory Holding Cost (Annual)

In order to determine the Total Inventory Holding Cost, the following formulas were used:

Calculation	Formula
Annual Demand, D Unit cost, C Holding cost, h Shipping cost, S	Determined above
Cycle inventory	$Q_L / 2$
Safety stock	$ss = \sqrt{L}\,\sigma_W\,F^{-1}[(CSL)_{desired}]$
Total inventory holding cost	$((Q_L / 2) + ss) \times (hC)$

Total Inventory Holding Cost results:

Year	Annual Demand, D	Sub system / Finished product	Unit cost, C	Holding cost, h	Cycle Inventory, QL/2	Safety stock, ss = √L σW F-1[(CSL)desired]	Total inventory holding cost = (QL / 2 + ss) (hC)
Year 1 - 2026	28878	Contact Sensor	21.99	0.2	14439.00	8398.58	$100,439.69
	28878	Speaker	3.52	0.2	14439.00	11877.39	$18,526.74
	28878	Motor	9.6	0.2	14439.00	11877.39	$50,527.47
	28878	Finished Product	115.28	0.2	14439.00	5938.69	$469,828.13
Year 2 - 2027	43188	Contact Sensor	21.99	0.2	21594.00	9416.81	$136,385.53
	43188	Speaker	3.52	0.2	21594.00	13317.38	$24,577.61
	43188	Motor	9.6	0.2	21594.00	13317.38	$67,029.84
	43188	Finished Product	115.28	0.2	21594.00	6658.69	$651,393.98
Year 3 - 2028	53555	Contact Sensor	21.99	0.2	26777.50	11270.76	$167,336.26
	53555	Speaker	3.52	0.2	26777.50	15939.26	$30,072.60
	53555	Motor	9.6	0.2	26777.50	15939.26	$82,016.19
	53555	Finished Product	115.28	0.2	26777.50	7969.63	$801,129.88
Year 4 - 2029	64254	Contact Sensor	21.99	0.2	32127.00	13265.90	$199,637.96
	64254	Speaker	3.52	0.2	32127.00	18760.81	$35,825.02
	64254	Motor	9.6	0.2	32127.00	18760.81	$97,704.60
	64254	Finished Product	115.28	0.2	32127.00	9380.41	$956,994.75

Step 4: Results & Interpretations

Annual Demand

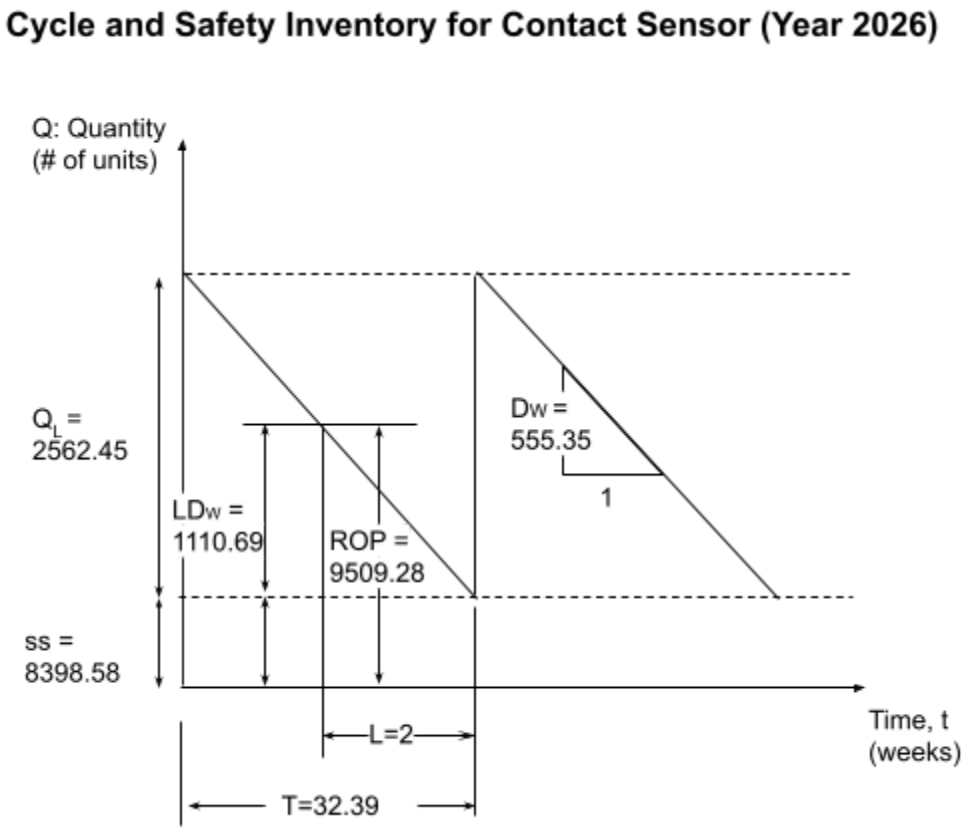
The demand for the Emergency Ready Lock is projected to grow immensely, rising from 28,878 units in 2206 to 64,254 units by 2029. Since this demand is calculated using Winter's method, it accurately accounts for the seasonal fluctuations that are prevalent in the home security market. Since there is almost a 122% increase in volume, we anticipate the necessity for a highly scalable supply chain so as to prevent shortages during peak purchasing periods. In conjunction with mitigating the Bullwhip Effect across its global T1 supplier network, Honeywell needs to proactively update its optimal lot size every six months.

Cycle Inventory Analysis for Contact Sensor

The Contact Sensor unit carries a moderate unit cost of $21.99, as it is a critical component utilized for the lock's primary intrusion detection function. Our analysis shows that the optimal lot size should increase from 2,562 units to 3,822 units over the 4 years to match the estimated rising demand. By maintaining a replenishment cycle of around 22-32 days, Honeywell ensures a high level of agility for these high-value components, and the increased frequency of replenishment in the strategy minimizes the risk of having defective or obsolete hardware, while keeping holding costs balanced.

Safety Inventory Analysis for Contact Sensor

With a two-week supplier lead time and a target 95% CSL, safety stock grows from 8,399 to 13,266 units over the four years, driving annual holding costs from $36,939 to $58,434. The relatively short lead time keeps safety stock manageable, but Honeywell should monitor demand variability closely as volume scales. Average fill rate over the four years is approximately 96.11%, which indicates high operational efficiency.



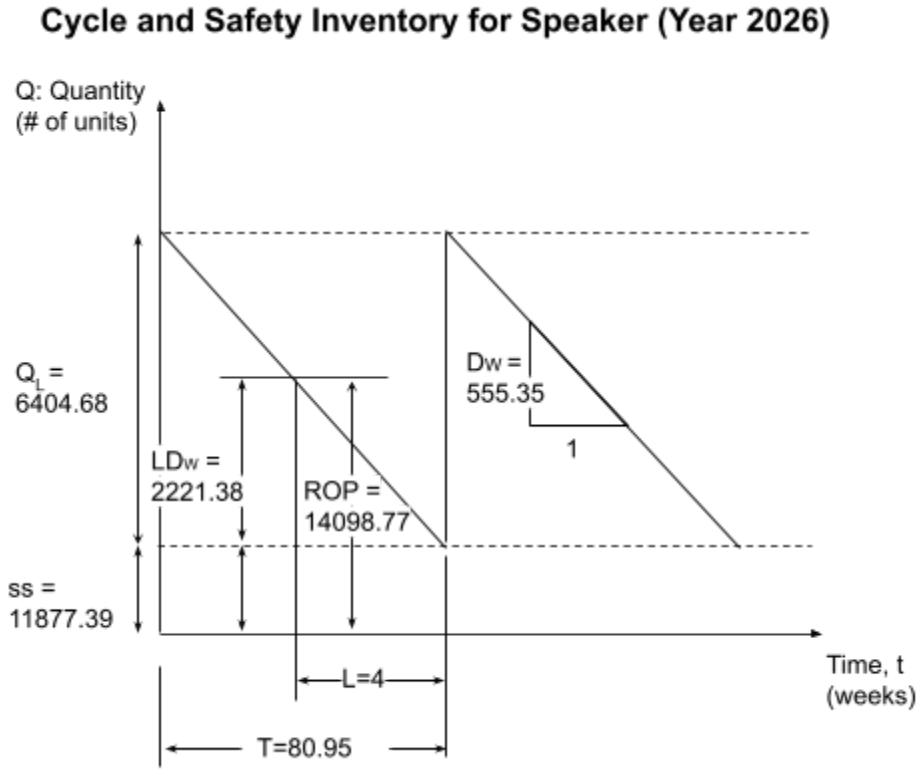
Cycle Inventory Analysis for Speaker

The Speaker unit represents a bulk-ordered component that has the lowest unit cost in the system at $3.52.Since the shipping cost per order is fixed at $500, the EOQ model generates the largest lot size, peaking at over 9,500 units by 2029. Through this, we get the longest average flow time of roughly 40 days, indicating that Honeywell is intentionally leveraging storage to minimize transportation expenses. Since these mechanical components have low technological obsolescence, holding a larger cycle inventory here is an effective strategy to lower the total cost of goods sold

Safety Inventory Analysis for Speaker

The speaker's 4 week lead time doubles demand exposure during replenishment, pushing safety stock from 11,877 to 18,761 units by year 4. Despite the low unit cost of $3.52, annual holding cost still reached $13,208, justifying close coordination with suppliers to potentially reduce lead time and buffer

requirements. Average fill rate over the four years is approximately 97.75%, which indicates high operational efficiency.

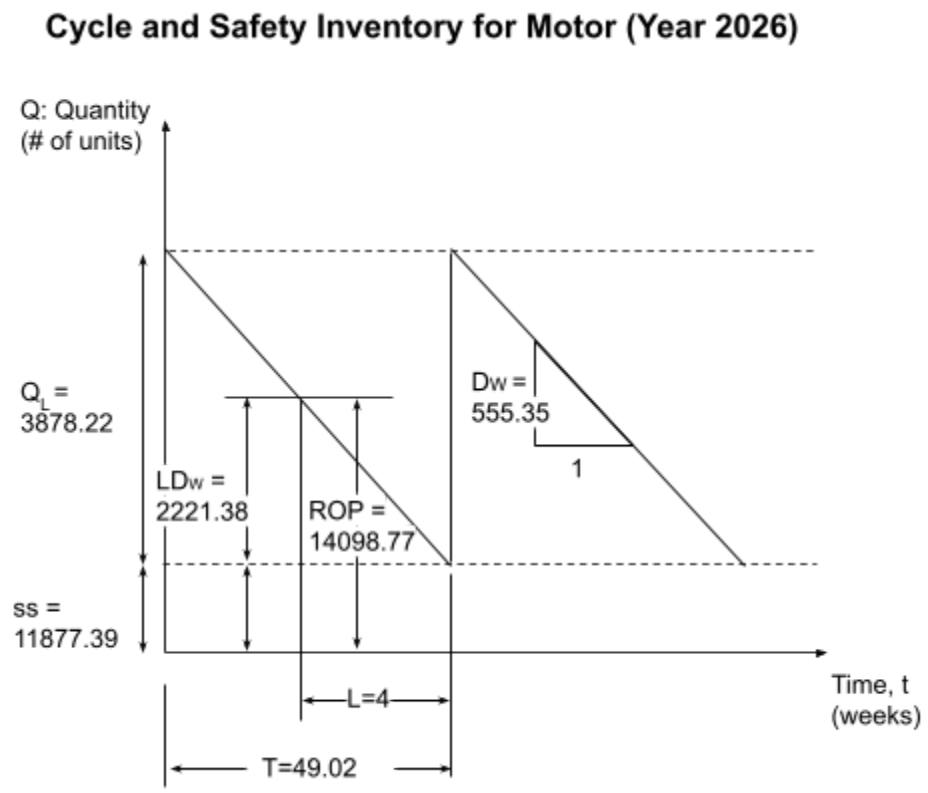


Cycle Inventory Analysis for Motor

The Motor unit serves as the mechanical core of the lock and sits at a mid-range unit cost of $9.60 per component. As demand scales, the optimal lot size grows from 3,878 to 5,785 units, allowing Honeywell to triple its output while only increasing the number of annual shipments from 7 to 11. This demonstrates a highly efficient use of economies of scale, where the company maximizes the value of each $500 shipping fee. By balancing a 33-to-49-day replenishment cycle, the company maintains enough stock to support assembly without incurring excessive financial burdens.

Safety Inventory Analysis for Motor

With a 4-week lead time and moderate unit cost of $9.60, Motor safety stock climbed from 11,877 to 18,761 units, resulting in annual holding costs reaching 36,020 by year 4. Given the motor's critical role in assembly continuity, maintaining this buffer is essential to prevent costly production stoppages. Average fill rate over the four years is approximately 96.37%, which indicates high operational efficiency.



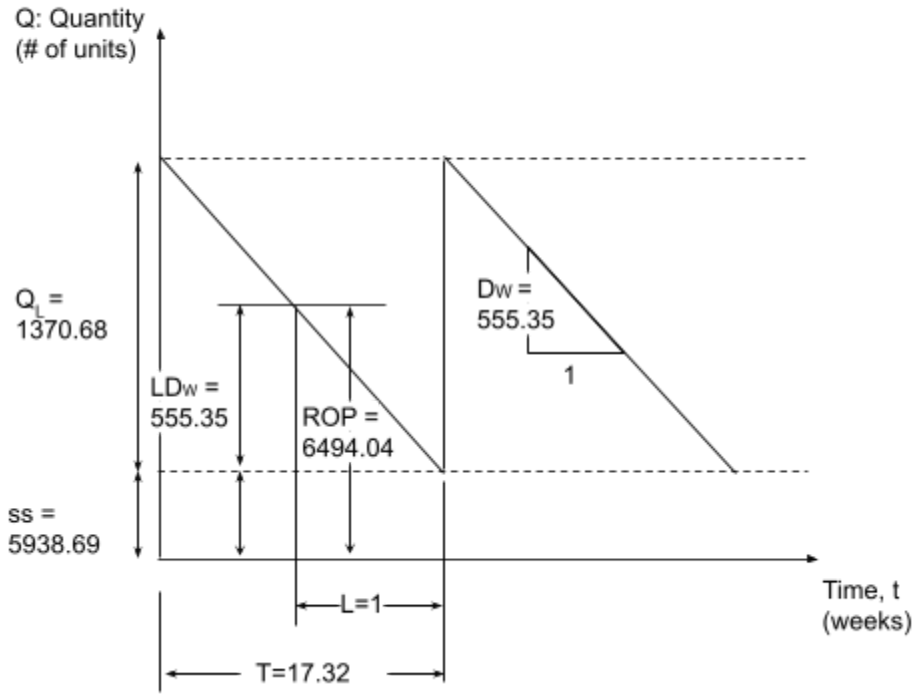
Cycle Inventory Analysis for Finished Product Interpretation

The Emergency Ready Lock is the most expensive item in the chain at $115.28, which demands a "Lean" inventory approach to avoid tying up excessive capital. To minimize high holding costs, the system utilizes the smallest relative lot sizes and a rapid replenishment cycle of just 11 to 17 days. This results in a very low average flow time of under 9 days, meaning the completed locks move through the Honeywell plant and out to distributors almost immediately. This strategy optimizes cash flow and ensures that the highest-value inventory is always moving towards the customer.

Safety Inventory Analysis for Finished Product Interpretation

The finished product benefits from a short 1-week lead time, keeping safety inventory stock relatively low at 5,393 to 9,380 units. However, at $115.28 per unit, holding costs are the highest in the supply chain, reaching ~$216, 275 annually by year 4, reinforcing the need for tight demand forecasting accuracy to avoid excess finished goods in inventory. Average fill rate over the four years is approximately 94.86%, which indicates high operational efficiency.

Cycle and Safety Inventory for Finished Product (Year 2026)



5. Financial Modeling

A financial model will allow Honeywell to determine the feasibility of the project. The calculation focuses on calculating the net present value (NPV) and considers sales volume per year, developmental cost, unit price, unit product cost, ramp-up cost, marketing cost, and annual discount factor. A negative NPV means the project is not feasible, and a positive NPV means that the project will be feasible. A simple step-wise process is the following:

Step 1: Explain the NPV setup used from the "Excel Tutorial".
Step 2: Use demand data processed from previous sections for sales volume.
Step 3: Define our NPV estimation process; the rationale for every numerical value.
Step 4: Explain the results for each NPV base case.
Step 5: Explain the NPV Sensitivity Results.

Step 1: NPV Setup

To estimate the product's base-case estimated profit, or Net Present Value, we referenced the "Excel Tutorial" written by R.Polany and S.Desa to create a modified version of the tutorial suitable for our product needs. The excel spreadsheet required the input of the project development cost, the unit price, unit product cost, ramp-up cost, marketing & support cost, annual discount factor to calculate the profit.

Inputs	Excel Formula & explanation
Development Cost	=(-1)*ScenarioParameters!$B$3/4*(1/1000) ● Multiplied by -1 to mathematically represent the cost. ● Total development cost is divided by 4 quarters to distribute development cost into 1 year. ● Divided by 1000 for scaling.
Ramp Up	=(-1)*ScenarioParameters!$B$6/2*(1/1000) ● Total ramp up cost is divided by 2 to represent 2 quarters of ramp up time. ● Divided by 1000 for scaling.
Marketing & Support Cost	=(-1)*ScenarioParameters!$B$7/4*(1/1000) ● Yearly marketing and support cost is divided by 4 to represent 4 quarters per year. ● Divided by 1000 for scaling.
Production Cost	=(-1)*(ScenarioParameters!$B$5)*ScenarioParameters!$G3*(1/1000) ● Unit cost multiplied by forecasted demand (G3) for the quarter. ● Divided by 1000 for scaling.
Production Volume	=ScenarioParameters!$G3 ● Forecasted demand (G3) for the quarter.
Unit Product Cost	=F9/F10 ● Production Cost divided by Production volume
Sales revenue	=(ScenarioParameters!$B$4*F10)*(1/1000)

	● Unit retail price multiplied by production volume. ● Divided by 1000 for scaling.
Forecasted Sales Volume	=F$10 ● Forecasted demand (F10) for the quarter.
Unit price	=F$12/F$13 ● Sales revenue divided by forecasted sales volume.
Period Cash Flow	=SUM(F6+F7+F8+F9+F12) ● All relevant costs and revenues added.
Unit Price	=F16/(1+(ScenarioParameters!$B$8/100)/4)^E3 ● Period cashflow divided by the discount factor for the quarter, where the discount factor is a growing exponential of the period.
Project NVP	=SUM(B17:U17) ● All unit prices for each quarter added.

After setting up the base case for the tutorial, we input known inputs from forecasting and performed online research to estimate the development cost, unit price, unit product cost, ramp up cost, marketing & support cost, and annual discount factor.

Step 2: NPV Input Estimation

The following table showcases how we were able to estimate each input value in the net present value calculation:

Net Present Values Estimations		
Sales & Product Volume (units/quarter)	106,825 units	The unit quantity for each quarter was imported from the forecasting model. For example, “Year 1, Projection Q1” in the financial modeling spreadsheet directly corresponds to Year 1 Q1 in our forecasting model (Winter’s).
Development Cost (total $)	$60,480	The development cost was determined using an estimation of estimated cost based on the team’s time commitment. Assuming a 9-hour commitment for 15 weeks by a team of 8 provides the following estimation: $= 9\ hours\ *\ 15\ weeks\ *\ 8\ members\ *\ \$56/hr$ $= \$60,480$ Where $56/hr is the average hourly wage of a research & development engineer according to ZipRecruiter (see appendix for additional information).

Unit price ($/unit)	$299	The unit price was determined by comparing competing product prices and identifying a competitive price. The following competing products were compared in MSRP: ● SimpliSafe Security ($250-$750, $440 is most comparable) ● Ring Alarm Pro ($430) ● EcoBee ($460) With this information, we establish our customer's willingness to pay up to $460 for similar competing products. Placing our product at $299 would considerably undercut competing products, with only SimpliSafe offering a more basic security system at this price range.
Unit Productions Cost ($/unit)	$115.28	The unit product cost was systematically determined by researching the cost of major subsystems identified in previous FAST diagrams, then providing an estimation of missing components. A comprehensive table for this result is found in the next section.
Ramp-up cost (total $)	$6,048	According to Tacto Management, Ramp-up cost is typically 5%-15% of annual cost. For our model, we utilized a 10% ramp up cost, which is between the typical 5%-15%, and applied this toward the current development cost. $= \$60,480 * 10\%$ $= \$6,048$
Marketing & support cost ($/year)	$840,000	According to Proto Marketing, marketing & support cost is typically 5%-15% of revenue. Our financial model without marketing & support determined an NPV of $28,000,000; or an NPV of 5,600,000 per year. Considering that NPV is profit and not actual revenue, we will consider using the NPV value with a higher marketing & support cost percentage to account for the difference between profit and revenue. $= \$5,600,000 * 0.15$ $= \$840\,000$
Annual Discount Factor (%)	7%	Our financial model will utilize a 7% discount factor which aligns with another discount factor Honeywell previously used when estimating the cost of an environmental clean-up project. This also aligns closely with the 10% estimation For Entrepreneurs recommends using for public companies.

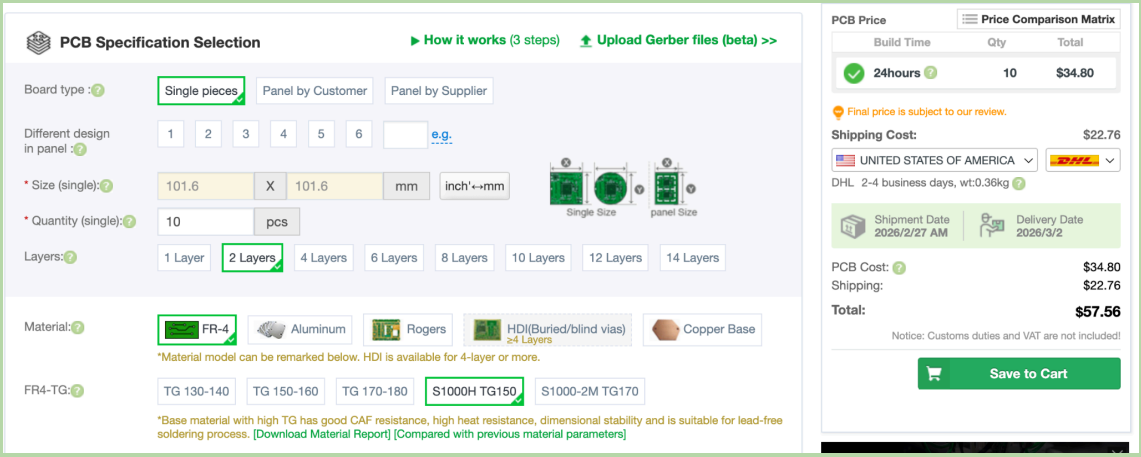
Determining Unit Price

The unit price was determined by identifying the cost for each subsystem in the product. This is assuming third-party components are utilized without bulk discounts. Actual cost will be less when using Honeywell-made components, and bulk discounts are applied. We are unable to obtain the actual component cost from Honeywell.

Components	Primary Source	Alternative Source
Battery * 4 batteries in series are needed to reach sufficient voltage to power the stepper motor.	 Panasonic BK-200AAB9B Unit cost: $4.29	 Panasonic HHR-210AAC4B Unit cost: $4.29
Contact Sensor	 ControlByWeb GRI-28C Unit Cost: $21.99	 Comet America SA200A Unit Cost: $19.20

Motor	 MIKROE 1530 Unit Cost: $9.60	 Seeed Technology 108990003 Unit Cost: $4.50
Speaker	 PUI Audio 668-1398-ND Unit cost: $3.52	 2056-W170-8OHM-ND Unit cost: $3.82
Wireless connectivity (Wifi + Bluetooth)	 Raspberry PI 2648-SC0020-ND Unit cost: $15.00	 SparkFun Electronics: 1568-11114-ND Unit cost: $11.25

Keypad	 Adafruit Industries 1528-1136-ND Unit cost: $3.95	 Knitter-Switch 3746-YFTKNI12B19-19 Unit cost: $14.44
Charging Port	 Adafruit Industries 1528-6071-ND Unit cost: $2.50	 Bulgin 708-2061-ND Unit cost: $23.16
Siren	 Floyd Bell 2339-M-80-ND Unit cost: $10.80	 Floyd Bell 2339-PM-80-ND Unit cost: $12.33

PCB	 Estimated Unit cost: $5.76 at 10 qty **Per PCB Way, assuming a standard 4x4-inch double-layered PCB.*
Enclosure * Will be manufactured in-house using strong metals.	 Unit Cost: $20-$30
Product Estimated Product Cost: $115.28	

Step 3: NPV Base Case Results

The net present value calculation revealed a unit cost of $115.28 and an NPV of $25,466,000. This means over 4 years, the product will generate an estimated baseline of 25 million dollars in net present value. As mentioned above, it is also assumed that the unit cost was determined using third-party components, assuming no bulk purchasing discount; it is strongly recommended to use first-party components where possible to further increase NVP. This showcases that the product is profitable under third-party and first-party component sourcing. Honeywell should pursue developing the product.

NPV Spreadsheet, input tab:

Scenario input parameters	
Sales & Product Volume (units/year)	Values imported from forecast
Development Cost (total $)	60,480.00
Unit price ($/unit)	299.00
Unit Productions Cost ($/unit)	115.28
Ramp-up cost (total S)	6,048.00
Marketing & suppost cost ($/year)	840,000.00
Annual Discount Factor (%)	7.00

Data imported from Winter's method

Year	Quarter	Forecast
1	1	3,107
1	2	7,938
1	3	6,160
1	4	11,873
2	1	6,518
2	2	12,978
2	3	8,761
2	4	15,586
3	1	8,261
3	2	16,216
3	3	10,767
3	4	18,944
4	1	10,119
4	2	19,669
4	3	12,945
4	4	22,591

NPV Spreadsheet, basecase tab:

BaseCase

	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection			
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Development Cost	-15.12	-15.12	-15.12	-15.12																
Ramp-up cost				-3.02	-3.02															
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26
Production volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit product cost					-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12
Sales revenue					928.98	2,373.48	1,841.95	3,550.00	1,948.96	3,880.32	2,619.63	4,660.26	2,469.99	4,848.48	3,219.41	5,664.35	3,025.62	5,881.10	3,870.41	6,754.63
Forecasted Sales Volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Period Cash Flow	-15	-15	-15	-18	358	1,248	922	1,971	988	2,174	1,400	2,653	1,308	2,769	1,768	3,270	1,649	3,404	2,168	3,940
Unit Price	-15	-15	-15	-17	334	1,145	831	1,746	860	1,860	1,177	2,192	1,062	2,210	1,387	2,521	1,249	2,534	1,587	2,834
Project NVP	25,466																			

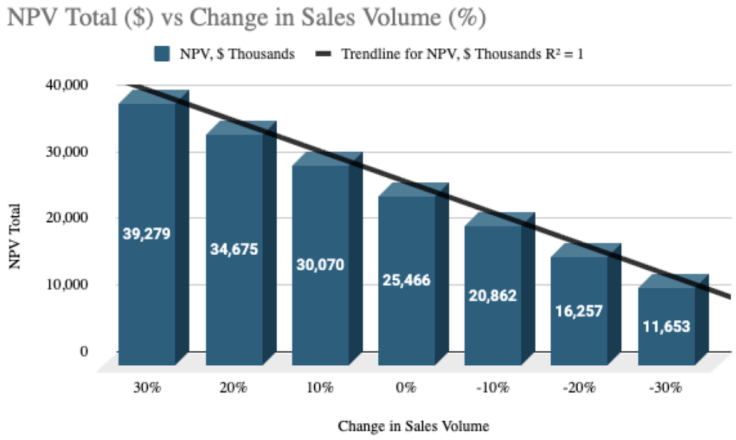
Step 4: NPV Sensitivity Analysis

This section presents a comprehensive financial model designed to understand the expected profitability of our product through changes in some variables. This is especially important considering the complexities of consumer demand, markets, and other unknown variables. The analysis will assist in identifying which variables in our financial model have a high impact on our product's profitability.

Sales Volume Sensitivity Results

Sales volume displays the revenue driver and the biggest source of uncertainty for any company's forecast. The analysis evaluates the exposure to demand-side risk, indicating the downside threshold at which the project becomes financially unviable.

Base Sales, Qty	Change in Sales Volume (%)	Sales Volume (Qty)	Change in Sales Volume (Qty)	Change in NPV, %	NPV, $ Thousands	Change in NPV, $ Thousands
25,466	30%	250,163	57,730	54.24%	39,279	13,813
25,466	20%	230,920	38,487	36.16%	34,675	9,209
25,466	10%	211,677	19,243	18.08%	30,070	4,604
25,466	0%	192,433	0	0.00%	25,466	0
25,466	-10%	173,190	-19,243	-18.08%	20,862	-4,604
25,466	-20%	153,947	-38,487	-36.16%	16,257	-9,209
25,466	-30%	134,703	-57,730	-54.24%	11,653	-13,813

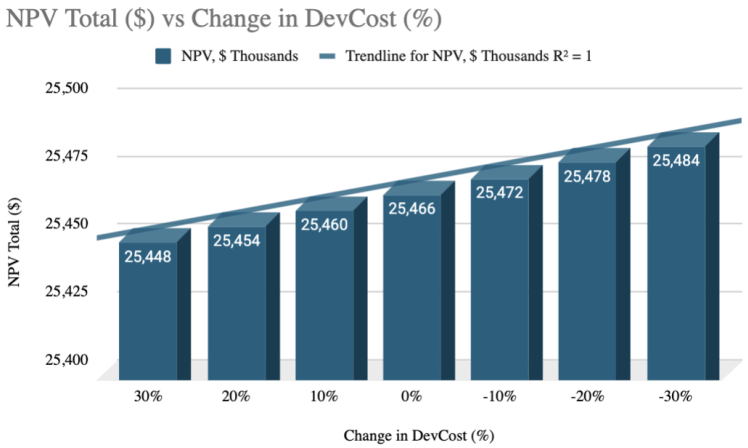


From the results, we can see that the NPV is highly elastic to sales volume fluctuations. Varying forecasted volume by ±10% to ±30% gave the results referenced in the screenshot above. A 30% volume shortfall drove NPV down to -$11,653 (in thousands), while 30% upside increased value to $39,279 (in thousands). The main risk factor here is demand realization; with this in mind, we should follow conservative forecasting.

Development Cost Sensitivity Results

Development cost shows the upfront investments required before it's possible to gain any new revenue. Our analysis evaluates the project's tolerance for R&D overruns or any factors that might come into play, causing a risk in the development. These indicators are then used to inform/guide decision thresholds.

Base Dev Cost, $	Change in Development Cost, %	Development Cost, $	Change in Development Cost, $	NPV, $ Thousands	Change in NPV, %	Change in NPV, $ Thousands
60,480	30%	78,624	18,144	25,448	-0.07%	-18
60,480	20%	72,576	12,096	25,454	-0.05%	-12
60,480	10%	66,528	6,048	25,460	-0.02%	-6
60,480	0%	60,480	0	25,466	0.00%	0
60,480	-10%	54,432	-6,048	25,472	0.02%	6
60,480	-20%	48,384	-12,096	25,478	0.05%	12
60,480	-30%	42,336	-18,144	25,484	0.07%	18

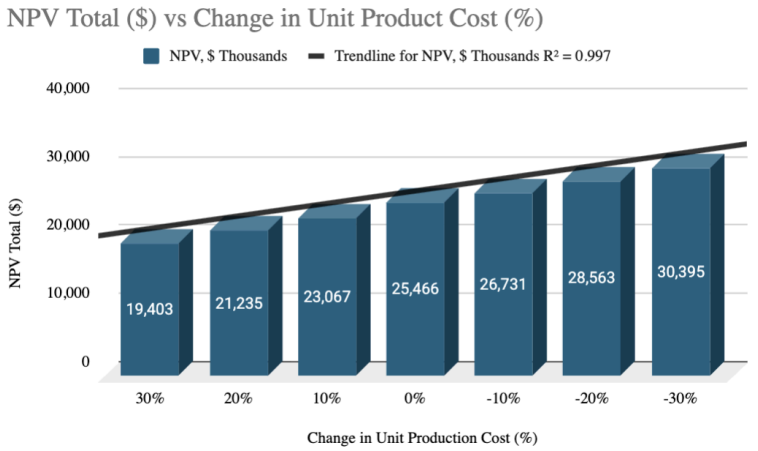


The results show extreme inelasticity to development cost variations. Adjusting the base dev cost of $60,480 (in thousands) by ±30% yields the results seen in the screenshot above. The NPV ranges from $25,448 (in thousands) to $25,484 (in thousands), showing that there's only a fluctuation of ±18, or less than 0.1% of the base NPV. This shows that the one-time development cost is financially insignificant relative to the huge recurring marketing expenditures ($840,000). From a risk factor viewpoint, development budget overruns aren't necessarily a huge concern; focusing should instead be on operating expenses to control the project's cost structure.

Unit Cost Sensitivity Results

Within this portion, it directly determines the gross margin and affects every unit sold across the entirety of the product's life cycle. The analysis quantifies exposure to supply chain volatility, raw material costs, and manufacturing efficiency variances to help strategize procurement methods.

Base Unit Cost, $	Change in Unit Production Cost, %	New Unit Cost	Unit Volume	Total Cost Product Cost	Change in Total Product Cost, $ Thousands	NPV, $ Thousands	Change in NPV, %	Change in NPV, $ Thousands
115	30%	150	192,433	-29,847	-7,663	19,403	-23.81%	-6,063
115	20%	138	192,433	-27,551	-5,367	21,235	-16.62%	-4,231
115	10%	127	192,433	-25,255	-3,071	23,067	-9.42%	-2,399
115	0%	115	192,433	-22,184	0	25,466	0.00%	0
115	-10%	104	192,433	-20,663	1,521	26,731	4.97%	1,265
115	-20%	92	192,433	-18,367	3,817	28,563	12.16%	3,097
115	-30%	81	192,433	-16,071	6,112	30,395	19.35%	4,929



The model shows the sensitivity towards unit product cost variations, as seen in the screenshot above. A 30% increase in unit product cost reduces the NPV to $19,403 (in thousands), a reduction of approximately $6,900 (in thousands) from the base case. Conversely, a 30% cost decrease improves the overall NPV to $30,395 (in thousands). The change is symmetrical, which shows that the unit product cost has a direct impact on overall project value, with every percent change shifting NPV by roughly $2,300 (in thousands).

Step 5: NPV Sensitivity Results

Across all sensitivity analyses tested, the project consistently maintained a positive NPV. This confirms the financial availability regardless of reasonable variations in key input parameters.

The Base Case NPV of $25,466,000 established a strong foundation. Within negative percentage scenarios, the project showed considerable resilience; even if there were significant margin compressions, it still generated well enough to be viable. Development cost had virtually no impact on viability, with NPV remaining stable across all percentage changes.

Conversely, positive scenarios revealed substantial value potential. Favorable movements in sales volume or unit cost drove NPV as high as $39,279,000, indicating that operational improvements would provide significant returns.

The consistent performance across all scenarios confirms that the project's overall economics are structurally sound, and this project can withstand demand shortfalls, inflation, and R&D overruns while remaining viable.

Scenario	Net Present Value
Base Case	$25,466,000
Sales Volume Cost	
+30% Sales Volume Cost	$39,279,000
+20% Sales Volume Cost	$34,675,000
+10% Sales Volume Cost	$30,070,000
-10% Sales Volume Cost	$20,862,000
-20% Sales Volume Cost	$16,257,000
-30% Sales Volume Cost	$11,653,000
Development Cost	
+30% Development Cost	$25,448,000
+20% Development Cost	$25,454,000
+10% Development Cost	$25,460,000
-10% Development Cost	$25,472,000
-20% Development Cost	$25,478,000
-30% Development Cost	$25,484,000
Unit Cost	

+30% Unit Cost	$19,403,000
+20% Unit Cost	$21,235,000
+10% Unit Cost	$23,067,000
-10% Unit Cost	$26,731,000
-20% Unit Cost	$28,563,000
-30% Unit Cost	$30,395,000

6. Phase 3 Conclusion

1. **Demand Estimation and Forecasting**
    a. Methodological Validation: The team successfully validated a top-down portfolio-based forecast by cross-checking it against an independent market-share build-up method. The close alignment between these two distinct approaches (estimating 42,917 vs 51,800 units by Year 3) confirmed that the base-case projections are a reliable foundation for operational planning.
    b. Optimal Model Selection: After testing five different forecasting models, Winter's Method was identified as the most effective for our product. It achieved the lowest MAPE of approximately 0.10, significantly outperforming simpler models like Static (0.33) and Simple Exponential Smoothing (0.35).
    c. Growth and Seasonality: The analysis confirmed that demand is highly seasonal, peaking in Q4 (35%) due to holiday sales and hitting a low in Q1 (15%). Total annual demand is projected to grow by 122% over four years, reaching 64,254 units by 2029.
2. **Inventory Management Strategies**
    a. Component-Specific Logic: The team established distinct inventory strategies based on component value and lead times:
        i. Low-cost Items (Speakers): Leveraged a bulk-ordering strategy with large lot sizes (9,500+) and longer flow times (40 days) to minimize fixed transportation costs.
        ii. High-value Items (Finished product): Implemented a lean approach for the $115.28 units, maintaining a rapid 11-17 day replenishment cycle to minimize capital tied up in holding costs.
    b. Operational Efficiency: The calculated safety stocks and re-order points resulted in high operational efficiency across the board, with average fill rates exceeding 94% for all analyzed subsystems.
    c. Bullwhip Mitigation: To address demand volatility and supply chain lag, the report concludes that Honeywell must proactively update its optimal lot sizes every six months.
3. **Financial Viability and Risk**
    a. Project Profitability: The financial model yielded a base-case NPV of $25,466,000, concluding that the Emergency Ready Lock project is highly feasible and financially sound.
    b. Competitive Positioning: By setting a unit price of $299, the team determined the product could significantly undercut major competitors (ranging from $430-$460) while still maintaining strong profit margins.
    c. Resilience to Volatility: Sensitivity analysis proved the project's robustness. Even with a 30% decrease in sales volume of a 30% increase in unit production costs, the project remains profitable. Notably, the NPV is highly insensitive to R&D overruns, with a 30% change in development costs shifting the final NPV by less than 0.1%.

Conclusion and Next Steps

The group's next steps focus on transitioning the conceptual design from TIM172P and the supply chain management strategy from TIM172Q into a market-ready product which requires actions related to prototyping (product platform and product lines), testing, and bringing to market.

Next Steps (Future Work)

- Inventory Component Optimization and Lead-Time Reduction
    - Establish an improved inventory optimization that is dynamic.
    - Focus on identifying which components have long lead times, creating a bottle neck in the supply chain.
- Prototyping: Product Platform and Product Lines
    - Creating a modular internal core that can be reused across different external product components and functionalities. Core would include the MEMS sensor module, redundant battery system, Z-wave/Wifi communication chip, and stepper motor logic.
    - Some possible product lines are: A base "Emergency Ready Lock" for common residential use. A higher-end version with enhanced features like facial recognition, biometric fingerprint access, etc. A cost-optimized lock that retains the core security and remote management capabilities suited for property managers.
- Testing and Validation:
    - Validate selected component work together and
    - Test the lock's compatibility with a wide range of existing smart home protocols to establish technical specifications and ensure the product is securing entry points.
- Refinement of Sales and Operations Planning Framework
    - Develop an Aggregate Planning strategy to manage the 35% seasonal demand spike identified in the forecasting model
    - This involves evaluating the various production strategies to determine the most cost-effective way to meet peak Q4 demand.

Appendix

Phase 1 Appendix

Demand Estimation and Forecasting Appendix

Spreadsheet:
https://docs.google.com/spreadsheets/d/1aeWA1VaXCp43i3FvtXSGujoB13OVoe5rzFyOn7FNsao/edit?gid=0#gid=0

Ring "alarm window and door contact sensor."

- https://ring.com/products/alarm-window-door-contact-sensor-v2?srsltid=AfmBOoqv_dAWrd7XUhbzhopTKVywUuqepn71Q7mMEvkS_oBwUOTZIijO
- Unit price: $14.99 - $19.99
- Requires Ring Alarm or Ring Alarm Pro
- Annual revenue from its online store in 2024: $420.9M
    - For ring.com, not the product
    - https://gripsintelligence.com/insights/retailers/ring.com

- Forecasting
    - Step 1: related products
    - Step 2: Find sales revenue for the products (find credible data, does not have to be recent but credible)
    - Step 3: consider the life cycle of the product is early
    - etc
    - Find similar companies & their supply chains
    - Use a stepwise process
    - Include the formulas we used in the processes
    - How do you estimate demand for a product that does not exist?
    - Static forecasting is not set up
    - Years 1 or 2?? When was this product designed? Designed in Yr 0 (2025), Yr 1 (2026), etc., forecasting into 2028

Key Takeaways on Security System Adoption:

- High Adoption Rates: Approximately 52% of U.S. households had a security system in 2024, up from 42% in 2023.
- Market Growth: The industry is booming, with roughly 13 million new systems expected in 2024.
- Popularity of Devices: Around 72% of U.S. households have at least one security device, with 68 million homes using cameras.
- Common Choices: Top companies include Ring, ADT, and SimpliSafe, with 36% of users choosing Ring.
- Average Cost: Homeowners can expect to pay around $475 for installation and $40 per month for monitoring.
- With over 13 million systems projected to be installed in 2024, this equates to roughly over 1 million systems sold per month in the United States alone.

- https://www.consumeraffairs.com/homeowners/home-security-industry-statistics.html\
- https://www.securityinformed.com/insights/is-there-a-seasonal-impact-on-security-system-sales.html

→ Spike in demand/sales during holiday seasons (black friday, Amazon Prime Day, Cyber Monday) around quarter 4 (Oct - Dec)

Approx 1+ million systems sold per month in the US

→ Spring and summer seasons are "moving seasons", high volume of new home purchases and renovations. High-quality security systems would be a top priority for new homeowners ⇒ around quarter 2 and 3

Related products:

- SimpliSafe smart lock
    - SimpliSafe's estimated annual revenue is currently $402.2M per year.(i)
    - SimpliSafe received $57.0M in venture funding in May 2014.
    - SimpliSafe's estimated revenue per employee is $350,350
    - SimpliSafe's total funding is $187M.
    - SimpliSafe's current valuation is $1B. (June 2018)
    - https://growjo.com/company/SimpliSafe
- Ring Alarm Pro
    - https://www.vox.com/recode/2020/1/21/21070402/amazon-ring-sales-jumpshot-data#:~:text=The%20security%20company%20Ring's%20online,has%20%E2%80%9Cmillions%E2%80%9D%20of%20customers.
    - "Ring Annual Revenue & Growth by Year: Ring's annual sales amounted to US$1,306m in 2024. This represents a change of 10-15% from the previous year. Ring is expected to see a growth rate of 5-10% in 2025 compared to 2024, indicating a projected shift in performance."
        - https://ecdb.com/resources/sample-data/retailer/ring
    - Annual sales increased/expected to increase by ~10% every year
    - Ring's estimated annual revenue is currently $635.2M per year.(i)
    - Ring's estimated revenue per employee is $350,350
    - Ring's total funding is $203.7M.
    - https://growjo.com/company/Ring
    - https://www.security.org/doorbell-camera/ring/prime-day/#:~:text=Final%20Words,Amazon%20Links%20to%20Amazon.com
- Ecobee smart security system
    - Ecobee's estimated annual revenue is currently $197.9M per year.(i)
    - Ecobee received $36.0M in venture funding in May 2018.
    - Ecobee's estimated revenue per employee is $311,150
    - Ecobee's total funding is $149.4M.
    - https://growjo.com/company/ecobee

https://investor.honeywell.com/static-files/d076962b-5bac-4811-adcc-5aefa2a972f7

Phase 2 Appendix

Forecasting:
Reasoning behind seasonality percentages:
(Q1 & Q3)
https://www.securityinformed.com/insights/is-there-a-seasonal-impact-on-security-system-sales.html

Q2: Typically Summer Vacation for families.
https://www.vivint.com/resources/article/summer-home-safety

Q4: Lowest price on products annually as Christmas, Black Friday, Cyber Monday deals take place.
https://blackfriday.com/categories/home-security
https://www.cnet.com/deals/the-best-home-security-deals-for-black-friday-and-cyber-monday/

(Other companies' success in sales during Q4)
https://news.adobe.com/news/2025/12/adobe-cyber-monday-hits-record#:~:text=What%20consumers%20are%20buying:%20Online%20sales%20of,(up%201%2C700%25)%2C%20home%20security%20systems%20(up%201%2C500%25)%2C

NPV:
Basecase followed “Excel Tutorial” by R.Polany and S.Desa

Discount factor source (see last page):
https://extapps.dec.ny.gov/data/DecDocs/915055/Report.HW.915055.2024-02-02.Site%20109%20Feasibility%20Study%20_Final.pdf

R&D engineer average wage source:
https://www.ziprecruiter.com/Salaries/R-D-Engineer-Salary

Phase 3 Appendix

Forecasting Appendix

Static and Adaptive Forecasting Process and Results

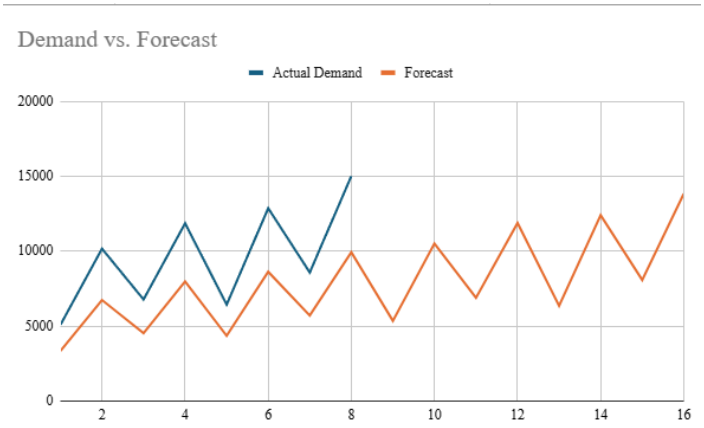
Forecasting

Year	Quarter	Period, t	Demand, D_t	Deseasonalized Demand,	Regressed deseasonalized demand, D_t^{-}	Seasonal Factor, S_i	Average Seasonal factor, S_i^{-}	Forecasts, F_{t+l}
1	1	1	5,086		7490.3	0.68	0.4466	3345.07
1	2	2	10,171		8053.6	1.26	0.8374	6743.80
1	3	3	6,781	8645.00	8616.9	0.79	0.5255	4528.30
1	4	4	11,866	9152.00	9180.1	1.29	0.8688	7975.61
2	1	5	6,438	9715.25	9743.4	0.66	0.4466	4351.31
2	2	6	12,875	10334.88	10306.7	1.25	0.8374	8630.52
2	3	7	8,583		10870.0	0.79	0.5255	5712.37
2	4	8	15,021		11433.3	1.31	0.8688	9933.12
3	1	9			11996.6		0.4466	5357.54
3	2	10			12559.9		0.8374	10517.23
3	3	11			13123.2		0.5255	6896.44
3	4	12			13686.4		0.8688	11890.63
4	1	13			14249.7		0.4466	6363.77
4	2	14			14813.0		0.8374	12403.95
4	3	15			15376.3		0.5255	8080.51
4	4	16			15939.6		0.8688	13848.15

Static Forecasting: Error Calculations

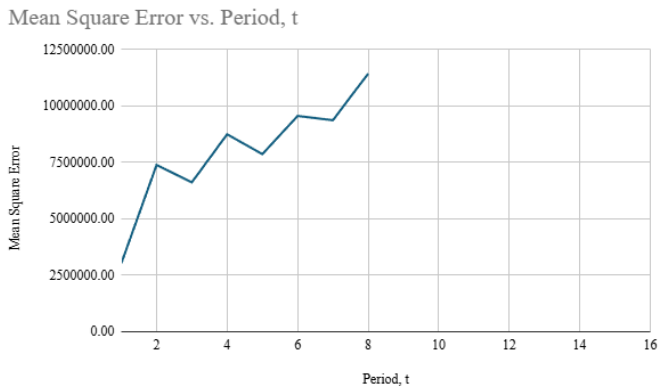
Forecast Error	Mean Square Error	Absolute Error or Deviation	Mean Absolute Deviation	Percent Error	Absolute Percent Error	Mean Absolute Percent Error	Bias	Tracking Signal
-1740.93	3030821.30	1740.93	1740.93	-34.23	34.23	34.23	-1740.93	-1.00
-3427.20	7388264.26	3427.20	2584.06	-33.70	33.70	33.96	-5168.13	-2.00
-2252.70	6617060.53	2252.70	2473.61	-33.22	33.22	33.72	-7420.83	-3.00
-3890.39	8746584.17	3890.39	2827.80	-32.79	32.79	33.48	-11311.22	-4.00
-2086.69	7868125.17	2086.69	2679.58	-32.41	32.41	33.27	-13397.91	-5.00
-4244.48	9559378.97	4244.48	2940.40	-32.97	32.97	33.22	-17642.40	-6.00
-2870.63	9370970.46	2870.63	2930.43	-33.45	33.45	33.25	-20513.03	-7.00
-5087.88	11435413.93	5087.88	3200.11	-33.87	33.87	33.33	-25600.91	-8.00

Demand vs. Forecasting



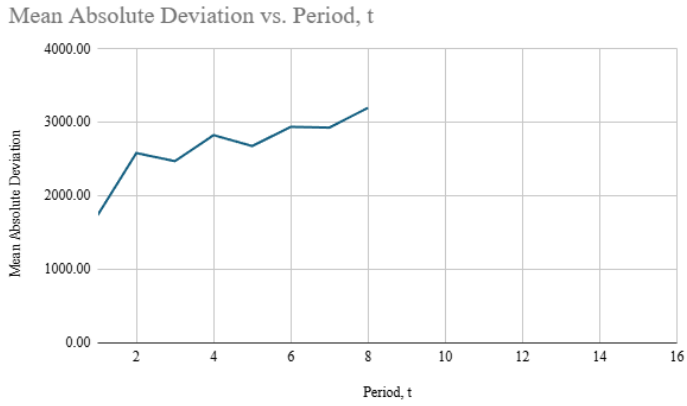
This graph compares actual demand with the forecasted demand across multiple time periods. The actual demand fluctuates significantly from period to period, showing noticeable peaks and drops, while the forecast follows a smoother and more gradual upward trend. Although the forecast captures the general growth pattern over time, it consistently underestimates higher demand spikes and does not fully reflect short term variability. Overall, the graph illustrates that the forecasting model tracks the overall trend but struggles to respond to rapid changes in demand.

MSE vs. Period



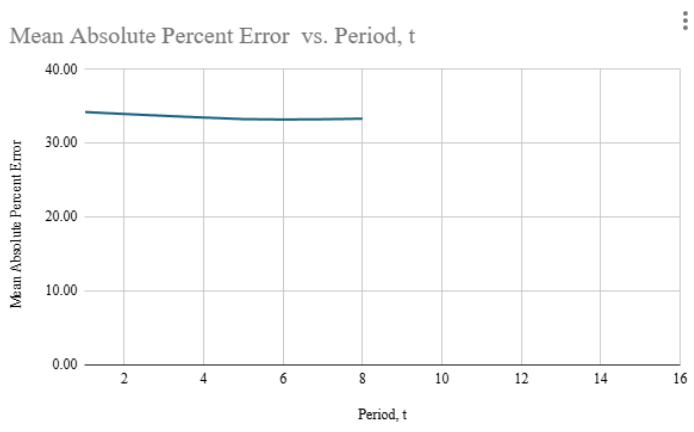
This graph shows how the Mean Squared Error changes across different time periods. The error increases overall as time progresses, indicating larger squared differences between actual demand and the forecast. There are small drops between some periods, but the general trend continues upward. This suggests that forecasting accuracy becomes slightly less consistent over time. Overall, the graph highlights increasing variation between predicted and actual values across later periods.

MAD vs. Period



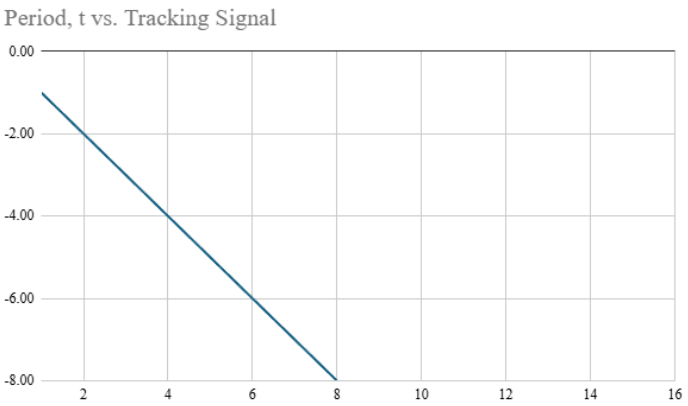
This graph shows how the Mean Absolute Deviation changes across different time periods. The deviation generally increases over time, indicating that the difference between forecasted and actual demand becomes slightly larger. Although there are small decreases between some periods, the overall trend moves upward. This suggests that forecasting errors gradually grow as time progresses. Overall, the graph indicates moderate variation in accuracy while remaining relatively stable without sudden spikes.

MAD vs. Period



This graph shows how the Mean Absolute Percent Error changes across different time periods. The error remains fairly stable, staying around the low 30 percent range throughout the observed periods. There is a slight downward trend at first, followed by minimal variation afterward. This indicates that the forecasting accuracy remains consistent over time without major improvements or declines. Overall, the graph suggests the forecasting model maintains steady performance across the periods analyzed.

Period vs Tracking Signal



This chart tracks whether the forecast is systematically biased over time. The line trends downward into negative territory, which indicates the forecast is generally underestimating demand. A strong negative tracking signal suggests repeated underforecasting. This aligns with the demand vs. forecast graph where actual demand is often higher than predicted. Overall, this chart shows the model has a consistent bias that should be corrected.

Adaptive Forecasting (Moving Average)

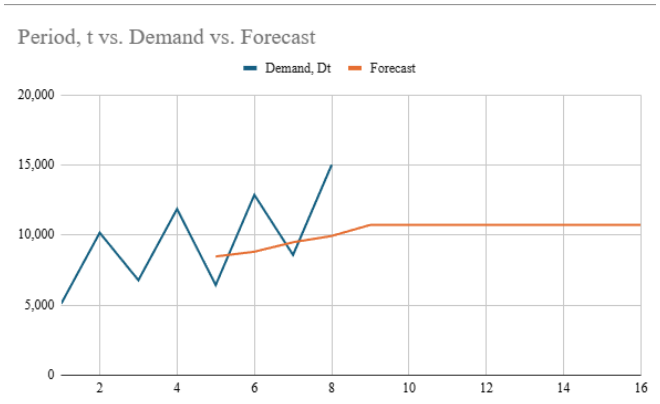
Forecast

Year	Quarter	Period, t	Demand, Dt	Level	Forecast
1	1	1	5,086		
1	2	2	10,171		
1	3	3	6,781		
1	4	4	11,866	8,476	
2	1	5	6,438	8,814	8,476
2	2	6	12,875	9,490	8,814
2	3	7	8,583	9,941	9,490
2	4	8	15,021	10,729	9,941
3	1	9			10,729
3	2	10			10,729
3	3	11			10,729
3	4	12			10,729
4	1	13			10,729
4	2	14			10,729
4	3	15			10,729
4	4	16			10,729

Moving Average: Errors

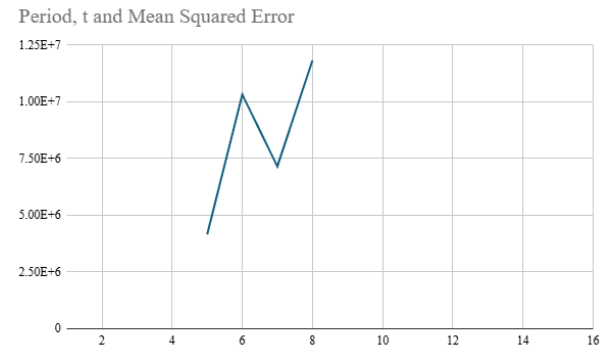
Forecast Error	Absolute Error	Squared Error	Mean Squared Error	Mean Absolute Deviation	Percent Error	Absolute Percent	Mean Absolute Percent Error	Bias	Tracking Signal
2,038	2038	4153444	4153444	2038	31.66%	31.66%	31.66%	2,038	1
-4,061	4061	16491721	10322582.5	3049.5	-31.54%	31.54%	31.60%	-2,023	-0.6633874406
907	907	822649	7155938	2335.333333	10.57%	10.57%	24.59%	-1,116	-0.4778761062
-5,081	5080.5	25811480.25	11819823.56	3021.625	-33.82%	33.82%	26.90%	-6,197	-2.050717743

Demand vs. Forecasting



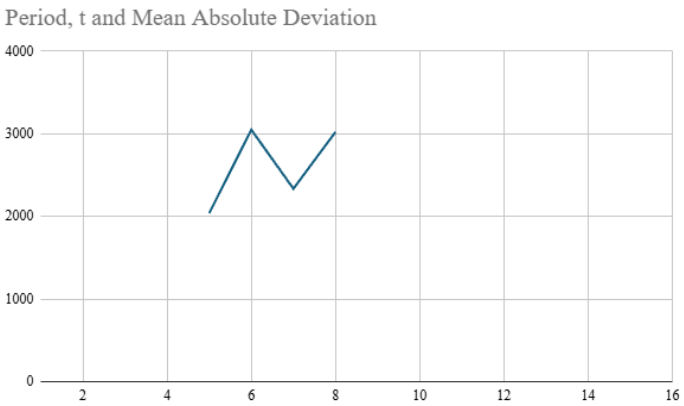
This graph compares actual demand with forecasted demand across different time periods. The demand values fluctuate significantly, showing several increases and decreases over time. In contrast, the forecast follows a smoother upward trend and eventually levels off in the later periods. This indicates that the forecasting model captures the general growth trend but smooths out short term changes in demand. Overall, the graph shows that the forecast provides a stable estimate even when actual demand varies from period to period.

Period and MSE



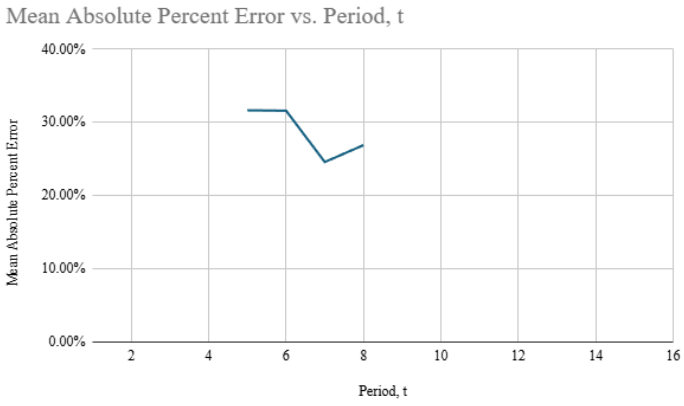
This graph shows how the Mean Squared Error changes across different time periods. The error increases sharply at first, indicating larger forecasting differences during those periods. It then decreases slightly before rising again toward the final period shown. These movements suggest that forecasting accuracy varies as demand patterns change over time. Overall, the graph indicates some instability in prediction error but remains within a similar overall range.

Period and MAD



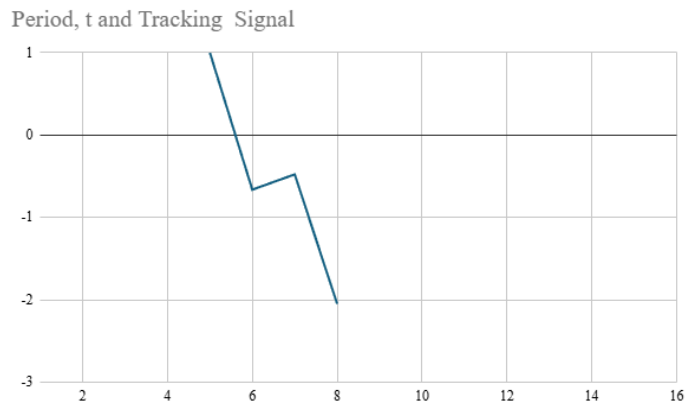
This graph shows how the Mean Absolute Deviation changes across several time periods. The deviation increases at first, indicating larger differences between the forecasted and actual demand. It then drops slightly before rising again toward the later periods. These fluctuations suggest that forecasting accuracy varies as new data is incorporated. Overall, the graph shows moderate variation in forecasting error but no extreme instability over time.

MAPE vs. Period



This graph shows how the Mean Absolute Percent Error changes over several time periods. The error remains around the low 30 percent range at first, indicating a moderate difference between the forecast and actual demand. Around the later periods, the error decreases noticeably, suggesting the forecast becomes more accurate. There is a small increase afterward, but the values still remain lower than earlier periods. Overall, the graph shows an improvement in forecasting accuracy as more data is incorporated over time.

Period vs. Tracking Signal



This graph shows how the tracking signal changes across different time periods. The tracking signal starts positive but quickly moves downward into negative values as time progresses. This indicates that the forecast begins to consistently overestimate or underestimate demand during later periods. The steady downward trend suggests a growing bias in the forecasting model. Overall, the graph helps show when the forecast may need adjustment to better align with actual demand.

Adaptive Forecasting (SES)

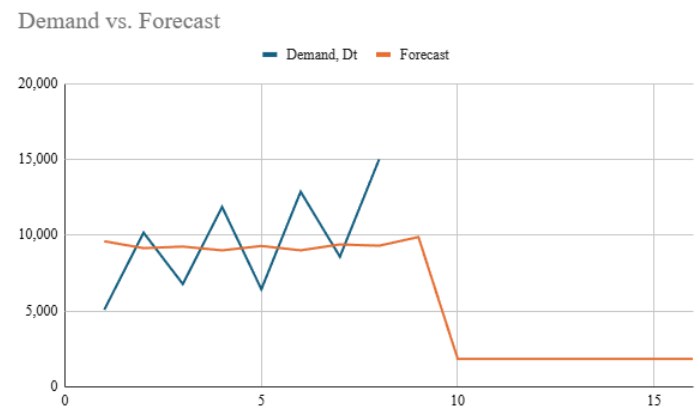
SES Forecasting

Year	Quarter	Period, t	Demand, Dt	Level	Forecast
		0		9,603	
1	1	1	5,086	9,151	9,603
1	2	2	10,171	9,253	9,151
1	3	3	6,781	9,006	9,253
1	4	4	11,866	9,292	9,006
2	1	5	6,438	9,006	9,292
2	2	6	12,875	9,393	9,006
2	3	7	8,583	9,312	9,393
2	4	8	15,021	9,883	9,312
3	1	9			9,883
3	2	10			1844
3	3	11			1844
3	4	12			1844
4	1	13			1844
4	2	14			1844
4	3	15			1844
4	4	16			1844

SES: Error Analysis

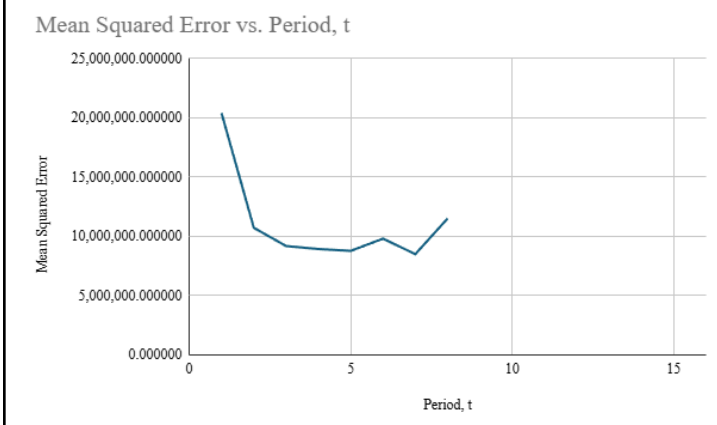
Forecast Error	Absolute Error	Squared Error	Mean Squared Error	Mean Absolute Deviation	Percent Error	Absolute Percent	Mean Absolute Percent Error	Bias	Tracking Signal
4,517	4516.625	20,399,901.390625	20,399,901.390625	4516.625	88.81%	88.81%	88.81%	4,517	1
-1,020	1020.0375	1,040,476.501406	10,720,188.946016	2768.33125	-10.03%	10.03%	49.42%	3,497	1.26306687
2,472	2471.96625	6,110,617.141139	9,183,665.011057	2669.542917	36.45%	36.45%	45.10%	5,969	2.235796141
-2,860	2860.230375	8,180,917.798073	8,932,978.207811	2717.214781	-24.10%	24.10%	39.85%	3,108	1.143937313
2,854	2853.792663	8144132.561	8,775,209.078356	2744.530358	44.33%	44.33%	40.74%	5,962	2.17236294
-3,869	3868.586604	14965962.31	9,807,001.283749	2931.873065	-30.05%	30.05%	38.96%	2,094	0.714058687
810	810.2720566	656540.8057	8,499,792.644035	2628.787207	9.44%	9.44%	34.74%	2,904	1.104616411
-5,709	5708.755149	32589885.35	11,511,054.232488	3013.7832	-38.01%	38.01%	35.15%	-2,805	-0.930708505

Demand vs. Forecast



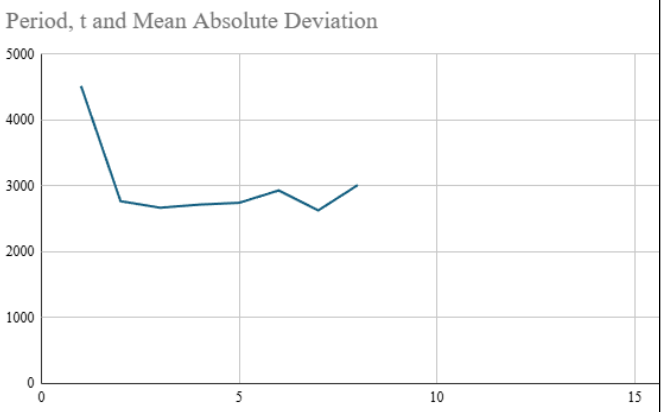
This graph compares actual demand with the forecasted values across multiple time periods. The demand line shows noticeable fluctuations, with several increases and decreases over time. The forecast remains relatively stable during the earlier periods, indicating the model is smoothing the demand pattern rather than reacting to every change. Toward the later periods, the forecast drops sharply and then stays constant, showing a major adjustment in the prediction. Overall, the graph highlights differences between real demand variability and how the forecasting model responds over time.

MSE vs. Period



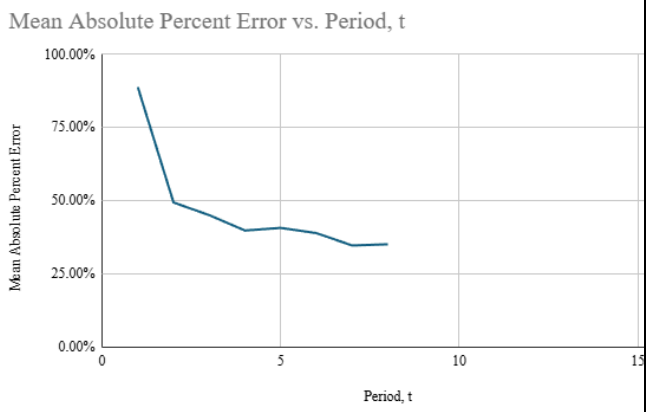
This graph shows how the Mean Squared Error changes over different time periods. The error starts very high in the first period but drops quickly as the forecasting model begins adjusting to the demand data. After the initial decline, the error remains relatively stable with only small fluctuations between periods. This indicates that the forecast becomes more accurate once the model incorporates more historical information. Overall, the graph suggests improved forecasting performance after the early adjustment phase.

MAD vs. Period



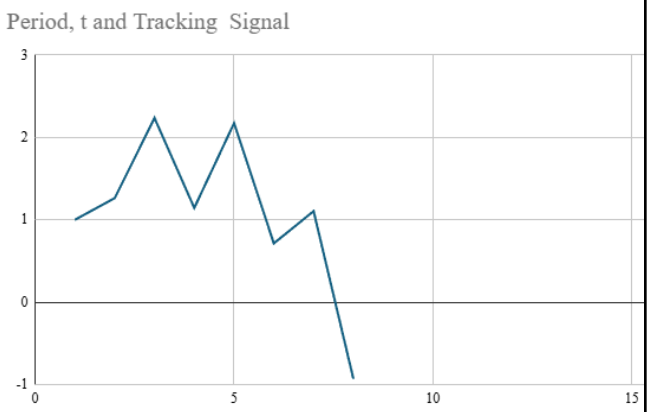
This graph shows how the Mean Absolute Deviation changes across different time periods. The error starts relatively high in the first period and then drops quickly as the forecast begins adjusting to the demand pattern. After the initial decrease, the deviation remains fairly stable with only small fluctuations between periods. This indicates that the forecasting model improves after the early periods and maintains more consistent accuracy. Overall, the graph suggests the forecast becomes more reliable once enough data is incorporated over time.

MAPE vs. Period



This chart shows percentage error dropping significantly over time. Early forecasts contain large proportional errors. As periods continue, the percent error stabilizes at a much lower level. This indicates the forecasting method adapts effectively. Overall, proportional forecasting accuracy improves substantially.

Tracking Signal vs. Period



This graph shows how the tracking signal changes across different time periods. The tracking signal begins with positive values, indicating the forecast is generally close to actual demand during the early periods. It fluctuates slightly but stays above zero for most of the timeline, suggesting limited forecasting bias at first. Toward the later periods, the tracking signal drops below zero, showing the forecast begins to consistently deviate from actual demand. Overall, the graph suggests the forecasting model performs reasonably well initially but may require adjustment as conditions change over time.

Adaptive Forecasting (Holt's Method)

Forecasting

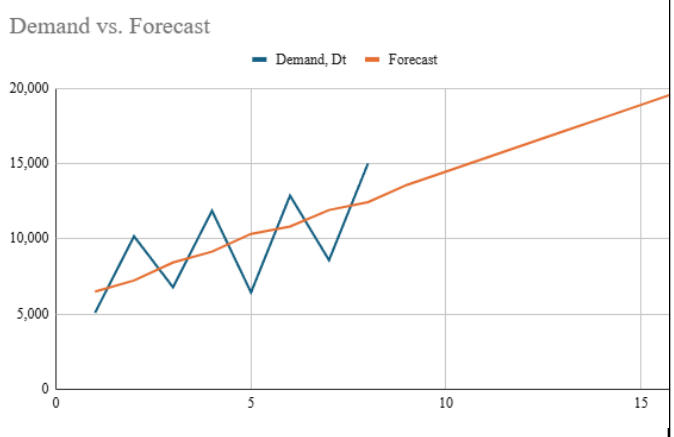
Year	Quarter	Period, t	Demand, Dt	Level	Trend	Forecast
		0		5,614	886	
1	1	1	5,086	6,359	872	6,500
1	2	2	10,171	7,525	902	7,231
1	3	3	6,781	8,262	885	8,427
1	4	4	11,866	9,419	912	9,147
2	1	5	6,438	9,942	873	10,332
2	2	6	12,875	11,022	894	10,816
2	3	7	8,583	11,582	861	11,916
2	4	8	15,021	12,701	887	12,443
3	1	9				13,587
3	2	10				14,474
3	3	11				15,360
3	4	12				16,247
4	1	13				17,134
4	2	14				18,020
4	3	15				18,907
4	4	16				19,793

a.

Holt's Method: Error Analysis

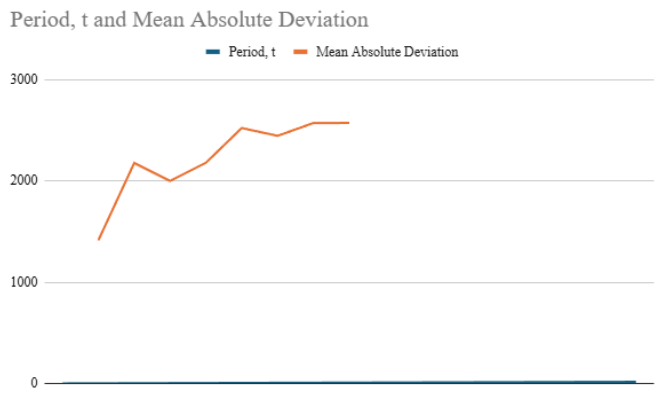
Forecast Error	Absolute Error	Squared Error	Mean Squared Error	Mean Absolute Deviation	Percent Error	Absolute Percent	Mean Absolute Percent Error	Bias	Tracking Signal
1,414	1414.166667	1,999,867.361111	1,999,867.361111	1414.166667	27.81%	27.81%	27.81%	1,414	1
-2,940	2939.975	8,643,453.000625	5,321,660.180868	2177.070833	-28.91%	28.91%	28.36%	-1,526	-0.7008537848
1,646	1645.69725	2,708,319.438658	4,450,546.600131	1999.946306	24.27%	24.27%	26.99%	120	0.05994606772
-2,719	2718.654698	7,391,083.364239	5,185,680.791158	2179.623404	-22.91%	22.91%	25.97%	-2,599	-1.192300366
3,894	3893.615097	15160238.52	7,180,592.337215	2522.421742	60.48%	60.48%	32.87%	1,295	0.5133357734
-2,059	2059.278239	4240626.867	6,690,598.092251	2445.231158	-15.99%	15.99%	30.06%	-764	-0.3126203103
3,333	3332.71054	11106959.55	7,321,506.871378	2572.013927	38.83%	38.83%	31.31%	2,568	0.9985488763
-2,578	2577.826663	6645190.305	7,236,967.300602	2572.740519	-17.16%	17.16%	29.54%	-10	-0.003710069557

Demand vs. Forecast



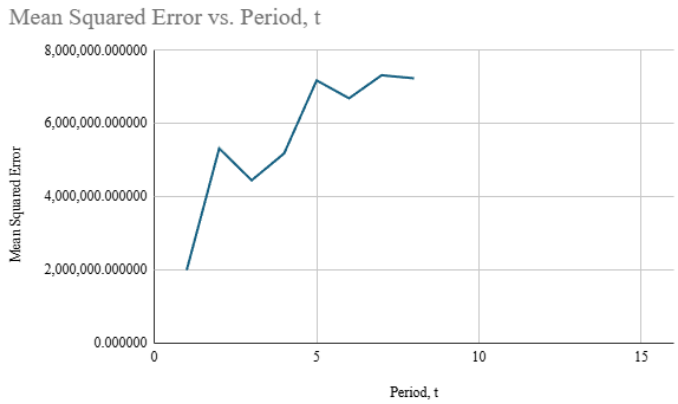
This graph compares actual demand with the forecasted values across different time periods. The demand line shows noticeable ups and downs, indicating fluctuations in sales over time. The forecast line increases more smoothly, showing a steady upward trend as the model predicts gradual growth. While the forecast generally follows the overall direction of demand, it does not capture every sudden change. Overall, the graph shows that the forecasting method reflects the long term trend but smooths out short term variability.

Period vs. MAD



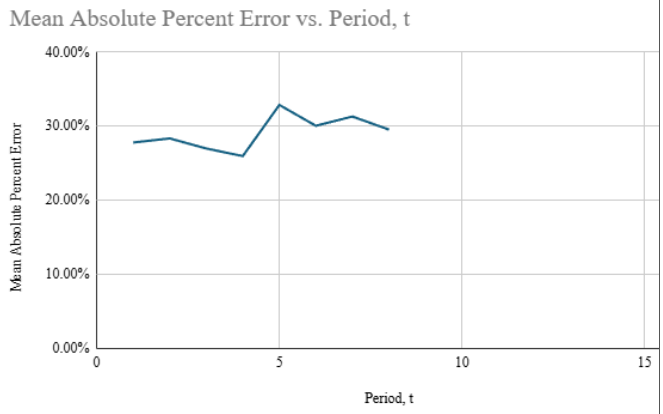
This graph shows how the Mean Absolute Deviation changes across different time periods. The deviation increases during the early periods, which means the forecast errors became larger at first. After reaching a higher level around the middle periods, the values begin to stabilize with only small changes. This suggests the forecasting model starts adjusting and producing more consistent prediction errors over time. Overall, the graph indicates that while errors grew initially, they eventually leveled out and remained relatively steady.

MSE vs. Period



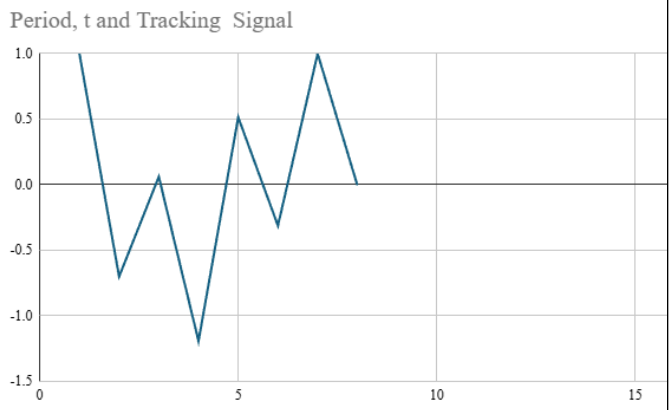
This graph displays how the Mean Squared Error changes as time progresses. The error increases during the early periods, showing that the difference between actual demand and forecast values became larger at first. After reaching a peak around the middle periods, the error begins to level off slightly. This suggests the forecasting method starts adjusting and producing more stable results over time. Overall, the trend shows initial instability followed by more consistent forecasting performance.

MAPE vs. Period



This graph shows how the Mean Absolute Percent Error changes across different time periods. The error percentage stays relatively consistent, mostly ranging between about 26 percent and 33 percent over time. There is a slight increase around the middle periods, which suggests the forecast became a bit less accurate during that time. After that point, the error stabilizes again and does not fluctuate heavily. Overall, the graph indicates that the forecasting model maintains fairly steady performance across periods, even though some variation exists.

Tracking Signal vs. Period



This chart shows the tracking signal moving downward into negative values. This confirms the forecast repeatedly underpredicts demand. The consistent direction indicates bias rather than random fluctuation. The model does not fully catch up to actual demand levels. Overall, this forecast shows persistent underestimation.

Adaptive Forecasting (Winter's Method)

Winter's Method: Forecasting

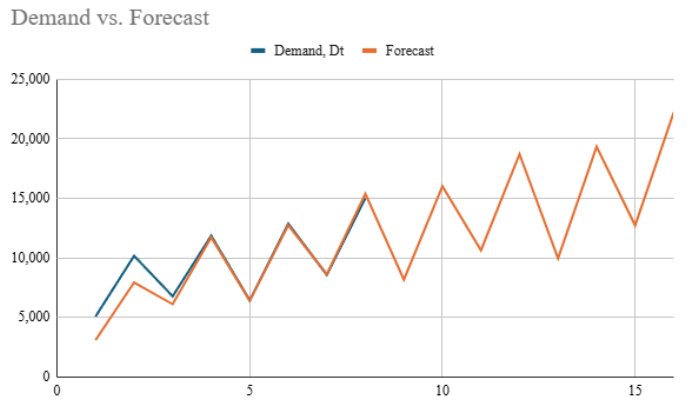
a.

Year	Quarter	Period, t	Demand, Dt	Level	Trend	Seasonal Factor	Forecast
		0		6926.9875	563.2875		
1	1	1	5,086	8,922	850	0.414800	3,107
1	2	2	10,171	10,596	932	0.812400	7,938
1	3	3	6,781	11,903	970	0.530600	6,117
1	4	4	11,866	12,922	975	0.910100	11,716
2	1	5	6,438	13,914	976	0.461383	6,412
2	2	6	12,875	14,932	980	0.856655	12,756
2	3	7	8,583	15,887	978	0.542321	8,630
2	4	8	15,021	16,743	966	0.912543	15,390
3	1	9				0.461778	8,178
3	2	10				0.858331	16,029
3	3	11				0.541705	10,639
3	4	12				0.907921	18,709
4	1	13					9,961
4	2	14					19,345
4	3	15					12,732
4	4	16					22,216

Winter's Method: Error Analysis

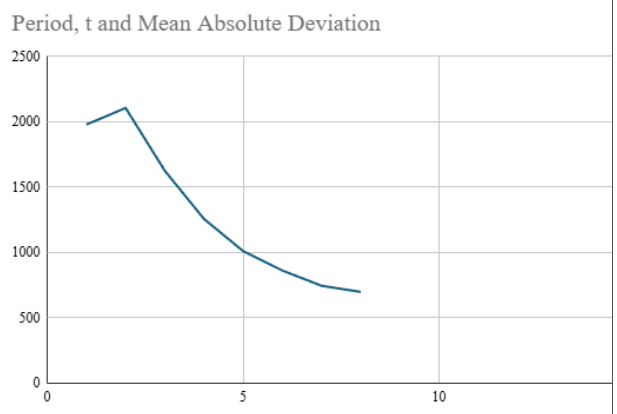
Forecast Error	Absolute Error	Squared Error	Mean Squared Error	Mean Absolute Deviation	Percent Error	Absolute Percent	Mean Absolute Percent Error	Bias	Tracking Signal
-1,979	1979.03393	3,916,575.296091	3,916,575.296091	1979.03393	-38.91%	38.91%	38.91%	-1,979	-1
-2,233	2232.923773	4,985,948.574587	4,451,261.935339	2105.978851	-21.95%	21.95%	30.43%	-4,212	-2
-664	664.3940857	441,419.501128	3,114,647.790602	1625.450596	-9.80%	9.80%	23.55%	-4,876	-3
-150	150.3422278	22,602.785449	2,341,636.539314	1256.673504	-1.27%	1.27%	17.98%	-5,027	-4
-26	26.14695571	683.6632927	1,873,445.964109	1010.568194	-0.41%	0.41%	14.47%	-5,053	-5
-119	119.19188	14206.70426	1,563,572.754134	862.0054753	-0.93%	0.93%	12.21%	-5,172	-6
47	46.62931356	2174.292883	1,340,515.831099	745.5231665	0.54%	0.54%	10.54%	-5,125	-6.874908479
369	368.5122782	135801.2992	1,189,926.514609	698.3968055	2.45%	2.45%	9.53%	-4,757	-6.811158389

Demand vs. Forecast



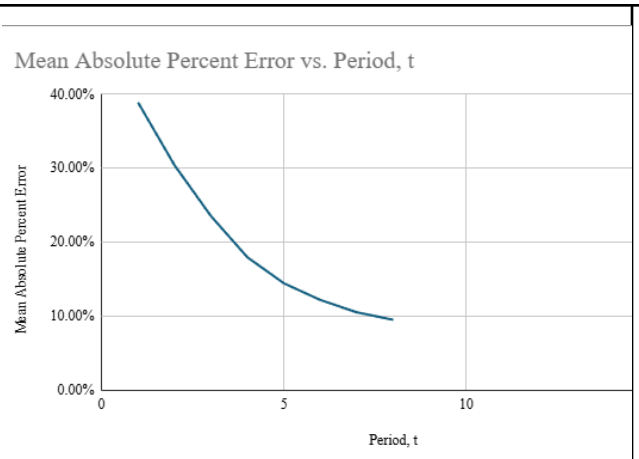
This graph compares actual demand with forecasted values across periods. Demand fluctuates with noticeable seasonal peaks, while the forecast follows a smoother upward pattern. The forecast captures overall growth but does not fully match sudden demand changes. Over time, the forecast begins aligning more closely with demand levels. Overall, the model tracks long term trends better than short term variation.

Period vs. MAD



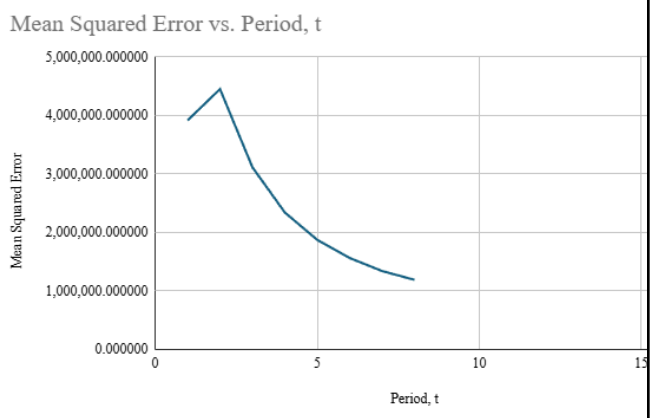
This chart shows mean absolute deviation decreasing as time progresses. Early periods have larger average forecasting errors. Over time, the error size becomes smaller and more stable. Since MAD reflects typical forecast deviation, this indicates steady improvement in performance. Overall, the forecasting model produces closer predictions in later periods.

MAPE vs. Period



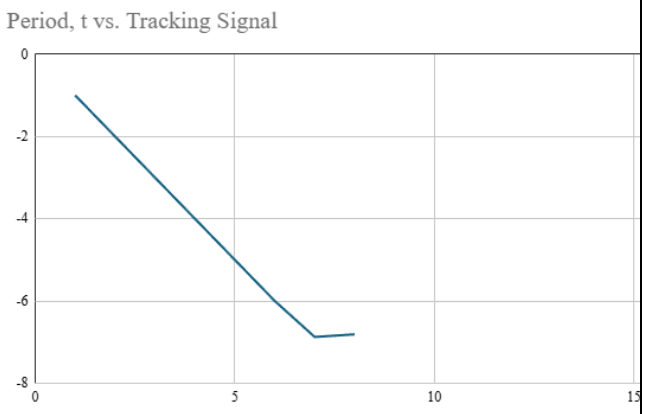
This graph tracks percentage forecasting error over time. The percent error starts high and steadily decreases each period. This means forecasts become more accurate relative to demand size. The consistent decline shows strong improvement in proportional accuracy. Overall, the forecasting model becomes more reliable over time.

MSE vs. Period



This chart shows mean squared error declining across periods. Error is highest early on and then drops steadily over time. Since MSE penalizes large mistakes more heavily, this decrease shows the model is reducing major forecasting errors. The improvement suggests the forecasting method learns as more data becomes available. Overall, forecast accuracy improves consistently across periods.

Period vs. Tracking Signal



This graph shows the tracking signal steadily decreasing over time. The values move further into negative territory as periods increase. This indicates the forecasting model is consistently underestimating demand. The strong downward trend suggests systematic bias rather than random error. Overall, the model needs adjustment because forecasts remain lower than actual demand.

Forecasting: Experimenting with Different Constants

Forecasting and Error Analysis given $\alpha = 0.1$ $\beta = 0.1$ $\gamma = 0.1$

Level	Trend	Seasonal Factor	Forecast	Forecast Error	Absolute Error	Squared Error	Mean Squared Error	Mean Absolute Deviation	Percent Error	Absolute Percent	Mean Absolute Percent Error	Bias	Tracking Signal
33252.325	750.5875												
35,711	921	0.414800	14,104	-7,086	7085.591895	50,205,612.502490	50,205,612.502490	7085.591895	-33.44%	33.44%	33.44%	-7,086	-1
38,186	1,077	0.812400	29,760	-12,620	12619.74543	159,257,974.670011	104,731,793.586250	9852.668662	-29.78%	29.78%	31.61%	-19,705	-2
40,661	1,217	0.530600	20,833	-7,420	7420.237429	55,059,923.498517	88,174,503.557006	9041.858251	-26.26%	26.26%	29.83%	-27,126	-3
43,123	1,341	0.910100	38,113	-11,330	11330.10091	128,371,186.701350	98,223,674.343092	9613.918916	-22.92%	22.92%	28.10%	-38,456	-4
45,331	1,428	0.432657	19,238	-3,753	3753.4482	14088373.39	81,396,614.153147	8441.824773	-16.33%	16.33%	25.74%	-42,209	-5
47,543	1,506	0.842143	39,378	-6,605	6605.141677	43627896.58	75,101,827.890261	8135.710924	-14.36%	14.36%	23.85%	-48,814	-6
49,749	1,576	0.547024	26,831	-3,824	3823.639804	14620221.35	66,461,598.384574	7519.700764	-12.47%	12.47%	22.22%	-52,638	-7
51,938	1,637	0.933747	47,924	-5,722	5721.592396	32736619.55	62,245,976.030441	7294.937218	-10.67%	10.67%	20.78%	-58,359	-8
		0.440109	23,579										
		0.854647	47,187										
		0.553941	31,492										
		0.943661	55,192										
			26,461										
			52,785										
			35,120										
			61,373										

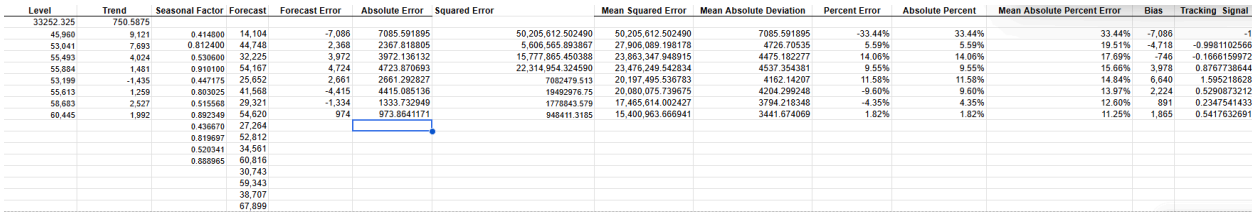
Forecasting and Error Analysis given $\alpha = 0.3$ $\beta = 0.3$ $\gamma = 0.3$

Level	Trend	Seasonal Factor	Forecast	Forecast Error	Absolute Error	Squared Error	Mean Squared Error	Mean Absolute Deviation	Percent Error	Absolute Percent	Mean Absolute Percent Error	Bias	Tracking Signal
33252.325	750.5875												
39,127	2,288	0.414800	14,104	-7,086	7085.591895	50,205,612.502490	50,205,612.502490	7085.591895	-33.44%	33.44%	33.44%	-7,086	-1
44,641	3,256	0.812400	33,646	-8,734	8734.080627	76,284,164.397091	63,244,888.449791	7909.836261	-20.61%	20.61%	27.02%	-15,820	-2
49,502	3,737	0.530600	25,414	-2,839	2839.224743	8,061,197.139200	44,850,324.679594	6219.632422	-10.05%	10.05%	21.37%	-18,659	-3
53,585	3,835	0.910100	48,453	-990	990.436609	980,964.676461	33,882,984.678811	4912.333468	-2.00%	2.00%	16.52%	-19,649	-4
55,412	3,239	0.452829	25,993	3,002	3001.507054	9009044.596	28,908,196.662196	4530.168186	13.06%	13.06%	15.83%	-16,648	-3.67488052
57,218	2,809	0.853487	50,057	4,074	4074.283093	16599782.72	26,856,794.338727	4454.187337	8.86%	8.86%	14.67%	-12,574	-2.822859205
58,967	2,491	0.542645	32,573	1,918	1918.396195	3680243.962	23,545,858.570609	4091.931459	6.26%	6.26%	13.47%	-10,655	-2.603940862
60,628	2,242	0.913983	56,171	2,525	2524.875555	6374996.568	21,399,500.820261	3896.049471	4.71%	4.71%	12.37%	-8,130	-2.086798958
		0.441454	27,754										
		0.838534	54,599										
		0.535813	36,090										
		0.905238	63,002										
			31,714										
			62,119										
			40,895										
			71,120										

Forecasting and Error Analysis given $\alpha = 0.5$ $\beta = 0.5$ $\gamma = 0.5$

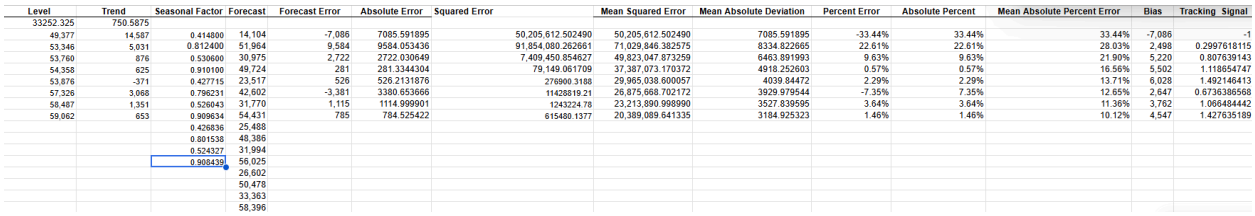
Level	Trend	Seasonal Factor	Forecast	Forecast Error	Absolute Error	Squared Error	Mean Squared Error	Mean Absolute Deviation	Percent Error	Absolute Percent	Mean Absolute Percent Error	Bias	Tracking Signal
33252.325	750.5875												
42,544	5,021	0.414800	14,104	-7,086	7085.591895	50,205,612.502490	50,205,612.502490	7085.591895	-33.44%	33.44%	33.44%	-7,086	-1
49,866	6,171	0.812400	38,642	-3,738	3738.225882	13,974,332.748512	32,089,972.625501	5411.908889	-8.82%	8.82%	21.13%	-10,824	-2
54,642	5,474	0.530600	29,733	1,480	1480.301162	2,191,291.529018	22,123,745.593340	4101.37298	5.24%	5.24%	15.83%	-9,344	-2.278143603
57,222	4,027	0.910100	54,712	5,269	5268.728388	27,759,498.830320	23,532,683.902585	4393.211832	10.66%	10.66%	14.54%	-4,075	-0.9275191781
55,809	1,307	0.456437	27,956	4,965	4964.971487	24650941.87	23,756,335.495437	4507.563763	21.60%	21.60%	15.95%	890	0.1974865595
56,221	859	0.831141	47,472	1,489	1489.050171	2217270.412	20,166,491.314812	4004.478164	3.24%	3.24%	13.83%	2,379	0.5941431899
57,801	1,220	0.523827	29,900	-755	754.7948206	569715.2213	17,366,951.872878	3540.237687	-2.46%	2.46%	12.21%	1,624	0.4588501546
59,748	1,583	0.887081	52,356	-1,290	1290.181516	1664568.344	15,404,153.931814	3258.980665	-2.40%	2.40%	10.98%	334	0.1025649208
		0.434196	26,630										
		0.824520	51,874										
		0.527092	33,996										
		0.892479	58,975										
			29,379										
			57,095										
			37,334										
			64,627										

Forecasting and Error Analysis given $\alpha = 0.7$ $\beta = 0.7$ $\gamma = 0.7$



Level	Trend	Seasonal Factor	Forecast	Forecast Error	Absolute Error	Squared Error	Mean Squared Error	Mean Absolute Deviation	Percent Error	Absolute Percent	Mean Absolute Percent Error	Bias	Tracking Signal
33252.325	750.5875												
45,960	9,121	0.414800	14,104	-7,086	7085.591895	50,205,612.502490	50,205,612.502490	7085.591895	-33.44%	33.44%	33.44%	-7,086	-1
53,041	7,693	0.812400	44,748	2,368	2367.818805	5,606,565.893867	27,906,089.198178	4726.70535	5.59%	5.59%	19.51%	-4,718	-0.9981102566
55,493	4,024	0.530600	32,225	3,972	3972.136132	15,777,865.450388	23,863,347.948915	4475.182277	14.06%	14.06%	17.69%	-746	-0.1666159972
55,884	1,481	0.910100	54,167	4,724	4723.870693	22,314,954.324590	23,476,249.542834	4537.354381	9.55%	9.55%	15.66%	3,978	0.8767738644
53,199	-1,435	0.447175	25,652	2,661	2661.292827	7082479.513	20,197,495.536783	4162.14207	11.58%	11.58%	14.84%	6,640	1.595218628
55,613	1,259	0.803025	41,568	-4,415	4415.085136	19492976.75	20,080,075.739675	4204.299248	-9.60%	9.60%	13.97%	2,224	0.5290873212
58,683	2,527	0.515568	29,321	-1,334	1333.732949	1778843.579	17,465,614.002427	3794.218348	-4.35%	4.35%	12.60%	891	0.2347541433
60,445	1,992	0.892349	54,620	974	973.8641171	948411.3185	15,400,963.666941	3441.674069	1.82%	1.82%	11.25%	1,865	0.5417632691
		0.436670	27,264										
		0.819697	52,812										
		0.520341	34,561										
		0.888965	60,816										
			30,743										
			59,343										
			38,707										
			67,899										

Forecasting and Error Analysis given $\alpha = 0.9$ $\beta = 0.9$ $\gamma = 0.9$



Level	Trend	Seasonal Factor	Forecast	Forecast Error	Absolute Error	Squared Error	Mean Squared Error	Mean Absolute Deviation	Percent Error	Absolute Percent	Mean Absolute Percent Error	Bias	Tracking Signal
33252.325	750.5875												
49,377	14,587	0.414800	14,104	-7,086	7085.591895	50,205,612.502490	50,205,612.502490	7085.591895	-33.44%	33.44%	33.44%	-7,086	-1
53,346	5,031	0.812400	51,964	9,584	9584.053436	91,854,080.262661	71,029,846.382575	8334.822665	22.61%	22.61%	28.03%	2,498	0.2997618115
53,760	876	0.530600	30,975	2,722	2722.030649	7,409,450.854627	49,823,047.873259	6463.891993	9.63%	9.63%	21.90%	5,220	0.807639143
54,358	625	0.910100	49,724	281	281.3344304	79,149.061709	37,387,073.170372	4918.252603	0.57%	0.57%	16.56%	5,502	1.118654747
53,876	-371	0.427715	23,517	526	526.2131876	276900.3188	29,965,038.600057	4039.84472	2.29%	2.29%	13.71%	6,028	1.492146413
57,326	3,068	0.796231	42,602	-3,381	3380.653666	11428819.21	26,875,668.702172	3929.979544	-7.35%	7.35%	12.65%	2,647	0.6736386568
58,487	1,351	0.526043	31,770	1,115	1114.999901	1243224.78	23,213,890.998990	3527.839595	3.64%	3.64%	11.36%	3,762	1.066484442
59,062	653	0.909634	54,431	785	784.525422	615480.1377	20,389,089.641335	3184.925323	1.46%	1.46%	10.12%	4,547	1.427635189
		0.426836	25,488										
		0.801538	48,386										
		0.524327	31,994										
		0.908439	56,025										
			26,602										
			50,478										
			33,363										
			58,396										

Financial Model Appendix

Inputs

Base Case followed “Excel Tutorial” by R.Polany and S.Desa

Discount factor source (see last page):
https://extapps.dec.ny.gov/data/DecDocs/915055/Report.HW.915055.2024-02-02.Site%20109%20Feasibility%20Study%20_Final.pdf

R&D engineer average wage source:
https://www.ziprecruiter.com/Salaries/R-D-Engineer-Salary

Compared security systems to determine our unit price:
Simplisafe: https://simplisafe.com/home-security-shop-packages
Ring: https://ring.com/products/security-system-alarm-pro-8
Ecobee: https://www.ecobee.com/en-us/bundles-exclusives/home-security-bundle/

Ramp up cost percentage:
https://en.tacto.ai/buyer-lexicon/ramp-up-management#:~:text=How%20are%20ramp%2Dup%20costs,phase%20enable%20early%20control%20measures.

Marketing percentage:
https://www.prontomarketing.com/blog/digital-marketing-cost-small-businesses/#:~:text=The%20Revenue%2DBased%20Approach,%2D25%25%20of%20revenue%20range.

Discount factor percentage:
https://www.forentrepreneurs.com/discount-rate-for-dcf/#:~:text=For%20SaaS%20companies%20using%20DCF,than%2040%25%20year%20on%20year)

Sales Volume Sensitivity (±10%, 20, 30)

Sales Volume Analysis:	10%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection					
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4		
Development Cost	-15.12	-15.12	-15.12	-15.12																		
Ramp-up cost				-3.02	-3.02																	
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00		
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26		
Production volume					3,418	8,732	6,776	13,060	7,170	14,275	9,637	17,145	9,087	17,837	11,844	20,839	11,131	21,636	14,239	24,850		
Unit product cost					-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10		
Sales revenue					1,021.88	2,610.83	2,026.14	3,905.00	2,143.85	4,268.35	2,881.60	5,126.28	2,716.99	5,333.33	3,541.35	6,230.79	3,328.18	6,469.21	4,257.45	7,430.09		
Forecasted Sales Volume					3,418	8,732	6,776	13,060	7,170	14,275	9,637	17,145	9,087	17,837	11,844	20,839	11,131	21,636	14,239	24,850		
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30		
Period Cash Flow	-15	-15	-15	-18	451	1,486	1,106	2,326	1,182	2,562	1,662	3,120	1,555	3,254	2,090	3,837	1,952	3,992	2,555	4,616		
Unit Price	-15	-15	-15	-17	420	1,362	997	2,060	1,029	2,192	1,397	2,578	1,262	2,597	1,639	2,958	1,479	2,972	1,870	3,320		

Sales Volume Analysis:	-10%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection					
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4		
Development Cost	-15.12	-15.12	-15.12	-15.12																		
Ramp-up cost				-3.02	-3.02																	
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00		
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26		
Production volume					2,796	7,144	5,544	10,686	5,866	11,680	7,885	14,028	7,435	14,594	9,691	17,050	9,107	17,702	11,650	20,332		
Unit product cost					-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13		
Sales revenue					836.08	2,136.14	1,657.75	3,195.00	1,754.06	3,492.28	2,357.67	4,194.23	2,222.99	4,363.63	2,897.47	5,097.92	2,723.06	5,292.99	3,483.37	6,079.16		
Forecasted Sales Volume					2,796	7,144	5,544	10,686	5,866	11,680	7,885	14,028	7,435	14,594	9,691	17,050	9,107	17,702	11,650	20,332		
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30		
Period Cash Flow	-15	-15	-15	-18	265	1,011	738	1,616	793	1,786	1,138	2,187	1,061	2,284	1,446	2,704	1,347	2,816	1,781	3,265		
Unit Price	-15	-15	-15	-17	247	927	665	1,431	690	1,528	956	1,807	861	1,823	1,134	2,084	1,020	2,096	1,303	2,348		

Sales Volume Analysis:	20%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection					
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4		
Development Cost	-15.12	-15.12	-15.12	-15.12																		
Ramp-up cost				-3.02	-3.02																	
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00		
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26		
Production volume					3,728	9,526	7,392	14,248	7,822	15,573	10,514	18,703	9,913	19,459	12,921	22,733	12,143	23,603	15,533	27,109		
Unit product cost					-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10		
Sales revenue					1,114.78	2,848.18	2,210.34	4,260.00	2,338.75	4,656.38	3,143.56	5,592.31	2,963.99	5,818.18	3,863.30	6,797.22	3,630.75	7,057.32	4,644.49	8,105.55		
Forecasted Sales Volume					3,728	9,526	7,392	14,248	7,822	15,573	10,514	18,703	9,913	19,459	12,921	22,733	12,143	23,603	15,533	27,109		
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30		
Period Cash Flow	-15	-15	-15	-18	544	1,723	1,290	2,681	1,377	2,950	1,924	3,586	1,802	3,739	2,412	4,403	2,254	4,580	2,942	5,291		
Unit Price	-15	-15	-15	-17	507	1,580	1,163	2,375	1,199	2,524	1,617	2,963	1,463	2,984	1,892	3,394	1,708	3,410	2,153	3,805		

Sales Volume Analysis:	-20%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection					
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4		
Development Cost	-15.12	-15.12	-15.12	-15.12																		
Ramp-up cost				-3.02	-3.02																	
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00		
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26		
Production volume					2,486	6,350	4,928	9,498	5,215	10,382	7,009	12,469	6,609	12,973	8,614	15,155	8,095	15,735	10,356	18,073		
Unit product cost					-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14		
Sales revenue					743.19	1,898.79	1,473.56	2,840.00	1,559.17	3,104.25	2,095.71	3,728.20	1,975.99	3,878.79	2,575.53	4,531.48	2,420.50	4,704.88	3,096.33	5,403.70		
Forecasted Sales Volume					2,486	6,350	4,928	9,498	5,215	10,382	7,009	12,469	6,609	12,973	8,614	15,155	8,095	15,735	10,356	18,073		
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30		
Period Cash Flow	-15	-15	-15	-18	172	774	553	1,261	598	1,398	876	1,721	814	1,799	1,124	2,138	1,044	2,227	1,394	2,589		
Unit Price	-15	-15	-15	-17	160	709	499	1,117	520	1,196	736	1,422	661	1,436	882	1,648	791	1,658	1,020	1,862		

Sales Volume Analysis:	30%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection					
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4		
Development Cost	-15.12	-15.12	-15.12	-15.12																		
Ramp-up cost				-3.02	-3.02																	
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00		
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26		
Production volume					4,039	10,319	8,008	15,435	8,474	16,871	11,390	20,262	10,739	21,080	13,997	24,628	13,155	25,570	16,828	29,368		
Unit product cost					-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09		
Sales revenue					1,207.68	3,085.53	2,394.53	4,615.00	2,533.65	5,044.41	3,405.52	6,058.33	3,210.98	6,303.03	4,185.24	7,363.66	3,933.31	7,645.43	5,031.54	8,781.01		
Forecasted Sales Volume					4,039	10,319	8,008	15,435	8,474	16,871	11,390	20,262	10,739	21,080	13,997	24,628	13,155	25,570	16,828	29,368		
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30		
Period Cash Flow	-15	-15	-15	-18	636	1,960	1,474	3,036	1,572	3,338	2,186	4,052	2,049	4,224	2,734	4,970	2,557	5,168	3,329	5,967		
Unit Price	-15	-15	-15	-17	594	1,798	1,329	2,689	1,368	2,856	1,837	3,348	1,664	3,371	2,144	3,831	1,937	3,848	2,436	4,291		

Sales Volume Analysis:	-30%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection					
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4		
Development Cost	-15.12	-15.12	-15.12	-15.12																		
Ramp-up cost				-3.02	-3.02																	
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00		
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26		
Production volume					2,175	5,557	4,312	8,311	4,563	9,084	6,133	10,910	5,783	11,351	7,537	13,261	7,083	13,768	9,061	15,814		
Unit product cost					-0.16	-0.16	-0.16	-0.16	-0.16	-0.16	-0.16	-0.16	-0.16	-0.16	-0.16	-0.16	-0.16	-0.16	-0.16	-0.16		
Sales revenue					650.29	1,661.44	1,289.36	2,485.00	1,364.27	2,716.22	1,833.74	3,262.18	1,728.99	3,393.94	2,253.59	3,965.05	2,117.94	4,116.77	2,709.29	4,728.24		
Forecasted Sales Volume					2,175	5,557	4,312	8,311	4,563	9,084	6,133	10,910	5,783	11,351	7,537	13,261	7,083	13,768	9,061	15,814		
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30		
Period Cash Flow	-15	-15	-15	-18	79	536	369	906	403	1,010	614	1,255	567	1,315	802	1,571	741	1,639	1,007	1,914		
Unit Price	-15	-15	-15	-17	74	492	333	803	351	864	516	1,037	460	1,049	629	1,211	562	1,221	737	1,377		

Development Cost Sensitivity (±10%, 20, 30)

Sales Volume Analysis:	10%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection					
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4		
Development Cost	-15.12	-15.12	-15.12	-15.12																		
Ramp-up cost				-3.02	-3.02																	
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00		
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26		
Production volume					3,418	8,732	6,776	13,060	7,170	14,275	9,637	17,145	9,087	17,837	11,844	20,839	11,131	21,636	14,239	24,850		
Unit product cost					-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10		
Sales revenue					1,021.88	2,610.83	2,026.14	3,905.00	2,143.85	4,268.35	2,881.60	5,126.28	2,716.99	5,333.33	3,541.35	6,230.79	3,328.18	6,469.21	4,257.45	7,430.09		
Forecasted Sales Volume					3,418	8,732	6,776	13,060	7,170	14,275	9,637	17,145	9,087	17,837	11,844	20,839	11,131	21,636	14,239	24,850		
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30		
Period Cash Flow	-15	-15	-15	-18	451	1,486	1,106	2,326	1,182	2,562	1,662	3,120	1,555	3,254	2,090	3,837	1,952	3,992	2,555	4,616		
Unit Price	-15	-15	-15	-17	420	1,362	997	2,060	1,029	2,192	1,397	2,578	1,262	2,597	1,639	2,958	1,479	2,972	1,870	3,320		
Project NVP	30,070																					

Sales Volume Analysis:	-10%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection					
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4		
Development Cost	-15.12	-15.12	-15.12	-15.12																		
Ramp-up cost				-3.02	-3.02																	
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00		
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26		
Production volume					2,796	7,144	5,544	10,686	5,866	11,680	7,885	14,028	7,435	14,594	9,691	17,050	9,107	17,702	11,650	20,332		
Unit product cost					-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13		
Sales revenue					836.08	2,136.14	1,657.75	3,195.00	1,754.06	3,492.28	2,357.67	4,194.23	2,222.99	4,363.63	2,897.47	5,097.92	2,723.06	5,292.99	3,483.37	6,079.16		
Forecasted Sales Volume					2,796	7,144	5,544	10,686	5,866	11,680	7,885	14,028	7,435	14,594	9,691	17,050	9,107	17,702	11,650	20,332		
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30		
Period Cash Flow	-15	-15	-15	-18	265	1,011	738	1,616	793	1,786	1,138	2,187	1,061	2,284	1,446	2,704	1,347	2,816	1,781	3,265		
Unit Price	-15	-15	-15	-17	247	927	665	1,431	690	1,528	956	1,807	861	1,823	1,134	2,084	1,020	2,096	1,303	2,348		

Development Cost Analysis:	20%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection					
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4		
Development Cost	-18.14	-18.14	-18.14	-18.14																		
Ramp-up cost				-3.02	-3.02																	
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00		
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26		
Production volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72		
Unit product cost					-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12		
Sales revenue					928.98	2,373.48	1,841.95	3,550.00	1,948.96	3,880.32	2,619.63	4,660.26	2,469.99	4,848.48	3,219.41	5,664.35	3,025.62	5,881.10	3,870.41	6,754.63		
Forecasted Sales Volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72		
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30		
Period Cash Flow	-18	-18	-18	-21	358	1,248	922	1,971	988	2,174	1,400	2,653	1,308	2,769	1,768	3,270	1,649	3,404	2,168	3,940		
Unit Price	-18	-18	-18	-20	334	1,145	831	1,746	860	1,860	1,177	2,192	1,062	2,210	1,387	2,521	1,249	2,534	1,587	2,834		

Development Cost Analysis:	-20%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection					
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4		
Development Cost	-12.10	-12.10	-12.10	-12.10																		
Ramp-up cost				-3.02	-3.02																	
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00		
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26		
Production volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72		
Unit product cost					-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12		
Sales revenue					928.98	2,373.48	1,841.95	3,550.00	1,948.96	3,880.32	2,619.63	4,660.26	2,469.99	4,848.48	3,219.41	5,664.35	3,025.62	5,881.10	3,870.41	6,754.63		
Forecasted Sales Volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72		
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30		
Period Cash Flow	-12	-12	-12	-15	358	1,248	922	1,971	988	2,174	1,400	2,653	1,308	2,769	1,768	3,270	1,649	3,404	2,168	3,940		
Unit Price	-12	-12	-12	-14	334	1,145	831	1,746	860	1,860	1,177	2,192	1,062	2,210	1,387	2,521	1,249	2,534	1,587	2,834		

Development Cost Analysis:	30%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection					
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4		
Development Cost	-19.66	-19.66	-19.66	-19.66																		
Ramp-up cost				-3.02	-3.02																	
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00		
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26		
Production volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72		
Unit product cost					-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12		
Sales revenue					928.98	2,373.48	1,841.95	3,550.00	1,948.96	3,880.32	2,619.63	4,660.26	2,469.99	4,848.48	3,219.41	5,664.35	3,025.62	5,881.10	3,870.41	6,754.63		
Forecasted Sales Volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72		
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30		
Period Cash Flow	-20	-20	-20	-23	358	1,248	922	1,971	988	2,174	1,400	2,653	1,308	2,769	1,768	3,270	1,649	3,404	2,168	3,940		
Unit Price	-20	-19	-19	-22	334	1,145	831	1,746	860	1,860	1,177	2,192	1,062	2,210	1,387	2,521	1,249	2,534	1,587	2,834		

Development Cost Analysis:	-30%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection					
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4		
Development Cost	-10.58	-10.58	-10.58	-10.58																		
Ramp-up cost				-3.02	-3.02																	
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00		
Production Cost					-358.17	-915.10	-710.17	-1,368.71	-751.42	-1,496.06	-1,010.00	-1,796.77	-952.31	-1,869.34	-1,241.25	-2,183.90	-1,166.53	-2,267.47	-1,492.24	-2,604.26		
Production volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72		
Unit product cost					-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12	-0.12		
Sales revenue					928.98	2,373.48	1,841.95	3,550.00	1,948.96	3,880.32	2,619.63	4,660.26	2,469.99	4,848.48	3,219.41	5,664.35	3,025.62	5,881.10	3,870.41	6,754.63		
Forecasted Sales Volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72		
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30		
Period Cash Flow	-11	-11	-11	-14	358	1,248	922	1,971	988	2,174	1,400	2,653	1,308	2,769	1,768	3,270	1,649	3,404	2,168	3,940		
Unit Price	-11	-10	-10	-13	334	1,145	831	1,746	860	1,860	1,177	2,192	1,062	2,210	1,387	2,521	1,249	2,534	1,587	2,834		

Unit Cost Sensitivity (±10%, 20, 30)

Unit Cost Analysis:	10%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection			
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Development Cost	-15.12	-15.12	-15.12	-15.12																
Ramp-up cost				-3.02	-3.02															
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00
Production Cost					-393.99	-1,006.61	-781.18	-1,505.58	-826.57	-1,645.67	-1,111.01	-1,976.45	-1,047.54	-2,056.28	-1,365.38	-2,402.29	-1,283.19	-2,494.21	-2,494.21	-2,864.68
Production volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit product cost					-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.13	-0.19	-0.13
Sales revenue					928.98	2,373.48	1,841.95	3,550.00	1,948.96	3,880.32	2,619.63	4,660.26	2,469.99	4,848.48	3,219.41	5,664.35	3,025.62	5,881.10	3,870.41	6,754.63
Forecasted Sales Volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Period Cash Flow	-15	-15	-15	-18	322	1,157	851	1,834	912	2,025	1,299	2,474	1,212	2,582	1,644	3,052	1,532	3,177	1,166	3,680
Unit Price	-15	-15	-15	-17	300	1,061	767	1,625	794	1,732	1,092	2,044	985	2,061	1,290	2,353	1,161	2,365	853	2,647

Unit Cost Analysis:	-10%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection			
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Development Cost	-15.12	-15.12	-15.12	-15.12																
Ramp-up cost				-3.02	-3.02															
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00
Production Cost					-322.35	-823.59	-639.15	-1,231.84	-676.28	-1,346.46	-909.00	-1,617.09	-857.08	-1,682.41	-1,117.13	-1,965.51	-1,049.88	-2,040.72	-2,040.72	-2,343.83
Production volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit product cost					-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.10	-0.16	-0.10
Sales revenue					928.98	2,373.48	1,841.95	3,550.00	1,948.96	3,880.32	2,619.63	4,660.26	2,469.99	4,848.48	3,219.41	5,664.35	3,025.62	5,881.10	3,870.41	6,754.63
Forecasted Sales Volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Period Cash Flow	-15	-15	-15	-18	394	1,340	993	2,108	1,063	2,324	1,501	2,833	1,403	2,956	1,892	3,489	1,766	3,630	1,620	4,201
Unit Price	-15	-15	-15	-17	367	1,229	895	1,867	925	1,988	1,262	2,341	1,139	2,359	1,484	2,689	1,338	2,703	1,185	3,021

Unit Cost Analysis:	20%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection			
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Development Cost	-15.12	-15.12	-15.12	-15.12																
Ramp-up cost				-3.02	-3.02															
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00
Production Cost					-429.81	-1,098.12	-852.20	-1,642.45	-901.71	-1,795.28	-1,212.01	-2,156.12	-1,142.77	-2,243.21	-1,489.50	-2,620.68	-1,399.84	-2,720.96	-2,720.96	-3,125.11
Production volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit product cost					-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.14	-0.21	-0.14
Sales revenue					928.98	2,373.48	1,841.95	3,550.00	1,948.96	3,880.32	2,619.63	4,660.26	2,469.99	4,848.48	3,219.41	5,664.35	3,025.62	5,881.10	3,870.41	6,754.63
Forecasted Sales Volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Period Cash Flow	-15	-15	-15	-18	286	1,065	780	1,698	837	1,875	1,198	2,294	1,117	2,395	1,520	2,834	1,416	2,950	939	3,420
Unit Price	-15	-15	-15	-17	267	977	703	1,503	729	1,604	1,007	1,896	907	1,912	1,192	2,184	1,073	2,197	687	2,459

Unit Cost Analysis:	-20%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection			
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Development Cost	-15.12	-15.12	-15.12	-15.12																
Ramp-up cost				-3.02	-3.02															
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00
Production Cost					-286.54	-732.08	-568.13	-1,094.97	-601.14	-1,196.85	-808.00	-1,437.42	-761.85	-1,495.47	-993.00	-1,747.12	-933.23	-1,813.97	-1,813.97	-2,083.41
Production volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit product cost					-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.09	-0.14	-0.09
Sales revenue					928.98	2,373.48	1,841.95	3,550.00	1,948.96	3,880.32	2,619.63	4,660.26	2,469.99	4,848.48	3,219.41	5,664.35	3,025.62	5,881.10	3,870.41	6,754.63
Forecasted Sales Volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Period Cash Flow	-15	-15	-15	-18	429	1,431	1,064	2,245	1,138	2,473	1,602	3,013	1,498	3,143	2,016	3,707	1,882	3,857	1,846	4,461
Unit Price	-15	-15	-15	-17	401	1,312	959	1,988	990	2,116	1,347	2,489	1,217	2,508	1,582	2,858	1,426	2,872	1,351	3,208

Unit Cost Analysis:	30%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection			
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Development Cost	-15.12	-15.12	-15.12	-15.12																
Ramp-up cost				-3.02	-3.02															
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00
Production Cost					-465.62	-1,189.63	-923.22	-1,779.32	-976.85	-1,944.88	-1,313.01	-2,335.80	-1,238.00	-2,430.14	-1,613.63	-2,839.07	-1,516.49	-2,947.71	-2,947.71	-3,385.54
Production volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit product cost					-0.15	-0.15	-0.15	-0.15	-0.15	-0.15	-0.15	-0.15	-0.15	-0.15	-0.15	-0.15	-0.15	-0.15	-0.23	-0.15
Sales revenue					928.98	2,373.48	1,841.95	3,550.00	1,948.96	3,880.32	2,619.63	4,660.26	2,469.99	4,848.48	3,219.41	5,664.35	3,025.62	5,881.10	3,870.41	6,754.63
Forecasted Sales Volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Period Cash Flow	-15	-15	-15	-18	250	974	709	1,561	762	1,725	1,097	2,114	1,022	2,208	1,396	2,615	1,299	2,723	713	3,159
Unit Price	-15	-15	-15	-17	234	893	639	1,382	663	1,476	922	1,747	830	1,762	1,095	2,016	984	2,028	522	2,272

Unit Cost Analysis:	-30%																					
	Year 0, Development				Year 1, Projection				Year 2, Projection				Year 3, Projection				Year 4, Projection			
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
($Value in thousands)	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Development Cost	-15.12	-15.12	-15.12	-15.12																
Ramp-up cost				-3.02	-3.02															
Marketing & support cost					-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00	-210.00
Production Cost					-250.72	-640.57	-497.12	-958.10	-526.00	-1,047.24	-707.00	-1,257.74	-666.62	-1,308.54	-868.88	-1,528.73	-816.57	-1,587.23	-1,587.23	-1,822.98
Production volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit product cost					-0.08	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08	-0.08	-0.12	-0.08
Sales revenue					928.98	2,373.48	1,841.95	3,550.00	1,948.96	3,880.32	2,619.63	4,660.26	2,469.99	4,848.48	3,219.41	5,664.35	3,025.62	5,881.10	3,870.41	6,754.63
Forecasted Sales Volume					3,106.97	7,938.08	6,160.36	11,872.92	6,518.26	12,977.65	8,761.32	15,586.14	8,260.83	16,215.66	10,767.27	18,944.32	10,119.14	19,669.22	12,944.52	22,590.72
Unit price					0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Period Cash Flow	-15	-15	-15	-18	465	1,523	1,135	2,382	1,213	2,623	1,703	3,193	1,593	3,330	2,141	3,926	1,999	4,084	2,073	4,722
Unit Price	-15	-15	-15	-17	434	1,396	1,023	2,110	1,056	2,244	1,431	2,638	1,294	2,658	1,679	3,026	1,515	3,041	1,517	3,396